

•

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Expand description

[§](#)Barter

[Barter](#) is an open-source Rust framework for building **event-driven live-trading & backtesting systems**. Algorithmic trade with the peace of mind that comes from knowing your strategies have been backtested with a near-identical trading Engine. It is:

- **Fast:** Barter provides a multi-threaded trading Engine framework built in high-performance Rust (in-rust-we-trust).
- **Easy:** Barter provides a modularised data architecture that focuses on simplicity.
- **Customisable:** A set of traits define how every Barter component communicates, providing a highly extensible framework for trading.

See [Readme](#).

[§](#)Overview

Barter is an open-source Rust framework for building **event-driven live-trading & backtesting systems**. It provides a high-performance, easy to customise, trading Engine that enables backtesting strategies on a near-identical system to live trading. The Engine can be **controlled by issuing Commands** over the Engine’s `command_tx`. Similarly, the **Engine’s Events can be listened to using the `event_rx`** (useful for event-sourcing). At a

high level, it provides several de-coupled components that interact via a set of traits:

- **Data:** Continuer & MarketGenerator traits govern the generation of a MarketEvents data feed that acts as the system heartbeat. For example, a LiveCandleHandler implementation is provided utilising [Barter-Data](#)'s WebSocket functionality to provide a live market Candle data feed to the system.
- **Strategy:** The SignalGenerator trait governs potential generation of SignalEvents after analysing incoming MarketEvents. SignalEvents are advisory signals sent to the Portfolio for analysis.
- **Portfolio:** MarketUpdater, OrderGenerator, and FillUpdater govern global state Portfolio implementations. A Portfolio may generate OrderEvents after receiving advisory SignalEvents from a Strategy. The Portfolio's state updates after receiving MarketEvents and FillEvents.
- **Execution:** The FillGenerator trait governs the generation of FillEvents after receiving OrderEvents from the Portfolio. For example, a SimulatedExecution handler implementation is provided for simulating any exchange execution behaviour required in dry-trading or backtesting runs.
- **Statistic:** Provides metrics such as Sharpe Ratio, Calmar Ratio, and Max Drawdown to analyse trading session performance. One-pass dispersion algorithms analyse each closed Position and efficiently calculates a trading summary.
- **Trader:** Capable of trading a single market pair using a customisable selection of it's own Data, Strategy & Execution instances, as well as shared access to a global Portfolio.
- **Engine:** Multi-threaded trading Engine capable of trading with an arbitrary number of Trader market pairs. Each contained Trader instance operates on its own thread.

[§Getting Started](#)

[§Data Handler](#)

```
use barter::{data::{Feed, historical, MarketGenerator}, test_util};
use barter_integration::model::Side;

let mut data = historical::MarketFeed::new([test_util::market_event_trade(
loop {
    let market_event = match data.next() {
        Feed::Next(market_event) => market_event,
        Feed::Finished => break,
        Feed::Unhealthy => continue,
    };
}
```

§Strategy

```
use barter::{
    strategy::{SignalGenerator, example::{Config as StrategyConfig, RSIStr
    test_util,
};
use barter_integration::model::Side;

let config = StrategyConfig {
    rsi_period: 14,
};

let mut strategy = RSIStrategy::new(config);

let market_event = test_util::market_event_trade(Side::Buy);

let signal_event = strategy.generate_signal(&market_event);
```

§Portfolio

```
use barter::{
    portfolio::{
        MarketUpdater, OrderGenerator, FillUpdater,
        portfolio::{PortfolioLego, MetaPortfolio},
        repository::in_memory::InMemoryRepository,
        allocator::DefaultAllocator,
        risk::DefaultRisk,
    },
    statistic::summary::{
        pnl::PnLReturnSummary,
        trading::{Config as StatisticConfig, TradingSummary},
    },
    event::Event,
    test_util,
};
use barter_integration::model::{Market, instrument::kind::InstrumentKind};
use std::marker::PhantomData;
use uuid::Uuid;

let components = PortfolioLego {
    engine_id: Uuid::new_v4(),
    markets: vec![Market::new("binance", ("btc", "usdt", InstrumentKind::S
    repository: InMemoryRepository::new(),
    allocator: DefaultAllocator{ default_order_value: 100.0 },
    risk: DefaultRisk{},
    starting_cash: 10000.0,
    statistic_config: StatisticConfig {
        starting_equity: 10000.0 ,
        trading_days_per_year: 365,
        risk_free_return: 0.0
    },
},
```

```

    _statistic_marker: PhantomData::<TradingSummary>::default()
};

let mut portfolio = MetaPortfolio::init(components).unwrap();

let some_event = Event::OrderNew(test_util::order_event());

match some_event {
    Event::Market(market) => {
        portfolio.update_from_market(&market);
    }
    Event::Signal(signal) => {
        portfolio.generate_order(&signal);
    }
    Event::SignalForceExit(signal) => {
        portfolio.generate_exit_order(signal);
    }
    Event::Fill(fill) => {
        portfolio.update_from_fill(&fill);
    }
    _ => {}
}

```

§Execution

```

use barter::{
    test_util,
    portfolio::OrderEvent,
    execution::{
        simulated::{Config as ExecutionConfig, SimulatedExecution},
        Fees, ExecutionClient,
    },
};

let config = ExecutionConfig {
    simulated_fees_pct: Fees {
        exchange: 0.1,
        slippage: 0.05, // Simulated slippage modelled as a Fee
        network: 0.0,
    },
};

let mut execution = SimulatedExecution::new(config);

let order_event = test_util::order_event();

let fill_event = execution.generate_fill(&order_event);

```

§Statistic

```
use barter::{
    test_util,
    portfolio::position::Position,
    statistic::summary::{
        trading::{Config as StatisticConfig, TradingSummary},
        Initialiser, PositionSummariser, TableBuilder
    }
};

// Do some automated trading with barter components that generates a vector
let positions = vec![test_util::position(), test_util::position()];

let config = StatisticConfig {
    starting_equity: 10000.0,
    trading_days_per_year: 253,
    risk_free_return: 0.5,
};

let mut trading_summary = TradingSummary::init(config);

trading_summary.generate_summary(&positions);

trading_summary
    .table("Total")
    .printstd();
```

§Engine & Traders

[See Readme Engine Example](#)

Modules§

- [data](#)
Defines a MarketEvent, and provides the Continuer and MarketGenerator traits for handling the generation of them. Contains implementations such as the (tick-by_tick) LiveTradeHandler, and HistoricalCandleHandler that generates a market feed and acts as the system heartbeat.
- [engine](#)
Multi-threaded trading Engine capable of trading with an arbitrary number market pairs. Contains a Trader for each Market pair that consists of it's own Data, Strategy & Execution components, as well as shared access to a global Portfolio.
- [event](#)
Defines an Event enum that contains variants that are vital to the trading event loop (eg/ MarketEvent). Other variants communicate work done by the system (eg/ FillEvent), as well as changes in system state (eg/ PositionUpdate).
- [execution](#)

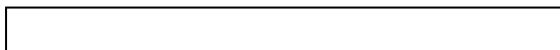
Defines a FillEvent, and provides a useful trait FillGenerator for handling the generation of them. Contains an example SimulatedExecution implementation that simulates live broker execution.

- [portfolio](#)
Defines useful data structures such as an OrderEvent and Position. The Portfolio must interact with MarketEvents, SignalEvents, OrderEvents, and FillEvents. The useful traits MarketUpdater, OrderGenerator, & FillUpdater are provided that define the interactions with these events. Contains a MetaPortfolio implementation that persists state in a generic Repository. This also contains example implementations of an OrderAllocator & OrderEvaluator, which help the Portfolio make decisions on whether to generate OrderEvents and of what size.
- [statistic](#)
Defines various iterative statistical methods that can be used to calculate trading performance metrics in one-pass. A trading performance summary implementation has been provided containing several key metrics such as Sharpe Ratio, Calmar Ratio, and Max Drawdown.
- [strategy](#)
Defines a SignalEvent and SignalForceExit, as well as the SignalGenerator trait for handling the generation of them. Contains an example RSIStrategy implementation that analyses a MarketEvent and may generate a new advisory SignalEvent to be analysed by the Portfolio OrderGenerator.
- [test_util](#)



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- [Structs](#)
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List of all items

Structs

- [data::MarketMeta](#)
- [data::historical::MarketFeed](#)
- [data::live::MarketFeed](#)
- [engine::Engine](#)
- [engine::EngineBuilder](#)

- [engine::EngineLego](#)
- [engine::trader::Trader](#)
- [engine::trader::TraderBuilder](#)
- [engine::trader::TraderLego](#)
- [event::EventTx](#)
- [execution::Fees](#)
- [execution::FillEvent](#)
- [execution::FillEventBuilder](#)
- [execution::simulated::Config](#)
- [execution::simulated::SimulatedExecution](#)
- [portfolio::Balance](#)
- [portfolio::OrderEvent](#)
- [portfolio::OrderEventBuilder](#)
- [portfolio::allocator::DefaultAllocator](#)
- [portfolio::portfolio::MetaPortfolio](#)
- [portfolio::portfolio::MetaPortfolioBuilder](#)
- [portfolio::portfolio::PortfolioLego](#)
- [portfolio::position::Position](#)
- [portfolio::position::PositionBuilder](#)
- [portfolio::position::PositionExit](#)
- [portfolio::position::PositionMeta](#)
- [portfolio::position::PositionUpdate](#)
- [portfolio::repository::in_memory::InMemoryRepository](#)
- [portfolio::repository::redis::Config](#)
- [portfolio::repository::redis::RedisRepository](#)
- [portfolio::repository::redis::RedisRepositoryBuilder](#)
- [portfolio::risk::DefaultRisk](#)
- [statistic::dispersion::Dispersion](#)
- [statistic::dispersion::Range](#)
- [statistic::metric::EquityPoint](#)
- [statistic::metric::drawdown::AvgDrawdown](#)
- [statistic::metric::drawdown::Drawdown](#)
- [statistic::metric::drawdown::MaxDrawdown](#)
- [statistic::metric::ratio::CalmarRatio](#)
- [statistic::metric::ratio::SharpeRatio](#)
- [statistic::metric::ratio::SortinoRatio](#)
- [statistic::summary::data::DataSummary](#)
- [statistic::summary::drawdown::DrawdownSummary](#)
- [statistic::summary::pnl::PnLReturnSummary](#)
- [statistic::summary::pnl::ProfitLossSummary](#)
- [statistic::summary::trading::Config](#)
- [statistic::summary::trading::TearSheet](#)
- [statistic::summary::trading::TradingSummary](#)
- [strategy::Signal](#)
- [strategy::SignalForceExit](#)
- [strategy::SignalStrength](#)
- [strategy::example::Config](#)
- [strategy::example::RSIStrategy](#)

Enums

- [data::Feed](#)

- [data::error::DataError](#)
- [engine::Command](#)
- [engine::error::EngineError](#)
- [event::Event](#)
- [execution::error::ExecutionError](#)
- [portfolio::OrderType](#)
- [portfolio::error::PortfolioError](#)
- [portfolio::repository::error::RepositoryError](#)
- [statistic::error::StatisticError](#)
- [strategy::Decision](#)

Traits

- [data::MarketGenerator](#)
- [event::MessageTransmitter](#)
- [execution::ExecutionClient](#)
- [portfolio::FillUpdater](#)
- [portfolio::MarketUpdater](#)
- [portfolio::OrderGenerator](#)
- [portfolio::allocator::OrderAllocator](#)
- [portfolio::position::PositionEnterer](#)
- [portfolio::position::PositionExiter](#)
- [portfolio::position::PositionUpdater](#)
- [portfolio::repository::BalanceHandler](#)
- [portfolio::repository::PositionHandler](#)
- [portfolio::repository::StatisticHandler](#)
- [portfolio::risk::OrderEvaluator](#)
- [statistic::metric::ratio::Ratio](#)
- [statistic::summary::Initialiser](#)
- [statistic::summary::PositionSummariser](#)
- [statistic::summary::TableBuilder](#)
- [strategy::SignalGenerator](#)

Functions

- [portfolio::portfolio::parse_signal_decisions](#)
- [portfolio::position::determine_position_id](#)
- [portfolio::repository::determine_exited_positions_id](#)
- [statistic::algorithm::welford_online::calculate_mean](#)
- [statistic::algorithm::welford_online::calculate_population_variance](#)
- [statistic::algorithm::welford_online::calculate_recurrence_relation_m](#)
- [statistic::algorithm::welford_online::calculate_sample_variance](#)
- [statistic::de_duration_from_secs](#)
- [statistic::metric::ratio::calculate_annual](#)
- [statistic::metric::ratio::calculate_daily](#)
- [statistic::se_duration_as_secs](#)
- [statistic::summary::combine](#)
- [statistic::summary::trading::calculate_trading_duration](#)
- [test_util::fill_event](#)
- [test_util::market_event_candle](#)
- [test_util::market_event_trade](#)

- [test_util::order_event](#)
- [test_util::position](#)
- [test_util::signal](#)

Type Aliases

- [execution::FeeAmount](#)
- [portfolio::BalanceId](#)
- [portfolio::position::PositionId](#)
- [portfolio::repository::ExitedPositionsId](#)

event

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[Module event](#)

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Module [barter::event](#)

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Expand description

Defines an Event enum that contains variants that are vital to the trading event loop (eg/ MarketEvent). Other variants communicate work done by the system (eg/ FillEvent), as well as changes in system state (eg/ PositionUpdate).

Structs§

- [EventTx](#)
Transmitter for sending Barter [Events](#) to an external sink. Useful for event-sourcing, real-time dashboards & general monitoring.

Enums§

- [Event](#)
Events that occur when bartering. [MarketEvent](#), [Signal](#), [OrderEvent](#), and [FillEvent](#) are vital to the [Trader](#) event loop, dictating the trading sequence. The [PositionExit](#) Event is a representation of work done by the system, and is useful for analysing performance & reconciliations.

Traits§

- [MessageTransmitter](#)
Message transmitter for sending Barter messages to downstream consumers.



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[Event](#)

[Variants](#)

- [Balance](#)
- [Fill](#)
- [Market](#)
- [OrderNew](#)
- [OrderUpdate](#)
- [PositionExit](#)
- [PositionNew](#)
- [PositionUpdate](#)
- [Signal](#)
- [SignalForceExit](#)

[Trait Implementations](#)

- [Clone](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [MessageTransmitter<Event>](#)
- [PartialEq](#)
- [Serialize](#)
- [StructuralPartialEq](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
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- [Unpin](#)
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Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
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- [Into<U>](#)
- [Same](#)
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- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::event`

[?](#)

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Enum `barter::event::Event`

Copy item path

[source](#) ·

[\[-\]](#)

```
pub enum Event {
    Market(MarketEvent<DataKind>),
    Signal(Signal),
    SignalForceExit(SignalForceExit),
    OrderNew(OrderEvent),
    OrderUpdate,
    Fill(FillEvent),
    PositionNew(Position),
    PositionUpdate(PositionUpdate),
    PositionExit(PositionExit),
    Balance(Balance),
}
```

Expand description

Events that occur when bartering. [MarketEvent](#), [Signal](#), [OrderEvent](#), and [FillEvent](#) are vital to the [Trader](#) event loop, dictating the trading sequence. The [PositionExit](#) Event is a representation of work done by the system, and is useful for analysing performance & reconciliations.

Variants§

§

Market([MarketEvent](#)<[DataKind](#)>)

§

Signal([Signal](#))

§

SignalForceExit([SignalForceExit](#))

§

OrderNew([OrderEvent](#))

§

OrderUpdate

§

Fill([FillEvent](#))

§

PositionNew([Position](#))

§

PositionUpdate([PositionUpdate](#))

§

PositionExit([PositionExit](#))

§

Balance([Balance](#))

Trait Implementations§

[source](#)§

impl [Clone](#) for [Event](#)

[source§](#)

fn [clone](#)(&self) -> [Event](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)

[source§](#)

impl [Debug](#) for [Event](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [Event](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [MessageTransmitter](#)<[Event](#)> for [EventTx](#)

[source§](#)

fn [send](#)(&mut self, message: [Event](#))

Attempts to send a message to an external message subscriber.

[source§](#)

fn [send_many](#)(&mut self, messages: [Vec](#)<[Event](#)>)

Attempts to send many messages to an external message subscriber.

[source§](#)

impl [PartialEq](#) for [Event](#)

[source§](#)

```
fn eq(&self, other: &Event) -> bool
```

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

```
fn ne(&self, other: &Rhs) -> bool
```

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [Serialize](#) for [Event](#)

[source§](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [StructuralPartialEq](#) for [Event](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Event](#)

§

impl [Send](#) for [Event](#)

§

impl [Sync](#) for [Event](#)

§

impl [Unpin](#) for [Event](#)

§

impl [UnwindSafe](#) for [Event](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



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[MessageTransmitter](#)

[Required Methods](#)

- [send](#)
- [send_many](#)

[Implementors](#)

[In `barter::event`](#)



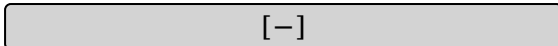
[?](#)

[Change settings](#)

Trait [barter::event::MessageTransmitter](#)



[source](#) ·



```
pub trait MessageTransmitter<Message> {  
    // Required methods
```

```

    fn send(&mut self, message: Message);
    fn send\_many(&mut self, messages: Vec<Message>);
}

```

Expand description

Message transmitter for sending Barter messages to downstream consumers.

Required Methods§

[source](#)

fn [send](#)(&mut self, message: Message)

Attempts to send a message to an external message subscriber.

[source](#)

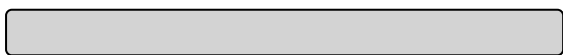
fn [send_many](#)(&mut self, messages: [Vec](#)<Message>)

Attempts to send many messages to an external message subscriber.

Implementors§

[source§](#)

impl [MessageTransmitter](#)<[Event](#)> for [EventTx](#)



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[EventTx](#)

[Methods](#)

- [new](#)

[Trait Implementations](#)

- [Clone](#)
- [Debug](#)
- [MessageTransmitter](#)<[Event](#)>

[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)

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Blanket Implementations

- [Any](#)
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In barter::event

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Struct [barter::event::EventTx](#)

Copy item path

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```
pub struct EventTx { /* private fields */ }
```

Expand description

Transmitter for sending Barter [Events](#) to an external sink. Useful for event-sourcing, real-time dashboards & general monitoring.

Implementations§

[source§](#)

impl [EventTx](#)

[source](#)

pub fn [new](#)(event_tx: [UnboundedSender](#)<[Event](#)>) -> Self

Constructs a new [EventTx](#) instance using the provided channel transmitter.

Trait Implementations§

[source§](#)

impl [Clone](#) for [EventTx](#)

[source§](#)

fn [clone](#)(&self) -> [EventTx](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [EventTx](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [MessageTransmitter](#)<[Event](#)> for [EventTx](#)

[source§](#)

fn [send](#)(&mut self, message: [Event](#))

Attempts to send a message to an external message subscriber.
[source§](#)

fn [send_many](#)(&mut self, messages: [Vec](#)<[Event](#)>)

Attempts to send many messages to an external message subscriber.

Auto Trait Implementations§

§

impl ![RefUnwindSafe](#) for [EventTx](#)

[§](#)

impl [Send](#) for [EventTx](#)

[§](#)

impl [Sync](#) for [EventTx](#)

[§](#)

impl [Unpin](#) for [EventTx](#)

[§](#)

impl ![UnwindSafe](#) for [EventTx](#)

Blanket Implementations[§](#)

[source](#)[§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)[§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)


```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source](#)

```
fn vzip(self) -> V
```

[source](#)

```
impl<T> WithSubscriber for T
```

[source](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

statistic

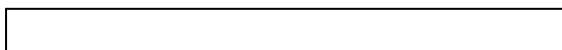


[barter](#)0.8.13

[Module statistic](#)

- [Modules](#)
- [Functions](#)

[In crate barter](#)



[?](#)
[Change settings](#)

Module [barter::statistic](#)

Copy item path

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Expand description

Defines various iterative statistical methods that can be used to calculate trading performance metrics in one-pass. A trading performance summary implementation has been provided containing several key metrics such as Sharpe Ratio, Calmar Ratio, and Max Drawdown.

Modules§

- [algorithm](#)
- [dispersion](#)
- [error](#)
- [metric](#)
- [summary](#)

Functions§

- [de_duration_from_secs](#)
Deserialize a number representing seconds into a [Duration](#)
- [se_duration_as_secs](#)
Serialize a [Duration](#) into a u64 representing the associated seconds.

[barter](#)0.8.13

In [barter::statistic](#)

?

[Change settings](#)

Function [barter::statistic::se_duration_as_secs](#)

Copy item path

[source](#) ·

```
pub fn se_duration_as_secs<S>(  
    duration: &Duration,  
    serializer: S  
) -> Result<S::Ok, S::Error>
```

where
S: [Serializer](#),

Expand description

Serialize a [Duration](#) into a u64 representing the associated seconds.

[barter](#)**0.8.13**

[In barter::statistic](#)

[?](#)
[Change settings](#)

Function [barter::statistic::de_duration_from_secs](#)

Copy item path

[source](#) ·

[-]

```
pub fn de_duration_from_secs<'de, D>(
    deserializer: D
) -> Result<Duration, D::Error>
where
    D: Deserializer<'de>,
```

Expand description

Deserialize a number representing seconds into a [Duration](#)

statistic - algorithm

[barter](#)**0.8.13**

[Module algorithm](#)

- [Modules](#)

[In barter::statistic](#)

[?](#)

[Change settings](#)

Module [barter::statistic::algorithm](#)

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[\[-\]](#)

Modules§

- [welford_online](#)

Grouping of [Welford Online](#) algorithms for calculating running values such as mean and variance in one pass through.

statistic - algorithm - welford_online

[barter](#)0.8.13

[In barter::statistic::algorithm::welford_online](#)

[?](#)

[Change settings](#)

Function [barter::statistic::algorithm::welford_online::calculate_sample_variance](#)

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[\[-\]](#)

```
pub fn calculate_sample_variance(recurrence_relation_m: f64, count: u64) -
```

Expand description

Calculates the next unbiased 'Sample' Variance using Bessel's correction (count - 1), and the Welford Online recurrence relation M.

[barter0.8.13](#)

[Module welford_online](#)

- [Functions](#)

[In barter::statistic::algorithm](#)

[?](#)

[Change settings](#)

Module [barter::statistic::algorithm::welford_online](#)

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[source](#) ·

[\[-\]](#)

Expand description

Grouping of [Welford Online](#) algorithms for calculating running values such as mean and variance in one pass through.

Functions§

- [calculate_mean](#)
Calculates the next mean.
- [calculate_population_variance](#)
Calculates the next biased 'Population' Variance using the Welford Online recurrence relation M.
- [calculate_recurrence_relation_m](#)
Calculates the next Welford Online recurrence relation M.
- [calculate_sample_variance](#)
Calculates the next unbiased 'Sample' Variance using Bessel's correction (count - 1), and the Welford Online recurrence relation M.

[barter0.8.13](#)

[In barter::statistic::algorithm::welford_online](#)

[?](#)

[Change settings](#)

Function [barter::statistic::algorithm::welford_online::calculate_population_variance](#)

Copy item path

[source](#) ·

```
pub fn calculate_population_variance(  
    recurrence_relation_m: f64,  
    count: u64  
) -> f64
```

Expand description

Calculates the next biased 'Population' Variance using the Welford Online recurrence relation M.

[barter0.8.13](#)

[In barter::statistic::algorithm::welford_online](#)

[?](#)
[Change settings](#)

Function [barter::statistic::algorithm::welford_online::calculate_mean](#)

Copy item path

[source](#) ·

```
pub fn calculate_mean<T>(prev_mean: T, next_value: T, count: T) -> T  
where  
    T: Copy + Sub<Output = T> + Div<Output = T> + AddAssign,
```

Expand description

Calculates the next mean.

[barter0.8.13](#)

[In barter::statistic::algorithm::welford_online](#)

?

[Change settings](#)

Function [barter::statistic::algorithm::welford_online::calculate_recurrence_relation_m](#)

Copy item path

[source](#) ·

[−]

```
pub fn calculate_recurrence_relation_m(  
    prev_m: f64,  
    prev_mean: f64,  
    new_value: f64,  
    new_mean: f64  
) -> f64
```

Expand description

Calculates the next Welford Online recurrence relation M.

statistic - dispersion

[barter](#)0.8.13

[Module dispersion](#)

- [Structs](#)

[In barter::statistic](#)

?

[Change settings](#)

Module [barter::statistic::dispersion](#)

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Structs§

- [Dispersion](#)
Representation of a dataset using measures of dispersion - range, variance & standard deviation.
- [Range](#)
Measure of dispersion providing the highest and lowest value of a dataset. Lazy evaluation is used when calculating the range between them via the `calculate()` function.



[barter](#)0.8.13

[Range](#)

[Fields](#)

- [activated](#)
- [high](#)
- [low](#)

[Methods](#)

- [calculate](#)
- [init](#)
- [update](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In [barter::statistic::dispersion](#)

[?](#)

[Change settings](#)

Struct [barter::statistic::dispersion::Range](#)

Copy item path

[source](#) ·

```
pub struct Range {  
    pub activated: bool,  
    pub high: f64,  
    pub low: f64,  
}
```

Expand description

Measure of dispersion providing the highest and lowest value of a dataset. Lazy evaluation is used when calculating the range between them via the `calculate()` function.

Fields§

§activated: [bool](#)§high: [f64](#)§low: [f64](#)

Implementations§

[source](#)§

impl [Range](#)

[source](#)

pub fn [init](#)(first_value: [f64](#)) -> Self

Initialises the Range with the provided first value of the dataset.

[source](#)

pub fn [update](#)(&mut self, new_value: [f64](#))

Iteratively updates the Range given the next value in the dataset.

[source](#)

pub fn [calculate](#)(&self) -> [f64](#)

Calculates the range between the highest and lowest value of a dataset.
Provided to allow lazy evaluation.

Trait Implementations§

[source§](#)

impl [Clone](#) for [Range](#)

[source§](#)

fn [clone](#)(&self) -> [Range](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [Range](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [Range](#)

[source§](#)

fn [default](#)() -> [Range](#)

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [Range](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [Range](#)

[source§](#)

fn [eq](#)(&self, other: &[Range](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [Range](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[Range](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [Range](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::[Ok](#), __S::[Error](#)>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Copy](#) for [Range](#)

[source§](#)

impl [StructuralPartialEq](#) for [Range](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Range](#)

§

impl [Send](#) for [Range](#)

§

impl [Sync](#) for [Range](#)

[§](#)

impl [Unpin](#) for [Range](#)

[§](#)

impl [UnwindSafe](#) for [Range](#)

Blanket Implementations

[source](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)

[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)

[source](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)

[source](#)

impl<T> [From](#)<T> for T

[source](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



[barter](#)0.8.13

[Dispersion](#)

[Fields](#)

- [range](#)
- [recurrence_relation_m](#)
- [std_dev](#)
- [variance](#)

[Methods](#)

- [update](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize](#)<'de>

- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::statistic::dispersion`

[?](#)

[Change settings](#)

Struct [barter::statistic::dispersion::Dispersion](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct Dispersion {
    pub range: Range,
    pub recurrence_relation_m: f64,
    pub variance: f64,
    pub std_dev: f64,
}
```

Expand description

Representation of a dataset using measures of dispersion - range, variance & standard deviation.

Fields§

§range: [Range](#)recurrence_relation_m: [f64](#)variance: [f64](#)std_dev: [f64](#)

Implementations§

[source](#)§

impl [Dispersion](#)

[source](#)

```
pub fn update( &mut self, prev_mean: f64, new_mean: f64,  
new_value: f64, value_count: u64 )
```

Iteratively updates the measures of Dispersion given the previous mean, new mean, new value, and the dataset count.

Trait Implementations§

[source](#)§

impl [Clone](#) for [Dispersion](#)

[source](#)§

```
fn clone(&self) -> Dispersion
```

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

```
fn clone\_from(&mut self, source: &Self)
```

Performs copy-assignment from source. [Read more](#)
[source](#)§

impl [Debug](#) for [Dispersion](#)

[source](#)§

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source](#)§

impl [Default](#) for [Dispersion](#)

[source§](#)

fn [default](#)() -> [Dispersion](#)

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [Dispersion](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [PartialEq](#) for [Dispersion](#)

[source§](#)

fn [eq](#)(&self, other: &[Dispersion](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.

1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [Dispersion](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[Dispersion](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Serialize](#) for [Dispersion](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [Copy](#) for [Dispersion](#)

[source](#)

impl [StructuralPartialEq](#) for [Dispersion](#)

Auto Trait Implementations

[§](#)

impl [RefUnwindSafe](#) for [Dispersion](#)

[§](#)

impl [Send](#) for [Dispersion](#)

[§](#)

impl [Sync](#) for [Dispersion](#)

[§](#)

impl [Unpin](#) for [Dispersion](#)

[§](#)

impl [UnwindSafe](#) for [Dispersion](#)

Blanket Implementations

[source](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)

[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)

[source](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)

[source](#)

impl<T> [From](#)<T> for T

[source](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
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impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

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Trait [barter::statistic::summary::Initialiser](#)

Copy item path

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[\[-\]](#)

```
pub trait Initialiser {  
    type Config: Copy;  
  
    // Required method  
    fn init(config: Self::Config) -> Self;  
}
```

Required Associated Types§

[source](#)

type [Config](#): [Copy](#)

Required Methods§

[source](#)

fn [init](#)(config: Self::Config) -> Self

Object Safety§

This trait is **not** [object safe](#).

Implementors§

[source§](#)

impl [Initialiser](#) for [PnLReturnSummary](#)

§

type [Config](#) = ()

[source§](#)

impl [Initialiser](#) for [TradingSummary](#)

§

type [Config](#) = [Config](#)

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Trait [barter::statistic::summary::PositionSummariser](#)

Copy item path

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```
pub trait PositionSummariser: Copy {  
    // Required method  
    fn update(&mut self, position: &Position);  
  
    // Provided method  
    fn generate\_summary(&mut self, positions: &[Position]) { ... }  
}
```

Required Methods§

[source](#)

fn [update](#)(&mut self, position: &[Position](#))

Provided Methods§

[source](#)

```
fn generate\_summary(&mut self, positions: &[Position])
```

Object Safety§

This trait is **not** [object safe](#).

Implementors§

[source§](#)

```
impl PositionSummariser for EquityPoint
```

[source§](#)

```
impl PositionSummariser for DrawdownSummary
```

[source§](#)

```
impl PositionSummariser for PnLReturnSummary
```

[source§](#)

```
impl PositionSummariser for ProfitLossSummary
```

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```
impl PositionSummariser for TradingSummary
```

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Function [barter::statistic::summary::combine](#)

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[−]

```
pub fn combine<Iter, T>(builders: Iter) -> Table
```

where

Iter: [IntoIterator](#)<Item = ([String](#), T)>,
T: [TableBuilder](#),

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[TableBuilder](#)

[Required Methods](#)

- [row](#)
- [titles](#)

[Provided Methods](#)

- [table](#)
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Trait [barter::statistic::summary::TableBuilder](#)

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```
pub trait TableBuilder {  
    // Required methods  
    fn titles(&self) -> Row;  
    fn row(&self) -> Row;  
  
    // Provided methods  
    fn table(&self, id_cell: &str) -> Table { ... }  
    fn table\_with<T: TableBuilder>(  
        &self,  
        id_cell: &str,  
        another: (T, &str)
```

```
    ) -> Table { ... }  
}
```

Required Methods§

[source](#)

```
fn titles(&self) -> Row
```

[source](#)

```
fn row(&self) -> Row
```

Provided Methods§

[source](#)

```
fn table(&self, id_cell: &str) -> Table
```

[source](#)

```
fn table\_with<T: TableBuilder>( &self, id_cell: &str, another: (T,  
&str) ) -> Table
```

Object Safety§

This trait is **not** [object safe](#).

Implementors§

[source§](#)

```
impl TableBuilder for DataSummary
```

[source§](#)

```
impl TableBuilder for DrawdownSummary
```

[source§](#)

```
impl TableBuilder for PnLReturnSummary
```

[source§](#)

```
impl TableBuilder for ProfitLossSummary
```

[source§](#)

impl [TableBuilder](#) for [TearSheet](#)

[source](#)[§]

impl [TableBuilder](#) for [TradingSummary](#)

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In barter::statistic::summary::data

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Struct [barter::statistic::summary::data::DataSummary](#)

Copy item path

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[–]

```
pub struct DataSummary {  
    pub count: u64,  
    pub sum: f64,  
    pub mean: f64,  
    pub dispersion: Dispersion,  
}
```

Fields§

§count: [u64](#)§sum: [f64](#)§mean: [f64](#)§dispersion: [Dispersion](#)

Implementations§

[source](#)§

impl [DataSummary](#)

[source](#)

pub fn [update](#)(&mut self, next_value: [f64](#))

Trait Implementations§

[source](#)§

impl [Clone](#) for [DataSummary](#)

[source](#)§

fn [clone](#)(&self) -> [DataSummary](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)§

impl [Debug](#) for [DataSummary](#)

[source](#)§

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl [Default](#) for [DataSummary](#)

[source](#)

fn [default](#)() -> [DataSummary](#)

Returns the “default value” for a type. [Read more](#)
[source](#)

impl<'de> [Deserialize](#)<'de> for [DataSummary](#)

[source](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl [PartialEq](#) for [DataSummary](#)

[source](#)

fn [eq](#)(&self, other: &[DataSummary](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)

impl [PartialOrd](#) for [DataSummary](#)

[source](#)

fn [partial_cmp](#)(&self, other: &[DataSummary](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Serialize](#) for [DataSummary](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [TableBuilder](#) for [DataSummary](#)

[source](#)

fn [titles](#)(&self) -> [Row](#)

[source](#)

fn [row](#)(&self) -> [Row](#)

[source](#)

```
fn table(&self, id_cell: &str) -> Table
```

[source](#)[§](#)

```
fn table\_with<T: TableBuilder>( &self, id_cell: &str, another: (T, &str) ) -> Table
```

[source](#)[§](#)

```
impl Copy for DataSummary
```

[source](#)[§](#)

```
impl StructuralPartialEq for DataSummary
```

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[§](#)

```
impl RefUnwindSafe for DataSummary
```

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```
impl Send for DataSummary
```

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```
impl Sync for DataSummary
```

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```
impl Unpin for DataSummary
```

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```
impl UnwindSafe for DataSummary
```

Blanket Implementations[§](#)

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```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source](#)[§](#)

```
fn type_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source§](#)

```
impl<T> Borrow<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> From<T> for T
```

[source§](#)

```
fn from(t: T) -> T
```

Returns the argument unchanged.

[source§](#)

```
impl<T> Instrument for T
```

[source§](#)

```
fn instrument(self, span: Span) -> Instrumented<Self>
```

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in_current_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source§](#)

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

§

```
type Owned = T
```

The resulting type after obtaining ownership.

[source§](#)

```
fn to\_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

```
fn clone\_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

```
impl<T, U> TryFrom<U> for T  
where U: Into<T>,
```

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)§

`impl<T> DeserializeOwned for T`
`where T: for<'de> Deserialize<'de>,`

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Struct [barter::statistic::summary::drawdown::DrawdownSummary](#)

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[–]

```
pub struct DrawdownSummary {  
    pub current_drawdown: Drawdown,  
    pub avg_drawdown: AvgDrawdown,  
    pub max_drawdown: MaxDrawdown,  
}
```

Fields§

§current_drawdown: [Drawdown](#)§avg_drawdown: [AvgDrawdown](#)§max_drawdown: [MaxDrawdown](#)

Implementations§

[source](#)§

impl [DrawdownSummary](#)

[source](#)

pub fn [new](#)(starting_equity: [f64](#)) -> Self

Trait Implementations§

[source](#)§

impl [Clone](#) for [DrawdownSummary](#)

[source](#)§

fn [clone](#)(&self) -> [DrawdownSummary](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)§

impl [Debug](#) for [DrawdownSummary](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [DrawdownSummary](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [PartialEq](#) for [DrawdownSummary](#)

[source§](#)

fn [eq](#)(&self, other: &[DrawdownSummary](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [DrawdownSummary](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[DrawdownSummary](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [PositionSummariser](#) for [DrawdownSummary](#)

[source§](#)

fn [update](#)(&mut self, position: [&Position](#))

[source§](#)

fn [generate_summary](#)(&mut self, positions: [&\[Position\]](#))

[source§](#)

impl [Serialize](#) for [DrawdownSummary](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
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impl [TableBuilder](#) for [DrawdownSummary](#)

[source§](#)

fn [titles](#)(&self) -> [Row](#)

[source§](#)

fn [row](#)(&self) -> [Row](#)

[source§](#)

fn [table](#)(&self, id_cell: &[str](#)) -> [Table](#)

[source§](#)

fn [table_with](#)<T: [TableBuilder](#)>(&self, id_cell: &[str](#), another: (T, &[str](#))) -> [Table](#)

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impl [StructuralPartialEq](#) for [DrawdownSummary](#)

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§

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impl [Send](#) for [DrawdownSummary](#)

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impl [Sync](#) for [DrawdownSummary](#)

§

impl [Unpin](#) for [DrawdownSummary](#)

§

impl [UnwindSafe](#) for [DrawdownSummary](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in\_current\_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

```
impl<T> Same for T
```

[§](#)

```
type Output = T
```

Should always be Self
[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

[§](#)

```
type Owned = T
```

The resulting type after obtaining ownership.
[source§](#)

```
fn to\_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

```
fn clone\_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de> ,

statistic - summary - pnl



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[ProfitLossSummary](#)

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- [long_pnl](#)
- [long_pnl_per_contract](#)
- [short_contracts](#)
- [short_pnl](#)
- [short_pnl_per_contract](#)
- [total_contracts](#)
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- [From<T>](#)
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- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::statistic::summary::pnl`

[?](#)

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Struct `barter::statistic::summary::pnl::ProfitLossSummary`

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct ProfitLossSummary {
    pub long_contracts: f64,
    pub long_pnl: f64,
    pub long_pnl_per_contract: f64,
    pub short_contracts: f64,
    pub short_pnl: f64,
    pub short_pnl_per_contract: f64,
    pub total_contracts: f64,
    pub total_pnl: f64,
    pub total_pnl_per_contract: f64,
}
```

Fields§

[§](#)long_contracts: [f64§](#)long_pnl: [f64§](#)long_pnl_per_contract: [f64§](#)short_contracts: [f64§](#)short_pnl: [f64§](#)short_pnl_per_contract: [f64§](#)total_contracts: [f64§](#)total_pnl: [f64§](#)total_pnl_per_contract: [f64§](#)

Implementations§

[source§](#)

impl [ProfitLossSummary](#)

[source](#)

pub fn [new](#)() -> Self

Trait Implementations§

[source§](#)

impl [Clone](#) for [ProfitLossSummary](#)

[source§](#)

fn [clone](#)(&self) -> [ProfitLossSummary](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [ProfitLossSummary](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [ProfitLossSummary](#)

[source§](#)

fn [default](#)() -> [ProfitLossSummary](#)

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [ProfitLossSummary](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [ProfitLossSummary](#)

[source§](#)

fn [eq](#)(&self, other: &[ProfitLossSummary](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [ProfitLossSummary](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[ProfitLossSummary](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [PositionSummariser](#) for [ProfitLossSummary](#)

[source](#)

fn [update](#)(&mut self, position: [&Position](#))

[source](#)

fn [generate_summary](#)(&mut self, positions: [&\[Position\]](#))

[source](#)

impl [Serialize](#) for [ProfitLossSummary](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [TableBuilder](#) for [ProfitLossSummary](#)

[source](#)

fn [titles](#)(&self) -> [Row](#)

[source](#)

fn [row](#)(&self) -> [Row](#)

[source](#)§

fn [table](#)(&self, id_cell: &[str](#)) -> [Table](#)

[source](#)§

fn [table_with](#)<T: [TableBuilder](#)>(&self, id_cell: &[str](#), another: (T, &[str](#))) -> [Table](#)

[source](#)§

impl [Copy](#) for [ProfitLossSummary](#)

[source](#)§

impl [StructuralPartialEq](#) for [ProfitLossSummary](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [ProfitLossSummary](#)

§

impl [Send](#) for [ProfitLossSummary](#)

§

impl [Sync](#) for [ProfitLossSummary](#)

§

impl [Unpin](#) for [ProfitLossSummary](#)

§

impl [UnwindSafe](#) for [ProfitLossSummary](#)

Blanket Implementations§

[source](#)§

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in\_current\_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

```
impl<T> Same for T
```

[§](#)

```
type Output = T
```

Should always be Self

[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

[§](#)

```
type Owned = T
```

The resulting type after obtaining ownership.

[source§](#)

```
fn to\_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

```
fn clone\_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de> ,

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Module [barter::statistic::summary::pnl](#)

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[-]

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- [PnLReturnSummary](#)
- [ProfitLossSummary](#)

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[PnLReturnSummary](#)

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- [losses](#)
- [time](#)

- [total](#)
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- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::statistic::summary::pnl

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Struct [barter::statistic::summary::pnl::PnLReturnSummary](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct PnLReturnSummary {  
    pub time: DateTime<Utc>,  
    pub duration: Duration,  
    pub trades_per_day: f64,  
    pub total: DataSummary,  
    pub losses: DataSummary,  
}
```

Fields§

[§](#)time: [DateTime<Utc>](#)[§](#)duration: [Duration](#)[§](#)trades_per_day: [f64](#)[§](#)total: [DataSummary](#)[§](#)losses: [DataSummary](#)

Implementations§

[source](#)§

impl [PnLReturnSummary](#)

[source](#)

pub fn [new](#)() -> Self

[source](#)

pub fn [update_trading_session_duration](#)(&mut self, position: [Position](#))

[source](#)

pub fn [update_trades_per_day](#)(&mut self)

Trait Implementations§

[source](#)§

impl [Clone](#) for [PnLReturnSummary](#)

[source§](#)

fn [clone](#)(&self) -> [PnLReturnSummary](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)

[source§](#)

impl [Debug](#) for [PnLReturnSummary](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl [Default](#) for [PnLReturnSummary](#)

[source§](#)

fn [default](#)() -> Self

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [PnLReturnSummary](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [Initialiser](#) for [PnLReturnSummary](#)

[§](#)

type [Config](#) = [\(\)](#)

[source§](#)

fn [init](#)(_: Self::Config) -> Self

[source§](#)

impl [PartialEq](#) for [PnLReturnSummary](#)

[source§](#)

fn [eq](#)(&self, other: &[PnLReturnSummary](#)) -> bool

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: [Rhs](#)) -> bool

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [PnLReturnSummary](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[PnLReturnSummary](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: [Rhs](#)) -> bool

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: [Rhs](#)) -> bool

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: [Rhs](#)) -> bool

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

```
fn ge(&self, other: &Rhs) -> bool
```

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

```
impl PositionSummariser for PnLReturnSummary
```

[source](#)

```
fn update(&mut self, position: &Position)
```

[source](#)

```
fn generate_summary(&mut self, positions: &\[Position\])
```

[source](#)

```
impl Serialize for PnLReturnSummary
```

[source](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

```
impl TableBuilder for PnLReturnSummary
```

[source](#)

```
fn titles(&self) -> Row
```

[source](#)

```
fn row(&self) -> Row
```

[source](#)

```
fn table(&self, id_cell: &str) -> Table
```

[source](#)

```
fn table_with<T: TableBuilder>( &self, id_cell: &str, another: (T,  
&str) ) -> Table
```

[source](#)

impl [Copy](#) for [PnLReturnSummary](#)

[source](#)§

impl [StructuralPartialEq](#) for [PnLReturnSummary](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [PnLReturnSummary](#)

§

impl [Send](#) for [PnLReturnSummary](#)

§

impl [Sync](#) for [PnLReturnSummary](#)

§

impl [Unpin](#) for [PnLReturnSummary](#)

§

impl [UnwindSafe](#) for [PnLReturnSummary](#)

Blanket Implementations§

[source](#)§

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)

[source](#)§

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)§


```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> From<T> for T
```

[source§](#)

```
fn from(t: T) -> T
```

Returns the argument unchanged.

[source§](#)

```
impl<T> Instrument for T
```

[source§](#)

```
fn instrument(self, span: Span) -> Instrumented<Self>
```

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in_current_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From<T>](#) for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be Self

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)

[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)

[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

```
impl<T, U> TryInto<U> for T
where U: TryFrom<T>,
```

[§](#)

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.
[source§](#)

```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source§](#)

```
impl<V, T> VZip<V> for T
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```

statistic - summary - trading



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[In barter::statistic::summary::trading](#)

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[Change settings](#)

Function [barter::statistic::summary::trading::calculate_trading_duration](#)

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```
pub fn calculate_trading_duration(  
    start_time: &DateTime<Utc>,  
    position: &Position  
) -> Duration
```

[barter](#)0.8.13

[Module trading](#)

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Structs§

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Configuration for initialising a [TradingSummary](#) via the `init()` constructor method.

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- [calculate_trading_duration](#)



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- [ToOwned](#)
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- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In `barter::statistic::summary::trading`](#)

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[Change settings](#)

Struct [barter::statistic::summary::trading::Config](#)

Copy item path

[source](#) ·

[−]

```
pub struct Config {
    pub starting_equity: f64,
    pub trading_days_per_year: usize,
    pub risk_free_return: f64,
}
```

Expand description

Configuration for initialising a [TradingSummary](#) via the `init()` constructor method.

Fields§

[starting_equity](#): [f64](#)[trading_days_per_year](#): [usize](#)[risk_free_return](#): [f64](#)

Trait Implementations§

[source](#)§

impl [Clone](#) for [Config](#)

[source](#)§

fn [clone](#)(&self) -> [Config](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)

impl [Debug](#) for [Config](#)

[source](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl<'de> [Deserialize](#)<'de> for [Config](#)

[source](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl [PartialEq](#) for [Config](#)

[source](#)

fn [eq](#)(&self, other: &[Config](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)

impl [PartialOrd](#) for [Config](#)

[source](#)

fn [partial_cmp](#)(&self, other: &[Config](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [Config](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Copy](#) for [Config](#)

[source§](#)

impl [StructuralPartialEq](#) for [Config](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Config](#)

§

impl [Send](#) for [Config](#)

[§](#)

impl [Sync](#) for [Config](#)

[§](#)

impl [Unpin](#) for [Config](#)

[§](#)

impl [UnwindSafe](#) for [Config](#)

Blanket Implementations[§](#)

[source](#)[§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)[§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



[barter](#)0.8.13

[TearSheet](#)

[Fields](#)

- [calmar_ratio](#)
- [sharpe_ratio](#)
- [sortino_ratio](#)

[Methods](#)

- [new](#)
- [update](#)

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- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)
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Auto Trait Implementations

- [RefUnwindSafe](#)
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- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

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- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::statistic::summary::trading`

[?](#)

[Change settings](#)

Struct `barter::statistic::summary::trading::TearSheet`

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct TearSheet {  
    pub sharpe_ratio: SharpeRatio,
```

```
    pub sortino_ratio: SortinoRatio,  
    pub calmar_ratio: CalmarRatio,  
}
```

Fields§

[\\$sharpe_ratio](#): [SharpeRatio](#)[\\$sortino_ratio](#): [SortinoRatio](#)[\\$calmar_ratio](#): [CalmarRatio](#)

Implementations§

[source§](#)

impl [TearSheet](#)

[source](#)

```
pub fn new(risk_free_return: f64) -> Self
```

[source](#)

```
pub fn update( &mut self, pnl_returns: &PnLReturnSummary,  
drawdown: &DrawdownSummary )
```

Trait Implementations§

[source§](#)

impl [Clone](#) for [TearSheet](#)

[source§](#)

```
fn clone(&self) -> TearSheet
```

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

```
fn clone\_from(&mut self, source: &Self)
```

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [TearSheet](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [TearSheet](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [TearSheet](#)

[source§](#)

fn [eq](#)(&self, other: &[TearSheet](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [TearSheet](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[TearSheet](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Serialize](#) for [TearSheet](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::[Ok](#), __S::[Error](#)>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [TableBuilder](#) for [TearSheet](#)

[source](#)

fn [titles](#)(&self) -> [Row](#)

[source](#)

fn [row](#)(&self) -> [Row](#)

[source](#)

fn [table](#)(&self, id_cell: &[str](#)) -> [Table](#)

[source](#)

fn [table_with](#)<T: [TableBuilder](#)>(&self, id_cell: &[str](#), another: (T, &[str](#))) -> [Table](#)

[source](#)

impl [Copy](#) for [TearSheet](#)

[source](#)§

impl [StructuralPartialEq](#) for [TearSheet](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [TearSheet](#)

§

impl [Send](#) for [TearSheet](#)

§

impl [Sync](#) for [TearSheet](#)

§

impl [Unpin](#) for [TearSheet](#)

§

impl [UnwindSafe](#) for [TearSheet](#)

Blanket Implementations§

[source](#)§

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)§

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)§

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> From<T> for T
```

[source§](#)

```
fn from(t: T) -> T
```

Returns the argument unchanged.

[source§](#)

```
impl<T> Instrument for T
```

[source§](#)

```
fn instrument(self, span: Span) -> Instrumented<Self>
```

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in_current_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From<T>](#) for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be Self

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)

[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)

[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

```
impl<T, U> TryInto<U> for T
where U: TryFrom<T>,
```

[§](#)

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.
[source§](#)

```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source§](#)

```
impl<V, T> VZip<V> for T
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```



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TradingSummary

Fields

- [drawdown](#)
- [pnl_returns](#)
- [tear_sheet](#)

Trait Implementations

- [Clone](#)
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- [Debug](#)
- [Deserialize<'de>](#)
- [Initialiser](#)
- [PartialEq](#)
- [PartialOrd](#)
- [PositionSummariser](#)
- [Serialize](#)
- [StructuralPartialEq](#)
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Auto Trait Implementations

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- [Send](#)
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- [UnwindSafe](#)

Blanket Implementations

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- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::statistic::summary::trading`

[?](#)

[Change settings](#)

Struct [barter::statistic::summary::trading::TradingSummary](#)

Copy item path

[source](#) ·

[−]

```
pub struct TradingSummary {  
    pub pnl_returns: PnLReturnSummary,  
    pub drawdown: DrawdownSummary,  
    pub tear_sheet: TearSheet,  
}
```

Fields§

[\\$pnl_returns](#): [PnLReturnSummary](#)[\\$drawdown](#): [DrawdownSummary](#)[\\$tear_sheet](#): [TearSheet](#)

Trait Implementations§

[source](#)§

impl [Clone](#) for [TradingSummary](#)

[source](#)§

fn [clone](#)(&self) -> [TradingSummary](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)§

impl [Debug](#) for [TradingSummary](#)

[source](#)§

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)§

impl<'de> [Deserialize](#)<'de> for [TradingSummary](#)

[source](#)§

```
fn deserialize<__D>(__deserializer: __D) -> Result<Self, __D::Error>
where __D: Deserializer<'de>,
```

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)§

impl [Initialiser](#) for [TradingSummary](#)

§

type [Config](#) = [Config](#)

[source](#)§

```
fn init(config: Self::Config) -> Self
```

[source](#)§

impl [PartialEq](#) for [TradingSummary](#)

[source](#)§

```
fn eq(&self, other: &TradingSummary) -> bool
```

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)§

```
fn ne(&self, other: &Rhs) -> bool
```

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)§

impl [PartialOrd](#) for [TradingSummary](#)

[source](#)§

```
fn partial\_cmp(&self, other: &TradingSummary) ->
Option<Ordering>
```

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source](#)§

fn lt(&self, other: &Rhs) -> bool

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn le(&self, other: &Rhs) -> bool

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn gt(&self, other: &Rhs) -> bool

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn ge(&self, other: &Rhs) -> bool

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl PositionSummariser for TradingSummary

[source](#)

fn update(&mut self, position: &Position)

[source](#)

fn generate_summary(&mut self, positions: &[Position])

[source](#)

impl Serialize for TradingSummary

[source](#)

fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok, __S::Error>
where __S: Serializer,

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [TableBuilder](#) for [TradingSummary](#)

[source§](#)

fn [titles](#)(&self) -> [Row](#)

[source§](#)

fn [row](#)(&self) -> [Row](#)

[source§](#)

fn [table](#)(&self, id_cell: &[str](#)) -> [Table](#)

[source§](#)

fn [table_with](#)<T: [TableBuilder](#)>(&self, id_cell: &[str](#), another: (T, &[str](#))) -> [Table](#)

[source§](#)

impl [Copy](#) for [TradingSummary](#)

[source§](#)

impl [StructuralPartialEq](#) for [TradingSummary](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [TradingSummary](#)

§

impl [Send](#) for [TradingSummary](#)

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impl [Sync](#) for [TradingSummary](#)

§

impl [Unpin](#) for [TradingSummary](#)

§

impl [UnwindSafe](#) for [TradingSummary](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

statistic - metric

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[Module metric](#)

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- [drawdown](#)
- [ratio](#)

Structs§

- [EquityPoint](#)
Total equity at a point in time - equates to [Balance.total](#).



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[EquityPoint](#)

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- [PartialOrd](#)
- [PositionSummariser](#)
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- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::statistic::metric](#)

[?](#)

[Change settings](#)

Struct [barter::statistic::metric::EquityPoint](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct EquityPoint {
    pub time: DateTime<Utc>,
    pub total: f64,
}
```

Expand description

Total equity at a point in time - equates to [Balance.total](#).

Fields§

[\\$time](#): [DateTime<Utc>](#)[\\$total](#): [f64](#)

Trait Implementations§

[source§](#)

impl [Clone](#) for [EquityPoint](#)

[source§](#)

fn [clone](#)(&self) -> [EquityPoint](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [EquityPoint](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [EquityPoint](#)

[source§](#)

fn [default](#)() -> Self

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [EquityPoint](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [From](#)<[Balance](#)> for [EquityPoint](#)

[source§](#)

fn [from](#)(balance: [Balance](#)) -> Self

Converts to this type from the input type.
[source§](#)

impl [PartialEq](#) for [EquityPoint](#)

[source§](#)

fn [eq](#)(&self, other: &[EquityPoint](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.

1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [EquityPoint](#)

[source§](#)

fn [partial_cmp](#)(&self, other: [&EquityPoint](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source§](#)

impl [PositionSummariser](#) for [EquityPoint](#)

[source§](#)

fn [update](#)(&mut self, position: [&Position](#))

Updates using the input [Position](#)'s PnL & associated timestamp.

[source§](#)

fn [generate_summary](#)(&mut self, positions: &[Position](#))

[source§](#)

impl [Serialize](#) for [EquityPoint](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Copy](#) for [EquityPoint](#)

[source§](#)

impl [StructuralPartialEq](#) for [EquityPoint](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [EquityPoint](#)

§

impl [Send](#) for [EquityPoint](#)

§

impl [Sync](#) for [EquityPoint](#)

§

impl [Unpin](#) for [EquityPoint](#)

§

impl [UnwindSafe](#) for [EquityPoint](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in\_current\_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

```
impl<T> Same for T
```

[§](#)

```
type Output = T
```

Should always be Self
[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

[§](#)

```
type Owned = T
```

The resulting type after obtaining ownership.
[source§](#)

```
fn to\_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

```
fn clone\_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more source](#)

```
impl<T> DeserializeOwned for T  
where T: for<'de> Deserialize<'de>,
```

statistic - metric - drawdown



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[AvgDrawdown](#)

[Fields](#)

- [count](#)
- [mean_drawdown](#)
- [mean_duration](#)
- [mean_duration_milliseconds](#)

[Methods](#)

- [init](#)
- [update](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::statistic::metric::drawdown

[?](#)

[Change settings](#)

Struct [barter::statistic::metric::drawdown::AvgDrawdown](#)

Copy item path

[source](#) ·

[−]

```
pub struct AvgDrawdown {
    pub count: u64,
    pub mean_drawdown: f64,
    pub mean_duration: Duration,
    pub mean_duration_milliseconds: i64,
}
```

Expand description

[AvgDrawdown](#) contains the average drawdown value and duration from a collection of [Drawdown](#)s within a specific period.

Fields§

§count: [u64](#)§mean_drawdown: [f64](#)§mean_duration: [Duration](#)§mean_duration_milliseconds: [i64](#)

Implementations§

[source§](#)

impl [AvgDrawdown](#)

[source](#)

pub fn [init](#)() -> Self

Initialises a new [AvgDrawdown](#) using the default method, providing zero values for all fields.

[source](#)

pub fn [update](#)(&mut self, drawdown: &[Drawdown](#))

Updates the [AvgDrawdown](#) using the latest input [Drawdown](#) of the Portfolio.

Trait Implementations§

[source§](#)

impl [Clone](#) for [AvgDrawdown](#)

[source§](#)

fn [clone](#)(&self) -> [AvgDrawdown](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: &[Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [AvgDrawdown](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [AvgDrawdown](#)

[source§](#)

fn [default](#)() -> Self

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [AvgDrawdown](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [PartialEq](#) for [AvgDrawdown](#)

[source§](#)

fn [eq](#)(&self, other: &[AvgDrawdown](#)) -> bool

This method tests for self and other values to be equal, and is used by ==.

1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> bool

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [AvgDrawdown](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[AvgDrawdown](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> bool

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [AvgDrawdown](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::[Ok](#), __S::[Error](#)>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Copy](#) for [AvgDrawdown](#)

[source§](#)

impl [StructuralPartialEq](#) for [AvgDrawdown](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [AvgDrawdown](#)

§

impl [Send](#) for [AvgDrawdown](#)

§

impl [Sync](#) for [AvgDrawdown](#)

[§](#)

impl [Unpin](#) for [AvgDrawdown](#)

[§](#)

impl [UnwindSafe](#) for [AvgDrawdown](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

[barter](#)0.8.13

[Module drawdown](#)

- [Structs](#)

[In barter::statistic::metric](#)

[?](#)

[Change settings](#)

Module [barter::statistic::metric::drawdown](#)

Copy item path

[source](#) ·

[-]

Structs§

- [AvgDrawdown](#)
[AvgDrawdown](#) contains the average drawdown value and duration from a collection of [Drawdown](#)s within a specific period.
- [Drawdown](#)
[Drawdown](#) is the peak-to-trough decline of the Portfolio, or investment, during a specific period. Drawdown is a measure of downside volatility.
- [MaxDrawdown](#)
[MaxDrawdown](#) is the largest peak-to-trough decline of the Portfolio, or investment. Max Drawdown is a measure of downside risk, with large values indicating down movements could be volatile.



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[MaxDrawdown](#)

[Fields](#)

- [drawdown](#)

[Methods](#)

- [init](#)
- [update](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::statistic::metric::drawdown`

[?](#)

[Change settings](#)

Struct [barter::statistic::metric::drawdown::MaxDrawdown](#)

Copy item path

[source](#) ·

```
pub struct MaxDrawdown {  
    pub drawdown: Drawdown,  
}
```

Expand description

[MaxDrawdown](#) is the largest peak-to-trough decline of the Portfolio, or investment. Max Drawdown is a measure of downside risk, with large values indicating down movements could be volatile.

See documentation: <https://www.investopedia.com/terms/m/maximum-drawdown-mdd.asp>

Fields§

§drawdown: [Drawdown](#)

Implementations§

[source§](#)

impl [MaxDrawdown](#)

[source](#)

pub fn [init](#)() -> Self

Initialises a new [MaxDrawdown](#) using the [Drawdown](#) default value.

[source](#)

pub fn [update](#)(&mut self, next_drawdown: &[Drawdown](#))

Updates the [MaxDrawdown](#) using the latest input [Drawdown](#) of the Portfolio. If the input drawdown is larger than the current [MaxDrawdown](#), it supersedes it.

Trait Implementations§

[source§](#)

impl [Clone](#) for [MaxDrawdown](#)

[source§](#)

fn [clone](#)(&self) -> [MaxDrawdown](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: &[Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [MaxDrawdown](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [MaxDrawdown](#)

[source§](#)

```
fn deserialize<__D>(__deserializer: __D) -> Result<Self, __D::Error>  
where __D: Deserializer<'de>,
```

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) **for** [MaxDrawdown](#)

[source§](#)

```
fn eq(&self, other: &MaxDrawdown) -> bool
```

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

```
fn ne(&self, other: &Rhs) -> bool
```

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) **for** [MaxDrawdown](#)

[source§](#)

```
fn partial\_cmp(&self, other: &MaxDrawdown) -> Option<Ordering>
```

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

```
fn lt(&self, other: &Rhs) -> bool
```

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

```
fn le(&self, other: &Rhs) -> bool
```

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

```
fn gt(&self, other: &Rhs) -> bool
```

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

```
fn ge(&self, other: &Rhs) -> bool
```

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

```
impl Serialize for MaxDrawdown
```

[source§](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

```
impl Copy for MaxDrawdown
```

[source§](#)

```
impl StructuralPartialEq for MaxDrawdown
```

Auto Trait Implementations§

§

```
impl RefUnwindSafe for MaxDrawdown
```

§

```
impl Send for MaxDrawdown
```

§

```
impl Sync for MaxDrawdown
```

§

```
impl Unpin for MaxDrawdown
```

§

```
impl UnwindSafe for MaxDrawdown
```

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in\_current\_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

```
impl<T> Same for T
```

[§](#)

```
type Output = T
```

Should always be Self

[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

[§](#)

```
type Owned = T
```

The resulting type after obtaining ownership.

[source§](#)

```
fn to\_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

```
fn clone\_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de> ,



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[Drawdown](#)

[Fields](#)

- [drawdown](#)
- [duration](#)
- [equity_range](#)
- [start_time](#)

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- [init](#)
- [is_waiting_for_peak](#)
- [update](#)

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- [PartialEq](#)
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- [From<T>](#)
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- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In [barter::statistic::metric::drawdown](#)

[?](#)

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Struct [barter::statistic::metric::drawdown::Drawdown](#)

Copy item path

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[\[-\]](#)

```
pub struct Drawdown {  
    pub equity_range: Range,  
    pub drawdown: f64,  
    pub start_time: DateTime<Utc>,  
    pub duration: Duration,  
}
```

Expand description

[Drawdown](#) is the peak-to-trough decline of the Portfolio, or investment, during a specific period. Drawdown is a measure of downside volatility.

See documentation: <https://www.investopedia.com/terms/d/drawdown.asp>

Fields§

§equity_range: [Range](#)§drawdown: [f64](#)§start_time: [DateTime<Utc>](#)§duration: [Duration](#)

Implementations§

[source§](#)

impl [Drawdown](#)

[source](#)

```
pub fn init(starting_equity: f64) -> Self
```

Initialises a new [Drawdown](#) using the starting equity as the first peak.

[source](#)

```
pub fn update(&mut self, current: EquityPoint) ->  
Option<Drawdown>
```

Updates the [Drawdown](#) using the latest input [EquityPoint](#) of the Portfolio. If the drawdown period has ended (investment recovers from a trough back above the previous peak), the function return `Some(Drawdown)`, else `None` is returned.

[source](#)

```
pub fn is\_waiting\_for\_peak(&self) -> bool
```

Determines if a [Drawdown](#) is waiting for the next equity peak. This is true if the new [EquityPoint](#) is higher than the previous peak.

[source](#)

```
pub fn calculate(&self) -> f64
```

Calculates the value of the [Drawdown](#) in the specific period. Uses the formula: $\text{Drawdown} = (\text{range_low} - \text{range_high}) / \text{range_high}$

Trait Implementations§

[source§](#)

impl [Clone](#) for [Drawdown](#)

[source§](#)

```
fn clone(&self) -> Drawdown
```

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [Drawdown](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [Drawdown](#)

[source§](#)

fn [default](#)() -> Self

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [Drawdown](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [Drawdown](#)

[source§](#)

fn [eq](#)(&self, other: &[Drawdown](#)) -> bool

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> bool

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [Drawdown](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[Drawdown](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source§](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source§](#)

fn [ge](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source§](#)

impl [Serialize](#) for [Drawdown](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::[Ok](#), __S::[Error](#)>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)

[source§](#)

impl [Copy](#) for [Drawdown](#)

[source§](#)

impl [StructuralPartialEq](#) for [Drawdown](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Drawdown](#)

§

impl [Send](#) for [Drawdown](#)

§

impl [Sync](#) for [Drawdown](#)

§

impl [Unpin](#) for [Drawdown](#)

§

impl [UnwindSafe](#) for [Drawdown](#)

Blanket Implementations§

[source](#)§

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)§

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)§

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)§

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)

[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)

[source§](#)

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

**impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,**

statistic - metric - ratio



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CalmarRatio

Fields

- [calmar_ratio_per_trade](#)
- [risk_free_return](#)
- [trades_per_day](#)

Methods

- [update](#)

Trait Implementations

- [Clone](#)
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- [PartialOrd](#)
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Auto Trait Implementations

- [RefUnwindSafe](#)
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Blanket Implementations

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- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::statistic::metric::ratio`

?

[Change settings](#)

Struct `barter::statistic::metric::ratio::CalmarRatio`

Copy item path

[source](#) ·

[–]

```
pub struct CalmarRatio {  
    pub risk_free_return: f64,  
    pub trades_per_day: f64,  
    pub calmar_ratio_per_trade: f64,  
}
```

Fields§

`risk_free_return`: [f64](#)`trades_per_day`: [f64](#)`calmar_ratio_per_trade`: [f64](#)

Implementations§

[source§](#)

impl `CalmarRatio`

[source](#)

```
pub fn update(&mut self, pnl_returns: &PnLReturnSummary,  
max_drawdown: f64)
```

Trait Implementations§

[source§](#)

impl `Clone` for `CalmarRatio`

[source§](#)

```
fn clone(&self) -> CalmarRatio
```

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [CalmarRatio](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [CalmarRatio](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [CalmarRatio](#)

[source§](#)

fn [eq](#)(&self, other: &[CalmarRatio](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [CalmarRatio](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[CalmarRatio](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Ratio](#) for [CalmarRatio](#)

[source](#)

fn [init](#)(risk_free_return: [f64](#)) -> Self

[source](#)

fn [ratio](#)(&self) -> [f64](#)

[source](#)

fn [trades_per_day](#)(&self) -> [f64](#)

[source](#)

fn [daily](#)(&self) -> [f64](#)

[source](#)

fn [annual](#)(&self, trading_days: [u32](#)) -> [f64](#)

[source](#)

impl [Serialize](#) for [CalmarRatio](#)

[source](#)§

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)§

impl [Copy](#) for [CalmarRatio](#)

[source](#)§

impl [StructuralPartialEq](#) for [CalmarRatio](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [CalmarRatio](#)

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impl [Send](#) for [CalmarRatio](#)

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impl [Sync](#) for [CalmarRatio](#)

§

impl [Unpin](#) for [CalmarRatio](#)

§

impl [UnwindSafe](#) for [CalmarRatio](#)

Blanket Implementations§

[source](#)§

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source§](#)

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

§

```
type Owned = T
```

The resulting type after obtaining ownership.

[source§](#)

```
fn to_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

```
fn clone_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

```
impl<T, U> TryFrom<U> for T  
where U: Into<T>,
```

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

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Structs§

- [CalmarRatio](#)
- [SharpeRatio](#)
- [SortinoRatio](#)

Traits§

- [Ratio](#)

Functions§

- [calculate_annual](#)
- [calculate_daily](#)

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SharpeRatio

Fields

- [risk_free_return](#)
- [sharpe_ratio_per_trade](#)
- [trades_per_day](#)

Methods

- [update](#)

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- [Ratio](#)
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- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

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Struct [barter::statistic::metric::ratio::SharpeRatio](#)

Copy item path

[source](#) ·

[−]

```
pub struct SharpeRatio {  
    pub risk_free_return: f64,  
    pub trades_per_day: f64,  
    pub sharpe_ratio_per_trade: f64,  
}
```

Fields§

[risk_free_return: \[f64\]\(#\)](#)[trades_per_day: \[f64\]\(#\)](#)[sharpe_ratio_per_trade: \[f64\]\(#\)](#)

Implementations§

[source§](#)

impl [SharpeRatio](#)

[source](#)

```
pub fn update(&mut self, pnl_returns: &PnLReturnSummary)
```

Trait Implementations§

[source§](#)

impl [Clone](#) for [SharpeRatio](#)

[source§](#)

```
fn clone(&self) -> SharpeRatio
```

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [SharpeRatio](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [SharpeRatio](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [SharpeRatio](#)

[source§](#)

fn [eq](#)(&self, other: &[SharpeRatio](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [SharpeRatio](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[SharpeRatio](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Ratio](#) for [SharpeRatio](#)

[source](#)

fn [init](#)(risk_free_return: [f64](#)) -> Self

[source](#)

fn [ratio](#)(&self) -> [f64](#)

[source](#)

fn [trades_per_day](#)(&self) -> [f64](#)

[source](#)

fn [daily](#)(&self) -> [f64](#)

[source](#)

fn [annual](#)(&self, trading_days: [u32](#)) -> [f64](#)

[source](#)

impl [Serialize](#) for [SharpeRatio](#)

[source§](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Copy](#) for [SharpeRatio](#)

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impl [StructuralPartialEq](#) for [SharpeRatio](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [SharpeRatio](#)

§

impl [Send](#) for [SharpeRatio](#)

§

impl [Sync](#) for [SharpeRatio](#)

§

impl [Unpin](#) for [SharpeRatio](#)

§

impl [UnwindSafe](#) for [SharpeRatio](#)

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source§](#)

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

§

```
type Owned = T
```

The resulting type after obtaining ownership.

[source§](#)

```
fn to_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

```
fn clone_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

```
impl<T, U> TryFrom<U> for T  
where U: Into<T>,
```

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```

[barter](#)0.8.13

[In barter::statistic::metric::ratio](#)

[?](#)

[Change settings](#)

Function [barter::statistic::metric::ratio::calculate_annual](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub fn calculate_annual(
    ratio_per_trade: f64,
    trades_per_day: f64,
    trading_days: u32
) -> f64
```

[barter](#)0.8.13

[SortinoRatio](#)

[Fields](#)

- [risk_free_return](#)
- [sortino_ratio_per_trade](#)
- [trades_per_day](#)

[Methods](#)

- [update](#)

Trait Implementations

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Ratio](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::statistic::metric::ratio`

[?](#)
[Change settings](#)

Struct `barter::statistic::metric::ratio::`

SortinoRatio

Copy item path

[source](#) ·

[−]

```
pub struct SortinoRatio {  
    pub risk_free_return: f64,
```

```
pub trades_per_day: f64,  
pub sortino_ratio_per_trade: f64,  
}
```

Fields§

[risk_free_return](#): [f64](#)[trades_per_day](#): [f64](#)[sortino_ratio_per_trade](#): [f64](#)

Implementations§

[source§](#)

impl [SortinoRatio](#)

[source](#)

```
pub fn update(&mut self, pnl_returns: &PnLReturnSummary)
```

Trait Implementations§

[source§](#)

impl [Clone](#) for [SortinoRatio](#)

[source§](#)

```
fn clone(&self) -> SortinoRatio
```

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

```
fn clone\_from(&mut self, source: &Self)
```

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [SortinoRatio](#)

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [SortinoRatio](#)

[source§](#)

```
fn deserialize<__D>(__deserializer: __D) -> Result<Self, __D::Error>  
where __D: Deserializer<'de>,
```

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [SortinoRatio](#)

[source§](#)

```
fn eq(&self, other: &SortinoRatio) -> bool
```

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

```
fn ne(&self, other: &Rhs) -> bool
```

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [SortinoRatio](#)

[source§](#)

```
fn partial\_cmp(&self, other: &SortinoRatio) -> Option<Ordering>
```

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

```
fn lt(&self, other: &Rhs) -> bool
```

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

```
fn le(&self, other: &Rhs) -> bool
```

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

```
fn gt(&self, other: &Rhs) -> bool
```

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

```
fn ge(&self, other: &Rhs) -> bool
```

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

```
impl Ratio for SortinoRatio
```

[source](#)

```
fn init(risk_free_return: f64) -> Self
```

[source](#)

```
fn ratio(&self) -> f64
```

[source](#)

```
fn trades_per_day(&self) -> f64
```

[source](#)

```
fn daily(&self) -> f64
```

[source](#)

```
fn annual(&self, trading_days: u32) -> f64
```

[source](#)

```
impl Serialize for SortinoRatio
```

[source](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [Copy](#) for [SortinoRatio](#)

[source](#)§

impl [StructuralPartialEq](#) for [SortinoRatio](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [SortinoRatio](#)

§

impl [Send](#) for [SortinoRatio](#)

§

impl [Sync](#) for [SortinoRatio](#)

§

impl [Unpin](#) for [SortinoRatio](#)

§

impl [UnwindSafe](#) for [SortinoRatio](#)

Blanket Implementations§

[source](#)§

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)§

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)§

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> From<T> for T
```

[source§](#)

```
fn from(t: T) -> T
```

Returns the argument unchanged.

[source§](#)

```
impl<T> Instrument for T
```

[source§](#)

```
fn instrument(self, span: Span) -> Instrumented<Self>
```

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in_current_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From<T>](#) for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be Self

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)

[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)

[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



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In barter::statistic::metric::ratio

[?](#)
[Change settings](#)

Function barter::statistic::metric::ratio::calculate_daily

Copy item path

[source](#) ·

[−]

pub fn calculate_daily(ratio_per_trade: [f64](#), trades_per_day: [f64](#)) -> [f64](#)

barter0.8.13

Ratio

Required Methods

- [init](#)
- [ratio](#)
- [trades_per_day](#)

Provided Methods

- [annual](#)
- [daily](#)

Object Safety

Implementors

In barter::statistic::metric::ratio

[?](#)
[Change settings](#)

Trait [barter::statistic::metric::ratio::Ratio](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub trait Ratio {  
    // Required methods  
    fn init(risk_free_return: f64) -> Self;  
    fn ratio(&self) -> f64;  
    fn trades\_per\_day(&self) -> f64;  
  
    // Provided methods  
    fn daily(&self) -> f64 { ... }  
    fn annual(&self, trading_days: u32) -> f64 { ... }  
}
```

Required Methods§

[source](#)

fn [init](#)(risk_free_return: [f64](#)) -> Self

[source](#)

fn [ratio](#)(&self) -> [f64](#)

[source](#)

fn [trades_per_day](#)(&self) -> [f64](#)

Provided Methods§

[source](#)

fn [daily](#)(&self) -> [f64](#)

[source](#)

fn [annual](#)(&self, trading_days: [u32](#)) -> [f64](#)

Object Safety§

This trait is **not** [object safe](#).

Implementors§

[source§](#)

impl [Ratio](#) for [CalmarRatio](#)

[source§](#)

impl [Ratio](#) for [SharpeRatio](#)

[source§](#)

impl [Ratio](#) for [SortinoRatio](#)

statistic - error

[barter](#)0.8.13

[Module error](#)

- [Enums](#)

[In barter::statistic](#)

²

[Change settings](#)

Module [barter::statistic::error](#)

Copy item path

[source](#) ·

[−]

Enums§

- [StatisticError](#)
All errors generated in the barter::statistic module.

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StatisticError

Variants

- [BuilderIncomplete](#)
- [BuilderNoMetricsProvided](#)

Trait Implementations

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Display](#)
- [Error](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [ToString](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::statistic::error

?

[Change settings](#)

Enum [barter::statistic::error::StatisticError](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub enum StatisticError {  
    BuilderIncomplete(&'static str),  
    BuilderNoMetricsProvided,  
}
```

Expand description

All errors generated in the barter::statistic module.

Variants§

§

BuilderIncomplete(&'static [str](#))

§

BuilderNoMetricsProvided

Trait Implementations§

[source](#)§

impl [Clone](#) for [StatisticError](#)

[source](#)§

fn [clone](#)(&self) -> [StatisticError](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)

[source](#)§

impl [Debug](#) for [StatisticError](#)

[source](#)§

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Display](#) for [StatisticError](#)

[source§](#)

fn [fmt](#)(&self, __formatter: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Error](#) for [StatisticError](#)

1.30.0 · [source§](#)

fn [source](#)(&self) -> [Option](#)<&(dyn [Error](#) + 'static)>

The lower-level source of this error, if any. [Read more](#)
1.0.0 · [source§](#)

fn [description](#)(&self) -> &[str](#)

Deprecated since 1.42.0: use the Display impl or to_string()
[Read more](#)
1.0.0 · [source§](#)

fn [cause](#)(&self) -> [Option](#)<&dyn [Error](#)>

Deprecated since 1.33.0: replaced by Error::source, which can support
downcasting
[source§](#)

fn [provide](#)<'a>(&'a self, request: &mut [Request](#)<'a>)

This is a nightly-only experimental API. (`error_generic_member_access`)
Provides type based access to context intended for error reports. [Read more](#)
[source§](#)

impl [Copy](#) for [StatisticError](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [StatisticError](#)

[§](#)

impl [Send](#) for [StatisticError](#)

[§](#)

impl [Sync](#) for [StatisticError](#)

[§](#)

impl [Unpin](#) for [StatisticError](#)

[§](#)

impl [UnwindSafe](#) for [StatisticError](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)

impl<T> [ToString](#) for T
where T: [Display](#) + ?[Sized](#),

[source](#)

default fn [to_string](#)(&self) -> [String](#)

Converts the given value to a String. [Read more](#)
[source](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

data



[barter](#)0.8.13

[Module data](#)

- [Modules](#)
- [Structs](#)
- [Enums](#)
- [Traits](#)

[In crate barter](#)

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Module [barter::data](#)

Copy item path

[source](#) ·

Expand description

Defines a MarketEvent, and provides the Continuer and MarketGenerator traits for handling the generation of them. Contains implementations such as the (tick-by_tick) LiveTradeHandler, and HistoricalCandleHandler that generates a market feed and acts as the system heartbeat.

Modules§

- [error](#)
Barter data module specific errors.
- [historical](#)
Historical market event feed for backtesting.
- [live](#)
Live market event feed for dry-trading & live-trading.

Structs§

- [MarketMeta](#)
Metadata detailing the [Candle](#) or [Trade](#) close price & it's associated timestamp. Used to propagate key market information in downstream Events.

Enums§

- [Feed](#)
Communicates the state of the [Feed](#) as well as the next event.

Traits§

- [MarketGenerator](#)
Generates the next Event. Acts as the system heartbeat.

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[MarketGenerator](#)

[Required Methods](#)

- [next](#)

[Implementors](#)

[In barter::data](#)

[?](#)

[Change settings](#)

Trait [barter::data::MarketGenerator](#)

Copy item path

[source](#) ·

[−]

```
pub trait MarketGenerator<Event> {  
    // Required method  
    fn next(&mut self) -> Feed<Event>;  
}
```

Expand description

Generates the next Event. Acts as the system heartbeat.

Required Methods§

[source](#)

fn [next](#)(&mut self) -> [Feed](#)<Event>

Return the next market Event.

Implementors§

[source§](#)

```
impl<Event> MarketGenerator<Event> for  
barter::data::live::MarketFeed<Event>
```

[source§](#)

```
impl<Iter, Event> MarketGenerator<Event> for  
barter::data::historical::MarketFeed<Iter, Event>  
where Iter: Iterator<Item = Event>,
```



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[MarketMeta](#)

[Fields](#)

- [close](#)
- [time](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)

- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::data](#)

[?](#)

[Change settings](#)

Struct [barter::data::MarketMeta](#)

Copy item path

[source](#) ·

```
pub struct MarketMeta {
    pub close: f64,
    pub time: DateTime<Utc>,
}
```

Expand description

Metadata detailing the [Candle](#) or [Trade](#) close price & it's associated timestamp. Used to propagate key market information in downstream Events.

Fields§

[\\$close](#): [f64](#)

Close value from the source market event.

[\\$time](#): [DateTime<Utc>](#)

Exchange timestamp from the source market event.

Trait Implementations§

[source§](#)

impl [Clone](#) for [MarketMeta](#)

[source§](#)

fn [clone](#)(&self) -> [MarketMeta](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [Self](#))

Performs copy-assignment from source. [Read more](#)

[source§](#)

impl [Debug](#) for [MarketMeta](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl [Default](#) for [MarketMeta](#)

[source§](#)

fn [default](#)() -> [Self](#)

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [MarketMeta](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<[Self](#), __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [PartialEq](#) for [MarketMeta](#)

[source§](#)

fn [eq](#)(&self, other: &[MarketMeta](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.

1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [MarketMeta](#)

[source§](#)

fn [partial_cmp](#)(&self, other: [&MarketMeta](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source§](#)

impl [Serialize](#) for [MarketMeta](#)

[source§](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

```
impl Copy for MarketMeta
```

[source](#)

```
impl StructuralPartialEq for MarketMeta
```

Auto Trait Implementations

[§](#)

```
impl RefUnwindSafe for MarketMeta
```

[§](#)

```
impl Send for MarketMeta
```

[§](#)

```
impl Sync for MarketMeta
```

[§](#)

```
impl Unpin for MarketMeta
```

[§](#)

```
impl UnwindSafe for MarketMeta
```

Blanket Implementations

[source](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)§

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be `Self`

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



[barter](#)0.8.13

[Feed](#)

[Variants](#)

- [Finished](#)
- [Next](#)
- [Unhealthy](#)

[Trait Implementations](#)

- [Clone](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [Eq](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [Equivalent<K>](#)
- [Equivalent<K>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)

- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::data](#)

[?](#)

[Change settings](#)

Enum [barter::data::Feed](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub enum Feed<Event> {
    Next(Event),
    Unhealthy,
    Finished,
}
```

Expand description

Communicates the state of the [Feed](#) as well as the next event.

Variants§

§

Next(Event)

§

Unhealthy

§

Finished

Trait Implementations§

[source§](#)

impl<Event: [Clone](#)> [Clone](#) for [Feed](#)<Event>

[source§](#)

fn [clone](#)(&self) -> [Feed](#)<Event>

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)

impl<Event: [Debug](#)> [Debug](#) for [Feed](#)<Event>

[source](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

**impl<'de, Event> [Deserialize](#)<'de> for [Feed](#)<Event>
where Event: [Deserialize](#)<'de>,**

[source](#)

**fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,**

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl<Event: [PartialEq](#)> [PartialEq](#) for [Feed](#)<Event>

[source](#)

fn [eq](#)(&self, other: &[Feed](#)<Event>) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)

impl<Event: [PartialOrd](#)> [PartialOrd](#) for [Feed](#)<Event>

[source](#)

fn [partial_cmp](#)(&self, other: &[Feed](#)<Event>) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

**impl<Event> [Serialize](#) for [Feed](#)<Event>
where Event: [Serialize](#),**

[source](#)

**fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok,
__S::Error>
where __S: [Serializer](#),**

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl<Event: [Eq](#)> [Eq](#) for [Feed](#)<Event>

[source](#)

impl<Event> [StructuralPartialEq](#) for [Feed](#)<Event>

Auto Trait Implementations§

[§](#)

impl<Event> [RefUnwindSafe](#) for [Feed](#)<Event>
where Event: [RefUnwindSafe](#),

[§](#)

impl<Event> [Send](#) for [Feed](#)<Event>
where Event: [Send](#),

[§](#)

impl<Event> [Sync](#) for [Feed](#)<Event>
where Event: [Sync](#),

[§](#)

impl<Event> [Unpin](#) for [Feed](#)<Event>
where Event: [Unpin](#),

[§](#)

impl<Event> [UnwindSafe](#) for [Feed](#)<Event>
where Event: [UnwindSafe](#),

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more source§](#)

```
impl<Q, K> Equivalent<K> for Q  
where Q: Eq + ?Sized, K: Borrow<Q> + ?Sized,
```

[source§](#)

```
fn equivalent(&self, key: &K) -> bool
```

Checks if this value is equivalent to the given key. [Read more source§](#)

```
impl<Q, K> Equivalent<K> for Q  
where Q: Eq + ?Sized, K: Borrow<Q> + ?Sized,
```

[source§](#)

```
fn equivalent(&self, key: &K) -> bool
```

Compare self to key and return true if they are equal. [source§](#)

```
impl<T> From<T> for T
```

[source§](#)

```
fn from(t: T) -> T
```

Returns the argument unchanged.

[source§](#)

```
impl<T> Instrument for T
```

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> **for** T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> **for** T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> **for** T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) **for** T

[source§](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```

data - live

[barter](#)0.8.13

[Module live](#)

- [Structs](#)

[In barter::data](#)

?

[Change settings](#)

Module [barter::data::live](#)

Copy item path

[source](#) ·

[-]

Expand description

Live market event feed for dry-trading & live-trading.

Structs

- [MarketFeed](#)
Live [Feed](#) of market events.



barter0.8.13

MarketFeed

Fields

- [market_rx](#)

Methods

- [new](#)

Trait Implementations

- [Debug](#)
- [MarketGenerator<Event>](#)

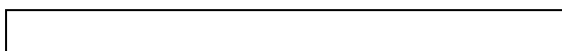
Auto Trait Implementations

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::data::live



[?](#)
[Change settings](#)

Struct [barter::data::live::MarketFeed](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct MarketFeed<Event> {  
    pub market_rx: UnboundedReceiver<Event>,  
}
```

Expand description

Live [Feed](#) of market events.

Fields§

`market_rx`: [UnboundedReceiver](#)<Event>

Implementations§

[source§](#)

impl<Event> [MarketFeed](#)<Event>

[source](#)

pub fn [new](#)(market_rx: [UnboundedReceiver](#)<Event>) -> Self

Initialises a live [MarketFeed](#) that yields market Events from the provided [mpsc::UnboundedReceiver](#).

Recommended use with the Barter-Data [Streams](#):

1. Initialise a [Streams](#) using the [StreamBuilder](#) or [MultiStreamBuilder](#).
2. Use [Streams::join](#) to join all exchange [mpsc::UnboundedReceiver](#) streams into a unified [mpsc::UnboundedReceiver](#).
3. Construct [Self](#) with the unified [mpsc::UnboundedReceiver](#).

Trait Implementations§

[source§](#)

impl<Event: [Debug](#)> [Debug](#) for [MarketFeed](#)<Event>

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

```
impl<Event> MarketGenerator<Event> for  
MarketFeed<Event>
```

[source§](#)

```
fn next(&mut self) -> Feed<Event>
```

Return the next market Event.

Auto Trait Implementations§

§

```
impl<Event> !RefUnwindSafe for MarketFeed<Event>
```

§

```
impl<Event> Send for MarketFeed<Event>  
where Event: Send,
```

§

```
impl<Event> Sync for MarketFeed<Event>  
where Event: Send,
```

§

```
impl<Event> Unpin for MarketFeed<Event>
```

§

```
impl<Event> !UnwindSafe for MarketFeed<Event>
```

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source§](#)

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source§](#)

```
impl<T, U> TryFrom<U> for T
where U: Into<T>,
```

§

```
type Error = Infallible
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<T, U> TryInto<U> for T
where U: TryFrom<T>,
```

§

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source](#)

```
fn vzip(self) -> V
```

[source](#)

```
impl<T> WithSubscriber for T
```

[source](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

data - historical

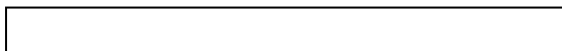


[barter](#)0.8.13

[Module historical](#)

- [Structs](#)

[In barter::data](#)



[?](#)
[Change settings](#)

Module [barter::data::historical](#)

Copy item path

[source](#) ·

[\[-\]](#)

Expand description

Historical market event feed for backtesting.

Structs§

- [MarketFeed](#)
Historical [Feed](#) of market events.

[barter](#)0.8.13

[MarketFeed](#)

[Fields](#)

- [market_iterator](#)

[Methods](#)

- [new](#)

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- [MarketGenerator<Event>](#)

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- [Instrument](#)

- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In [barter::data::historical](#)

[?](#)

[Change settings](#)

Struct [barter::data::historical::MarketFeed](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct MarketFeed<Iter, Event>
where
    Iter: Iterator<Item = Event>,
{
    pub market_iterator: Iter,
}
```

Expand description

Historical [Feed](#) of market events.

Fields§

§market_iterator: Iter

Implementations§

[source§](#)

```
impl<Iter, Event> MarketFeed<Iter, Event>
where Iter: Iterator<Item = Event>,
```

[source](#)

```
pub fn new<IntoIter>(market_iterator: IntoIter) -> Self
where IntoIter: IntoIterator<Item = Event, IntoIter = Iter>,
```

Construct a historical [MarketFeed](#) that yields market events from the IntoIterator provided.

Trait Implementations§

[source§](#)

```
impl<Iter, Event: Debug> Debug for MarketFeed<Iter, Event>  
where Iter: Iterator<Item = Event> + Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

```
impl<Iter, Event> MarketGenerator<Event> for  
MarketFeed<Iter, Event>  
where Iter: Iterator<Item = Event>,
```

[source§](#)

```
fn next(&mut self) -> Feed<Event>
```

Return the next market Event.

Auto Trait Implementations§

§

```
impl<Iter, Event> RefUnwindSafe for MarketFeed<Iter, Event>  
where Iter: RefUnwindSafe,
```

§

```
impl<Iter, Event> Send for MarketFeed<Iter, Event>  
where Iter: Send,
```

§

```
impl<Iter, Event> Sync for MarketFeed<Iter, Event>  
where Iter: Sync,
```

§

```
impl<Iter, Event> Unpin for MarketFeed<Iter, Event>  
where Iter: Unpin,
```

[§](#)

```
impl<Iter, Event> UnwindSafe for MarketFeed<Iter,  
Event>  
where Iter: UnwindSafe,
```

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source§](#)

```
impl<T> Borrow<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow\_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> From<T> for T
```

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

data - error

[barter](#)**0.8.13**

[Module error](#)

- [Enums](#)

[In barter::data](#)

[?](#)

[Change settings](#)

Module [barter::data::error](#)

Copy item path

[source](#) ·

[-]

Expand description

Barter data module specific errors.

Enums§

- [DataError](#)
All errors generated in the barter::data module.

[barter](#)**0.8.13**

[DataError](#)

[Variants](#)

- [BuilderAttributesInvalid](#)
- [BuilderIncomplete](#)
- [Data](#)
- [Socket](#)

Trait Implementations

- [Debug](#)
- [Display](#)
- [Error](#)
- [From<DataError>](#)
- [From<SocketError>](#)

Auto Trait Implementations

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToString](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::data::error`

?

[Change settings](#)

Enum `barter::data::error::DataError`

Copy item path

[source](#) ·

[−]

```
pub enum DataError {
    BuilderAttributesInvalid,
    BuilderIncomplete(&'static str),
    Socket(SocketError),
    Data(DataError),
}
```

Expand description

All errors generated in the `barter::data` module.

Variants§

§

BuilderAttributesInvalid

§

BuilderIncomplete(&'static [str](#))

§

Socket([SocketError](#))

§

Data([DataError](#))

Trait Implementations§

[source](#)§

impl [Debug](#) for [DataError](#)

[source](#)§

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source](#)§

impl [Display](#) for [DataError](#)

[source](#)§

```
fn fmt(&self, __formatter: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source](#)§

impl [Error](#) for [DataError](#)

[source](#)§

fn [source](#)(&self) -> [Option](#)<&(dyn [Error](#) + 'static)>

The lower-level source of this error, if any. [Read more](#)
1.0.0 · [source§](#)

fn [description](#)(&self) -> &[str](#)

Deprecated since 1.42.0: use the Display impl or to_string()
[Read more](#)
1.0.0 · [source§](#)

fn [cause](#)(&self) -> [Option](#)<&dyn [Error](#)>

Deprecated since 1.33.0: replaced by Error::source, which can support
downcasting
[source§](#)

fn [provide](#)<'a>(&'a self, request: &mut [Request](#)<'a>)

This is a nightly-only experimental API. (error_generic_member_access)
Provides type based access to context intended for error reports. [Read more](#)
[source§](#)

impl [From](#)<[DataError](#)> for [DataError](#)

[source§](#)

fn [from](#)(source: [DataError](#)) -> Self

Converts to this type from the input type.
[source§](#)

impl [From](#)<[SocketError](#)> for [DataError](#)

[source§](#)

fn [from](#)(source: [SocketError](#)) -> Self

Converts to this type from the input type.

Auto Trait Implementations§

§

impl ![RefUnwindSafe](#) for [DataError](#)

§

impl [Send](#) for [DataError](#)

[§](#)

impl [Sync](#) for [DataError](#)

[§](#)

impl [Unpin](#) for [DataError](#)

[§](#)

impl [!UnwindSafe](#) for [DataError](#)

Blanket Implementations

[source](#)

impl<T> [Any](#) for T
where T: 'static + [?Sized](#),

[source](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: [?Sized](#),

[source](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [BorrowMut](#)<T> for T
where T: [?Sized](#),

[source](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T> [ToString](#) for T
where T: [Display](#) + ?[Sized](#),

[source](#)§

default fn [to_string](#)(&self) -> [String](#)

Converts the given value to a String. [Read more](#)
[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)§

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

§

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)§

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)§

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)§

fn [vzip](#)(self) -> V

[source](#)§

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

execution

[barter](#)0.8.13

[FeeAmount](#)

[In barter::execution](#)

[?](#)
[Change settings](#)

Type Alias [barter::execution::FeeAmount](#)

Copy item path

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[-]

pub type FeeAmount = [f64](#);

Expand description

Communicative type alias for Fee amount as f64.

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[Module execution](#)

- [Modules](#)
- [Structs](#)
- [Traits](#)
- [Type Aliases](#)

[In crate barter](#)

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Module [barter::execution](#)

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Expand description

Defines a FillEvent, and provides a useful trait FillGenerator for handling the generation of them. Contains an example SimulatedExecution implementation that simulates live broker execution.

Modules§

- [error](#)
Barter execution module specific errors.
- [simulated](#)
Handlers for simulated and live [OrderEvent](#) execution.

Structs§

- [Fees](#)
All potential fees incurred by a [FillEvent](#).
- [FillEvent](#)
Fills are journals of work done by an Execution handler. These are sent back to the portfolio so it can apply updates.
- [FillEventBuilder](#)
Builder to construct [FillEvent](#) instances.

Traits§

- [ExecutionClient](#)
Generates a result [FillEvent](#) by executing an [OrderEvent](#).

Type Aliases

- [FeeAmount](#)
Communicative type alias for Fee amount as f64.



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[FillEventBuilder](#)

Fields

- [decision](#)
- [exchange](#)
- [fees](#)
- [fill_value_gross](#)
- [instrument](#)
- [market_meta](#)
- [quantity](#)
- [time](#)

Methods

- [build](#)
- [decision](#)
- [exchange](#)
- [fees](#)
- [fill_value_gross](#)
- [instrument](#)
- [market_meta](#)
- [new](#)
- [quantity](#)
- [time](#)

[Trait Implementations](#)

- [Debug](#)
- [Default](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::execution

[?](#)

[Change settings](#)

Struct [barter::execution::FillEventBuilder](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct FillEventBuilder {
    pub time: Option<DateTime<Utc>>,
    pub exchange: Option<Exchange>,
    pub instrument: Option<Instrument>,
    pub market_meta: Option<MarketMeta>,
    pub decision: Option<Decision>,
    pub quantity: Option<f64>,
    pub fill_value_gross: Option<f64>,
    pub fees: Option<Fees>,
}
```

Expand description

Builder to construct [FillEvent](#) instances.

Fields

[\\$time](#): [Option<DateTime<Utc>>](#)[\\$exchange](#): [Option<Exchange>](#)[\\$instrument](#):
[Option<Instrument>](#)[\\$market_meta](#): [Option<MarketMeta>](#)[\\$decision](#):
[Option<Decision>](#)[\\$quantity](#): [Option<f64>](#)[\\$fill_value_gross](#):
[Option<f64>](#)[\\$fees](#): [Option<Fees>](#)

Implementations§

[source§](#)

impl [FillEventBuilder](#)

[source](#)

pub fn [new](#)() -> Self

[source](#)

pub fn [time](#)(self, value: [DateTime](#)<[Utc](#)>) -> Self

[source](#)

pub fn [exchange](#)(self, value: [Exchange](#)) -> Self

[source](#)

pub fn [instrument](#)(self, value: [Instrument](#)) -> Self

[source](#)

pub fn [market_meta](#)(self, value: [MarketMeta](#)) -> Self

[source](#)

pub fn [decision](#)(self, value: [Decision](#)) -> Self

[source](#)

pub fn [quantity](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [fill_value_gross](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [fees](#)(self, value: [Fees](#)) -> Self

[source](#)

```
pub fn build(self) -> Result<FillEvent, ExecutionError>
```

Trait Implementations§

[source§](#)

```
impl Debug for FillEventBuilder
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_,>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

```
impl Default for FillEventBuilder
```

[source§](#)

```
fn default() -> FillEventBuilder
```

Returns the “default value” for a type. [Read more](#)

Auto Trait Implementations§

§

```
impl RefUnwindSafe for FillEventBuilder
```

§

```
impl Send for FillEventBuilder
```

§

```
impl Sync for FillEventBuilder
```

§

```
impl Unpin for FillEventBuilder
```

§

```
impl UnwindSafe for FillEventBuilder
```

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)


```
fn in\_current\_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

```
impl<T> Same for T
```

[§](#)

```
type Output = T
```

Should always be Self

[source§](#)

```
impl<T, U> TryFrom<U> for T  
where U: Into<T>,
```

[§](#)

```
type Error = Infallible
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try\_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<T, U> TryInto<U> for T  
where U: TryFrom<T>,
```

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



[barter](#)0.8.13

[Fees](#)

[Fields](#)

- [exchange](#)
- [network](#)
- [slippage](#)

Methods

- [calculate_total_fees](#)

Trait Implementations

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::execution

[?](#)

[Change settings](#)

Struct [barter::execution::Fees](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct Fees {  
    pub exchange: FeeAmount,  
    pub slippage: FeeAmount,  
    pub network: FeeAmount,  
}
```

Expand description

All potential fees incurred by a [FillEvent](#).

Fields§

[\\$exchange](#): [FeeAmount](#)

Fee taken by the exchange/broker (eg/ commission).

[\\$slippage](#): [FeeAmount](#)

Order book slippage modelled as a fee.

[\\$network](#): [FeeAmount](#)

Fee incurred by any required network transactions (eg/ GAS).

Implementations§

[source§](#)

impl [Fees](#)

[source](#)

```
pub fn calculate\_total\_fees(&self) -> f64
```

Calculates the sum of every [FeeAmount](#) in [Fees](#).

Trait Implementations§

[source§](#)

impl [Clone](#) for [Fees](#)

[source§](#)

fn [clone](#)(&self) -> [Fees](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)

impl [Debug](#) for [Fees](#)

[source](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl [Default](#) for [Fees](#)

[source](#)

fn [default](#)() -> [Fees](#)

Returns the “default value” for a type. [Read more](#)
[source](#)

impl<'de> [Deserialize](#)<'de> for [Fees](#)

[source](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl [PartialEq](#) for [Fees](#)

[source](#)

fn [eq](#)(&self, other: &[Fees](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source](#)

impl [PartialOrd](#) for [Fees](#)

[source](#)

fn [partial_cmp](#)(&self, other: [&Fees](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source](#)

impl [Serialize](#) for [Fees](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)

[source§](#)

impl [Copy](#) for [Fees](#)

[source§](#)

impl [StructuralPartialEq](#) for [Fees](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Fees](#)

§

impl [Send](#) for [Fees](#)

§

impl [Sync](#) for [Fees](#)

§

impl [Unpin](#) for [Fees](#)

§

impl [UnwindSafe](#) for [Fees](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> From<T> for T
```

[source§](#)

```
fn from(t: T) -> T
```

Returns the argument unchanged.

[source§](#)

```
impl<T> Instrument for T
```

[source§](#)

```
fn instrument(self, span: Span) -> Instrumented<Self>
```

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in_current_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From<T>](#) for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be Self

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)

[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)

[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

```
impl<T, U> TryInto<U> for T
where U: TryFrom<T>,
```

[§](#)

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.
[source§](#)

```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source§](#)

```
impl<V, T> VZip<V> for T
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```



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FillEvent

Fields

- [decision](#)
- [exchange](#)
- [fees](#)
- [fill_value_gross](#)
- [instrument](#)
- [market_meta](#)
- [quantity](#)
- [time](#)

Associated Constants

- [EVENT_TYPE](#)

Methods

- [builder](#)

Trait Implementations

- [Clone](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)

- [Intro<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::execution](#)

[?](#)

[Change settings](#)

Struct [barter::execution::FillEvent](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct FillEvent {
    pub time: DateTime<Utc>,
    pub exchange: Exchange,
    pub instrument: Instrument,
    pub market_meta: MarketMeta,
    pub decision: Decision,
    pub quantity: f64,
    pub fill_value_gross: f64,
    pub fees: Fees,
}
```

Expand description

Fills are journals of work done by an Execution handler. These are sent back to the portfolio so it can apply updates.

Fields

[\\$time](#): [DateTime<Utc>](#)[\\$exchange](#): [Exchange](#)[\\$instrument](#): [Instrument](#)[\\$market_meta](#): [MarketMeta](#)

Metadata propagated from source MarketEvent

[\\$decision](#): [Decision](#)

LONG, CloseLong, SHORT or CloseShort

[\\$quantity](#): [f64](#)

+ve or -ve Quantity depending on Decision

`$fill_value_gross: f64`

`abs(Quantity) * ClosePrice, excluding TotalFees`

`$fees: Fees`

All fee types incurred when executing an [OrderEvent](#), and their associated [FeeAmount](#).

Implementations§

[source§](#)

impl [FillEvent](#)

[source](#)

pub const [EVENT_TYPE](#): &'static [str](#) = "Fill"

[source](#)

pub fn [builder](#)() -> [FillEventBuilder](#)

Returns a [FillEventBuilder](#) instance.

Trait Implementations§

[source§](#)

impl [Clone](#) for [FillEvent](#)

[source§](#)

fn [clone](#)(&self) -> [FillEvent](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [FillEvent](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [FillEvent](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [FillEvent](#)

[source§](#)

fn [eq](#)(&self, other: &[FillEvent](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [FillEvent](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[FillEvent](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Serialize](#) for [FillEvent](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [StructuralPartialEq](#) for [FillEvent](#)

Auto Trait Implementations

[§](#)

impl [RefUnwindSafe](#) for [FillEvent](#)

[§](#)

impl [Send](#) for [FillEvent](#)

[§](#)

impl [Sync](#) for [FillEvent](#)

[§](#)

impl [Unpin](#) for [FillEvent](#)

[§](#)

impl [UnwindSafe](#) for [FillEvent](#)

Blanket Implementations[§](#)

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

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[ExecutionClient](#)

[Required Methods](#)

- [generate_fill](#)

[Implementors](#)

[In barter::execution](#)

[?](#)

[Change settings](#)

Trait [barter::execution::ExecutionClient](#)

Copy item path

[source](#) ·

[-]

```
pub trait ExecutionClient {
    // Required method
    fn generate\_fill(
        &self,
        order: &OrderEvent
    ) -> Result<FillEvent, ExecutionError>;
}
```

Expand description

Generates a result [FillEvent](#) by executing an [OrderEvent](#).

Required Methods§

[source](#)

```
fn generate\_fill(&self, order: &OrderEvent) -> Result<FillEvent,  
ExecutionError>
```

Return a [FillEvent](#) from executing the input [OrderEvent](#).

Implementors§

[source§](#)

impl [ExecutionClient](#) for [SimulatedExecution](#)

execution - simulated



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[SimulatedExecution](#)

[Methods](#)

- [new](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize](#)<'de>
- [ExecutionClient](#)

- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::execution::simulated

?

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Struct [barter::execution::simulated::SimulatedExecution](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct SimulatedExecution { /* private fields */ }
```

Expand description

Simulated execution handler that executes [OrderEvents](#) to generate [FillEvents](#) via a simulated broker interaction.

Implementations§

[source§](#)

impl [SimulatedExecution](#)

[source](#)

pub fn [new](#)(cfg: [Config](#)) -> Self

Constructs a new [SimulatedExecution](#) component.

Trait Implementations§

[source§](#)

impl [Clone](#) for [SimulatedExecution](#)

[source§](#)

fn [clone](#)(&self) -> [SimulatedExecution](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [SimulatedExecution](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [SimulatedExecution](#)

[source§](#)

fn [default](#)() -> [SimulatedExecution](#)

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [SimulatedExecution](#)

[source§](#)

```
fn deserialize<__D>(__deserializer: __D) -> Result<Self, __D::Error>  
where __D: Deserializer<'de>,
```

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [ExecutionClient](#) for [SimulatedExecution](#)

[source§](#)

```
fn generate\_fill(&self, order: &OrderEvent) -> Result<FillEvent,  
ExecutionError>
```

Return a [FillEvent](#) from executing the input [OrderEvent](#).
[source§](#)

impl [PartialEq](#) for [SimulatedExecution](#)

[source§](#)

```
fn eq(&self, other: &SimulatedExecution) -> bool
```

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

```
fn ne(&self, other: &Rhs) -> bool
```

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [SimulatedExecution](#)

[source§](#)

```
fn partial\_cmp(&self, other: &SimulatedExecution) ->  
Option<Ordering>
```

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Serialize](#) for [SimulatedExecution](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [Copy](#) for [SimulatedExecution](#)

[source](#)

impl [StructuralPartialEq](#) for [SimulatedExecution](#)

Auto Trait Implementations

§

impl [RefUnwindSafe](#) for [SimulatedExecution](#)

§

impl [Send](#) for [SimulatedExecution](#)

[§](#)

impl [Sync](#) for [SimulatedExecution](#)

[§](#)

impl [Unpin](#) for [SimulatedExecution](#)

[§](#)

impl [UnwindSafe](#) for [SimulatedExecution](#)

Blanket Implementations[§](#)

[source](#)[§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)[§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

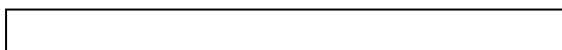


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[Module simulated](#)

- [Structs](#)

[In barter::execution](#)



[?](#)

[Change settings](#)

Module [barter::execution::simulated](#)

Copy item path

[source](#) ·

Expand description

Handlers for simulated and live [OrderEvent](#) execution.

Structs§

- [Config](#)
Configuration for constructing a [SimulatedExecution](#) via the new() constructor method.
- [SimulatedExecution](#)
Simulated execution handler that executes [OrderEvents](#) to generate [FillEvents](#) via a simulated broker interaction.

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[Config](#)

[Fields](#)

- [simulated_fees_pct](#)

[Trait Implementations](#)

- [Clone](#)
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- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::execution::simulated`

[?](#)

[Change settings](#)

Struct `barter::execution::simulated::Config`

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct Config {  
    pub simulated_fees_pct: Fees,  
}
```

Expand description

Configuration for constructing a [SimulatedExecution](#) via the `new()` constructor method.

Fields§

`simulated_fees_pct`: [Fees](#)

Simulated fee percentage to be used for each [Fees](#) field in decimal form (eg/ 0.01 for 1%)

Trait Implementations§

[source§](#)

impl [Clone](#) for [Config](#)

[source§](#)

fn [clone](#)(&self) -> [Config](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)

[source§](#)

impl [Debug](#) for [Config](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl [Default](#) for [Config](#)

[source§](#)

fn [default](#)() -> [Config](#)

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [Config](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [PartialEq](#) for [Config](#)

[source§](#)

fn [eq](#)(&self, other: &[Config](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.

1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for `!=`. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [Config](#)

[source§](#)

fn [partial_cmp](#)(&self, other: [&Config](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between `self` and `other` values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for `self` and `other`) and is used by the `<` operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for `self` and `other`) and is used by the `<=` operator. [Read more](#)

1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for `self` and `other`) and is used by the `>` operator. [Read more](#)

1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for `self` and `other`) and is used by the `>=` operator. [Read more](#)

[source§](#)

impl [Serialize](#) for [Config](#)

[source§](#)


```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

```
impl Copy for Config
```

[source](#)

```
impl StructuralPartialEq for Config
```

Auto Trait Implementations

§

```
impl RefUnwindSafe for Config
```

§

```
impl Send for Config
```

§

```
impl Sync for Config
```

§

```
impl Unpin for Config
```

§

```
impl UnwindSafe for Config
```

Blanket Implementations

[source](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)§

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be `Self`

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```

execution - error

[barter](#)0.8.13

[Module error](#)

- [Enums](#)

[In barter::execution](#)

[?](#)
[Change settings](#)

Module [barter::execution::error](#)

Copy item path

[source](#) ·

[-]

Expand description

Barter execution module specific errors.

Enums§

- [ExecutionError](#)
All errors generated in the barter::execution module.

[barter](#)0.8.13

[ExecutionError](#)

[Variants](#)

- [BuilderIncomplete](#)

Trait Implementations

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Display](#)
- [Error](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [ToString](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::execution::error`

[?](#)
[Change settings](#)

Enum `barter::execution::error::`

ExecutionError

Copy item path

[source](#) ·

[−]

```
pub enum ExecutionError {
    BuilderIncomplete(&'static str),
}
```

Expand description

All errors generated in the `barter::execution` module.

Variants§

§

BuilderIncomplete(&'static [str](#))

Trait Implementations§

[source§](#)

impl [Clone](#) for [ExecutionError](#)

[source§](#)

fn [clone](#)(&self) -> [ExecutionError](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [ExecutionError](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Display](#) for [ExecutionError](#)

[source§](#)

fn [fmt](#)(&self, __formatter: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Error](#) for [ExecutionError](#)

1.30.0 · [source§](#)

fn [source](#)(&self) -> [Option](#)<&(dyn [Error](#) + 'static)>

The lower-level source of this error, if any. [Read more](#)
1.0.0 · [source](#)§

fn [description](#)(&self) -> &str

Deprecated since 1.42.0: use the Display impl or to_string()
[Read more](#)
1.0.0 · [source](#)§

fn [cause](#)(&self) -> [Option](#)<&dyn [Error](#)>

Deprecated since 1.33.0: replaced by Error::source, which can support downcasting
[source](#)§

fn [provide](#)<'a>(&'a self, request: &mut [Request](#)<'a>)

This is a nightly-only experimental API. (error_generic_member_access)
Provides type based access to context intended for error reports. [Read more](#)
[source](#)§

impl [Copy](#) for [ExecutionError](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [ExecutionError](#)

§

impl [Send](#) for [ExecutionError](#)

§

impl [Sync](#) for [ExecutionError](#)

§

impl [Unpin](#) for [ExecutionError](#)

§

impl [UnwindSafe](#) for [ExecutionError](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T> [ToString](#) for T
where T: [Display](#) + ?[Sized](#),

[source§](#)

default fn [to_string](#)(&self) -> [String](#)

Converts the given value to a String. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

engine



[barter](#)0.8.13

[EngineBuilder](#)

[Methods](#)

- [build](#)
- [command_rx](#)
- [engine_id](#)
- [portfolio](#)
- [statistics_summary](#)
- [trader_command_txs](#)
- [traders](#)

[Trait Implementations](#)

- [Debug](#)
- [Default](#)

Auto Trait Implementations

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::engine`

[?](#)

[Change settings](#)

Struct `barter::engine::EngineBuilder`

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct EngineBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution> {  
    where
```

```
    EventTx: MessageTransmitter<Event>,  
    Statistic: Serialize + Send,  
    Portfolio: MarketUpdater + OrderGenerator + FillUpdater + Send,  
    Data: MarketGenerator<MarketEvent<DataKind>> + Send,  
    Strategy: SignalGenerator + Send,  
    Execution: ExecutionClient + Send,  
    { /* private fields */ }
```

Expand description

Builder to construct [Engine](#) instances.

Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> EngineBuilder<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event>, Statistic:
PositionSummariser + Serialize + Send, Portfolio:
PositionHandler + StatisticHandler<Statistic> +
MarketUpdater + OrderGenerator + FillUpdater + Send,
Data: MarketGenerator<MarketEvent<DataKind>> +
Send, Strategy: SignalGenerator + Send, Execution:
ExecutionClient + Send,
```

[source](#)

```
pub fn engine\_id(self, value: Uuid) -> Self
```

[source](#)

```
pub fn command\_rx(self, value: Receiver<Command>) -> Self
```

[source](#)

```
pub fn portfolio(self, value: Arc<Mutex<Portfolio>>) -> Self
```

[source](#)

```
pub fn traders( self, value: Vec<Trader<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>> ) -> Self
```

[source](#)

```
pub fn trader\_command\_txs(self, value: HashMap<Market,
Sender<Command>>) -> Self
```

[source](#)

```
pub fn statistics\_summary(self, value: Statistic) -> Self
```

[source](#)

```
pub fn build( self ) -> Result<Engine<EventTx, Statistic, Portfolio, Data, Strategy, Execution>, EngineError>
```

Trait Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Debug for EngineBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>  
where EventTx: MessageTransmitter<Event> + Debug,  
Statistic: Serialize + Send + Debug, Portfolio:  
MarketUpdater + OrderGenerator + FillUpdater + Send + Debug, Data: MarketGenerator<MarketEvent<DataKind>>  
+ Send + Debug, Strategy: SignalGenerator + Send + Debug, Execution: ExecutionClient + Send + Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Default for EngineBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>  
where EventTx: MessageTransmitter<Event> + Default,  
Statistic: Serialize + Send + Default, Portfolio:  
MarketUpdater + OrderGenerator + FillUpdater + Send + Default, Data:  
MarketGenerator<MarketEvent<DataKind>> + Send + Default, Strategy: SignalGenerator + Send + Default,  
Execution: ExecutionClient + Send + Default,
```

[source§](#)

```
fn default( ) -> EngineBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
```

Returns the “default value” for a type. [Read more](#)

Auto Trait Implementations§

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !RefUnwindSafe for EngineBuilder<EventTx,
Statistic, Portfolio, Data, Strategy, Execution>
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Send for EngineBuilder<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where EventTx: Send,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Sync for EngineBuilder<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where Data: Sync, EventTx: Sync, Execution: Sync,
Statistic: Sync, Strategy: Sync,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Unpin for EngineBuilder<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where Data: Unpin, EventTx: Unpin, Execution: Unpin,
Statistic: Unpin, Strategy: Unpin,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !UnwindSafe for EngineBuilder<EventTx,
Statistic, Portfolio, Data, Strategy, Execution>
```

Blanket Implementations[§](#)

[source](#)[§](#)

```
impl<T> Any for T
where T: 'static + ?Sized,
```

[source](#)[§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)§

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be `Self`

[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

§

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source](#)§

```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.

[source](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source](#)

```
fn vzip(self) -> V
```

[source](#)

```
impl<T> WithSubscriber for T
```

[source](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[barter0.8.13](#)

[Module engine](#)

- [Modules](#)
- [Structs](#)
- [Enums](#)

[In crate barter](#)

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Module [barter::engine](#)

Copy item path

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[−]

Expand description

Multi-threaded trading Engine capable of trading with an arbitrary number market pairs. Contains a Trader for each Market pair that consists of it's own Data, Strategy & Execution components, as well as shared access to a global Portfolio.

Modules§

- [error](#)
Barter Engine module specific errors.
- [trader](#)
Contains the trading event loop for a Trader capable of trading a single market pair. A Trader has it's own Data handler, Strategy & Execution handler, as well as shared access to a global Portfolio instance.

Structs§

- [Engine](#)
Multi-threaded Trading Engine capable of trading with an arbitrary number of [Traders](#), one for each unique [Market](#).
- [EngineBuilder](#)
Builder to construct [Engine](#) instances.
- [EngineLego](#)
Lego components for constructing an [Engine](#) via the new() constructor method.

Enums§

- [Command](#)
Commands that can be actioned by an [Engine](#) and it's associated [Traders](#).

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[Engine](#)

[Methods](#)

- [builder](#)

- [new](#)
- [run](#)

[Trait Implementations](#)

- [Debug](#)

[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
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[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::engine](#)

[?](#)

[Change settings](#)

Struct [barter::engine::Engine](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct Engine<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
where
```

```
    EventTx: MessageTransmitter<Event>,
    Statistic: PositionSummariser + Serialize + Send,
    Portfolio: PositionHandler + StatisticHandler<Statistic> + MarketUpdate,
    Data: MarketGenerator<MarketEvent<DataKind>> + Send + 'static,
    Strategy: SignalGenerator + Send,
    Execution: ExecutionClient + Send,
{ /* private fields */ }
```

Expand description

Multi-threaded Trading Engine capable of trading with an arbitrary number of [Traders](#), one for each unique [Market](#).

Each [Trader](#) operates on it's own thread and has it's own Data handler, Strategy & Execution Handler, as well as shared access to a global Portfolio instance. A graceful remote shutdown is made possible by sending a [Command::Terminate](#) to the Engine's broadcast::Receiver termination_rx.

Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Engine<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event> + Send + 'static, Statistic: PositionSummariser + TableBuilder + Serialize + Send + 'static, Portfolio: PositionHandler + StatisticHandler<Statistic> + MarketUpdater + OrderGenerator + FillUpdater + Send + 'static, Data: MarketGenerator<MarketEvent<DataKind>> + Send, Strategy: SignalGenerator + Send + 'static, Execution: ExecutionClient + Send + 'static,
```

[source](#)

```
pub fn new( lego: EngineLego<EventTx, Statistic, Portfolio, Data, Strategy, Execution> ) -> Self
```

Constructs a new trading [Engine](#) instance using the provided [EngineLego](#).

[source](#)

```
pub fn builder( ) -> EngineBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
```

Builder to construct [Engine](#) instances.

[source](#)

```
pub async fn run(self)
```

Run the trading [Engine](#). Spawns a thread for each [Trader](#) to run on. Asynchronously receives [Commands](#) via the command_rx and actions them (eg/ terminate_traders, fetch_open_positions). If all of the [Traders](#) stop organically (eg/ due to a finished [MarketGenerator](#)), the [Engine](#) terminates & prints a summary for the trading session.

Trait Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Debug for Engine<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event> + Debug,
Statistic: PositionSummariser + Serialize + Send + Debug,
Portfolio: PositionHandler + StatisticHandler<Statistic> +
MarketUpdater + OrderGenerator + FillUpdater + Send +
'static + Debug, Data:
MarketGenerator<MarketEvent<DataKind>> + Send +
'static + Debug, Strategy: SignalGenerator + Send +
Debug, Execution: ExecutionClient + Send + Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

Auto Trait Implementations§

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !RefUnwindSafe for Engine<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Send for Engine<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where EventTx: Send,
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Sync for Engine<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where Data: Sync, EventTx: Sync, Execution: Sync,
Statistic: Sync, Strategy: Sync,
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Unpin for Engine<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where Data: Unpin, EventTx: Unpin, Execution: Unpin,
Statistic: Unpin, Strategy: Unpin,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !UnwindSafe for Engine<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
```

Blanket Implementations§

[source§](#)

```
impl<T> Any for T
where T: 'static + ?Sized,
```

[source§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source§](#)

```
impl<T> Borrow<T> for T
where T: ?Sized,
```

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T
where T: ?Sized,
```

[source§](#)

```
fn borrow\_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

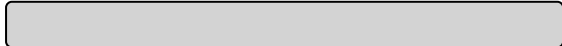
fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



[barter](#)0.8.13

[EngineLego](#)

[Fields](#)

- [command_rx](#)
- [engine_id](#)
- [portfolio](#)
- [statistics_summary](#)
- [trader_command_txs](#)
- [traders](#)

[Trait Implementations](#)

- [Debug](#)

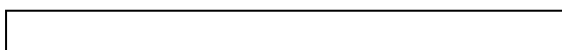
[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::engine](#)



[?](#)
[Change settings](#)

Struct `barter::engine::EngineLego`

Copy item path

[source](#) · [\[-\]](#)

```
pub struct EngineLego<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
where
    EventTx: MessageTransmitter<Event> + Send,
    Statistic: Serialize + Send,
    Portfolio: MarketUpdater + OrderGenerator + FillUpdater + Send,
    Data: MarketGenerator<MarketEvent<DataKind>> + Send,
    Strategy: SignalGenerator + Send,
    Execution: ExecutionClient + Send,
{
    pub engine_id: Uuid,
    pub command_rx: Receiver<Command>,
    pub portfolio: Arc<Mutex<Portfolio>>,
    pub traders: Vec<Trader<EventTx, Statistic, Portfolio, Data, Strategy,
    pub trader_command_txs: HashMap<Market, Sender<Command>>,
    pub statistics_summary: Statistic,
}
```

Expand description

Lego components for constructing an [Engine](#) via the `new()` constructor method.

Fields§

`§engine_id`: [Uuid](#)

Unique identifier for an [Engine](#) in Uuid v4 format. Used as a unique identifier seed for the Portfolio, Trader & Positions associated with this [Engine](#).

`§command_rx`: [Receiver](#)<[Command](#)>

mpsc::Receiver for receiving [Commands](#) from a remote source.

`§portfolio`: [Arc](#)<[Mutex](#)<[Portfolio](#)>>

Shared-access to a global Portfolio instance.

`§traders`: [Vec](#)<[Trader](#)<[EventTx](#), [Statistic](#), [Portfolio](#), [Data](#), [Strategy](#), [Execution](#)>>

Collection of [Trader](#) instances that can concurrently trade a market pair on it's own thread.

`strader_command_txs: HashMap<Market, Sender<Command>>`

HashMap containing a [Command](#) transmitter for every [Trader](#) associated with this [Engine](#).

`statistics_summary: Statistic`

Uses trading session's exited [Positions](#) to calculate an average statistical summary across all [Markets](#) traded.

Trait Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Debug for EngineLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event> + Send +
Debug, Statistic: Serialize + Send + Debug, Portfolio:
MarketUpdater + OrderGenerator + FillUpdater + Send +
Debug, Data: MarketGenerator<MarketEvent<DataKind>>
+ Send + Debug, Strategy: SignalGenerator + Send +
Debug, Execution: ExecutionClient + Send + Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

Auto Trait Implementations§

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !RefUnwindSafe for EngineLego<EventTx,
Statistic, Portfolio, Data, Strategy, Execution>
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Send for EngineLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Sync for EngineLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where Data: Sync, EventTx: Sync, Execution: Sync,
Statistic: Sync, Strategy: Sync,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Unpin for EngineLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where Data: Unpin, EventTx: Unpin, Execution: Unpin,
Statistic: Unpin, Strategy: Unpin,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !UnwindSafe for EngineLego<EventTx,
Statistic, Portfolio, Data, Strategy, Execution>
```

Blanket Implementations

[source](#)[§](#)

```
impl<T> Any for T
where T: 'static + ?Sized,
```

[source](#)[§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

```
impl<T> Borrow<T> for T
where T: ?Sized,
```

[source](#)[§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



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[Command](#)

[Variants](#)

- [ExitAllPositions](#)
- [ExitPosition](#)
- [FetchOpenPositions](#)
- [Terminate](#)

[Trait Implementations](#)

- [Debug](#)

[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)

- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In [barter::engine](#)

[?](#)
[Change settings](#)

Enum [barter::engine::Command](#)

Copy item path

[source](#) ·

```
pub enum Command {
    FetchOpenPositions(Sender<Result<Vec<Position>, EngineError>>),
    Terminate(String),
    ExitAllPositions,
    ExitPosition(Market),
}
```

Expand description

Commands that can be actioned by an [Engine](#) and it's associated [Traders](#).

Variants§

§

FetchOpenPositions([Sender](#)<[Result](#)<[Vec](#)<[Position](#)>, [EngineError](#)>>)

Fetches all the [Engine](#)'s open [Positions](#) and sends them on the provided `oneshot::Sender`. Involves the [Engine](#) only.

§

Terminate([String](#))

Terminate every running [Trader](#) associated with this [Engine](#). Involves all [Traders](#).

§

ExitAllPositions

Exit every open [Position](#) associated with this [Engine](#). Involves all [Traders](#).

[§](#)

ExitPosition([Market](#))

Exit a [Position](#). Uses the [Market](#) provided to route this [Command](#) to the relevant [Trader](#) instance. Involves one [Trader](#).

Trait Implementations§

[source§](#)

impl [Debug](#) for [Command](#)

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

Auto Trait Implementations§

[§](#)

impl ![RefUnwindSafe](#) for [Command](#)

[§](#)

impl [Send](#) for [Command](#)

[§](#)

impl [Sync](#) for [Command](#)

[§](#)

impl [Unpin](#) for [Command](#)

[§](#)

impl ![UnwindSafe](#) for [Command](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in\_current\_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

```
impl<T> Same for T
```

[§](#)

```
type Output = T
```

Should always be Self

[source§](#)

```
impl<T, U> TryFrom<U> for T  
where U: Into<T>,
```

[§](#)

```
type Error = Infallible
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try\_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<T, U> TryInto<U> for T  
where U: TryFrom<T>,
```

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

engine - trader



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[Module trader](#)

- [Structs](#)

[In barter::engine](#)

[?](#)

[Change settings](#)

Module [barter::engine::trader](#)

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Expand description

Contains the trading event loop for a Trader capable of trading a single market pair. A Trader has it's own Data handler, Strategy & Execution handler, as well as shared access to a global Portfolio instance.

Structs§

- [Trader](#)
Trader instance capable of trading a single market pair with it's own Data Handler, Strategy & Execution Handler, as well as shared access to a global Portfolio instance. It has a many-to-1 relationship with an Engine/Portfolio. A graceful remote shutdown is made possible by sending a [Command::Terminate](#) to the Trader's mpsc::Receiver command_rx.
- [TraderBuilder](#)
Builder to construct [Trader](#) instances.
- [TraderLego](#)
Lego components for constructing a [Trader](#) via the new() constructor method.

[barter](#)0.8.13

[TraderLego](#)

[Fields](#)

- [command_rx](#)
- [data](#)
- [engine_id](#)
- [event_tx](#)
- [execution](#)
- [market](#)
- [portfolio](#)
- [strategy](#)

Trait Implementations

- [Debug](#)

Auto Trait Implementations

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::engine::trader`

[?](#)

[Change settings](#)

Struct `barter::engine::trader::TraderLego`

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[\[-\]](#)

```
pub struct TraderLego<EventTx, Statistic, Portfolio, Data, Strategy, Execution> where
```

```
    EventTx: MessageTransmitter<Event>,
    Statistic: Serialize + Send,
    Portfolio: MarketUpdater + OrderGenerator + FillUpdater,
    Data: MarketGenerator<MarketEvent<DataKind>>,
    Strategy: SignalGenerator,
    Execution: ExecutionClient,
{
    pub engine_id: Uuid,
    pub market: Market,
    pub command_rx: Receiver<Command>,
```



```

    pub event_tx: EventTx,
    pub portfolio: Arc<Mutex<Portfolio>>,
    pub data: Data,
    pub strategy: Strategy,
    pub execution: Execution,
    /* private fields */
}

```

Expand description

Lego components for constructing a [Trader](#) via the new() constructor method.

Fields§

[\\$engine_id](#): [Uuid](#)

Identifier for the [Engine](#) this [Trader](#) is associated with (1-to-many relationship).

[\\$market](#): [Market](#)

Communicates the unique [Market](#) this [Trader](#) is bartering on.

[\\$command_rx](#): [Receiver<Command>](#)

mpsc::Receiver for receiving [Commands](#) from a remote source.

[\\$event_tx](#): EventTx

[Event](#) transmitter for sending every [Event](#) the [Trader](#) encounters to an external sink.

[\\$portfolio](#): [Arc<Mutex<Portfolio>>](#)

Shared-access to a global Portfolio instance that implements [MarketUpdater](#), [OrderGenerator](#) & [FillUpdater](#).

[\\$data](#): Data

Data handler that implements [MarketGenerator](#).

[\\$strategy](#): Strategy

Strategy that implements [SignalGenerator](#).

[\\$execution](#): Execution

Execution handler that implements [ExecutionClient](#).

Trait Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Debug for TraderLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event> + Debug,
Statistic: Serialize + Send + Debug, Portfolio:
MarketUpdater + OrderGenerator + FillUpdater + Debug,
Data: MarketGenerator<MarketEvent<DataKind>> +
Debug, Strategy: SignalGenerator + Debug, Execution:
ExecutionClient + Debug,
```

[source](#)§

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

Auto Trait Implementations§

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !RefUnwindSafe for TraderLego<EventTx,
Statistic, Portfolio, Data, Strategy, Execution>
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Send for TraderLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where Data: Send, EventTx: Send, Execution: Send,
Portfolio: Send, Strategy: Send,
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Sync for TraderLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where Data: Sync, EventTx: Sync, Execution: Sync,
Portfolio: Send, Statistic: Sync, Strategy: Sync,
```

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Unpin for TraderLego<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
where Data: Unpin, EventTx: Unpin, Execution: Unpin,
Statistic: Unpin, Strategy: Unpin,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !UnwindSafe for TraderLego<EventTx,
Statistic, Portfolio, Data, Strategy, Execution>
```

Blanket Implementations[§](#)

[source§](#)

```
impl<T> Any for T
where T: 'static + ?Sized,
```

[source§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source§](#)

```
impl<T> Borrow<T> for T
where T: ?Sized,
```

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T
where T: ?Sized,
```

[source§](#)

```
fn borrow\_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

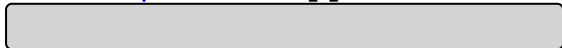
fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



[barter](#)0.8.13

[Trader](#)

[Methods](#)

- [builder](#)
- [new](#)
- [run](#)

[Trait Implementations](#)

- [Debug](#)

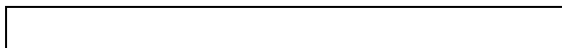
[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
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[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
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- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::engine::trader](#)



[?](#)
[Change settings](#)

Struct [barter::engine::trader::Trader](#)

Copy item path

[source](#) ·

[−]

```
pub struct Trader<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
where
```

```
    EventTx: MessageTransmitter<Event>,
    Statistic: Serialize + Send,
    Portfolio: MarketUpdater + OrderGenerator + FillUpdater,
    Data: MarketGenerator<MarketEvent<DataKind>> + Send,
    Strategy: SignalGenerator + Send,
    Execution: ExecutionClient + Send,
{ /* private fields */ }
```

Expand description

Trader instance capable of trading a single market pair with it's own Data Handler, Strategy & Execution Handler, as well as shared access to a global Portfolio instance. It has a many-to-1 relationship with an Engine/Portfolio. A graceful remote shutdown is made possible by sending a [Command::Terminate](#) to the Trader's mpsc::Receiver command_rx.

Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Trader<EventTx, Statistic, Portfolio, Data,
Strategy, Execution>
where EventTx: MessageTransmitter<Event>, Statistic:
Serialize + Send, Portfolio: MarketUpdater +
OrderGenerator + FillUpdater, Data:
MarketGenerator<MarketEvent<DataKind>> + Send,
Strategy: SignalGenerator + Send, Execution:
ExecutionClient + Send,
```

[source](#)

```
pub fn new( lego: TraderLego<EventTx, Statistic, Portfolio, Data,
Strategy, Execution> ) -> Self
```

Constructs a new [Trader](#) instance using the provided [TraderLego](#).

[source](#)

```
pub fn builder( ) -> TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
```

Builder to construct [Trader](#) instances.

[source](#)

```
pub fn run(self)
```

Run the trading event-loop for this [Trader](#) instance. Loop will run until [Trader](#) receives a [Command::Terminate](#) via the mpsc::Receiver command_rx, or the [MarketGenerator](#) yields [Feed::Finished](#).

Trait Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Debug for Trader<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event> + Debug,
Statistic: Serialize + Send + Debug, Portfolio:
MarketUpdater + OrderGenerator + FillUpdater + Debug,
Data: MarketGenerator<MarketEvent<DataKind>> + Send
+ Debug, Strategy: SignalGenerator + Send + Debug,
Execution: ExecutionClient + Send + Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

Auto Trait Implementations§

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> !RefUnwindSafe for Trader<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
```

§


```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Send for Trader<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where EventTx: Send, Portfolio: Send,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Sync for Trader<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where Data: Sync, EventTx: Sync, Execution: Sync,
Portfolio: Send, Statistic: Sync, Strategy: Sync,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> Unpin for Trader<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where Data: Unpin, EventTx: Unpin, Execution: Unpin,
Statistic: Unpin, Strategy: Unpin,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> !UnwindSafe for Trader<EventTx, Statistic,
Portfolio, Data, Strategy, Execution>
```

Blanket Implementations[§](#)

[source](#)[§](#)

```
impl<T> Any for T
where T: 'static + ?Sized,
```

[source](#)[§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

```
impl<T> Borrow<T> for T
where T: ?Sized,
```

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From<T>](#) for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be Self

[source](#)§

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

§

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source](#)§

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)§

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



[barter0.8.13](#)

[TraderBuilder](#)

[Methods](#)

- [build](#)
- [command_rx](#)
- [data](#)
- [engine_id](#)
- [event_tx](#)
- [execution](#)
- [market](#)
- [portfolio](#)
- [strategy](#)

[Trait Implementations](#)

- [Debug](#)
- [Default](#)

[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)

- [Unpin](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::engine::trader

[?](#)

[Change settings](#)

Struct [barter::engine::trader::TraderBuilder](#)

Copy item path

[source](#) ·

[\[-\]](#)

pub struct TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution> {
 where

```
    EventTx: MessageTransmitter<Event>,  

    Statistic: Serialize + Send,  

    Portfolio: MarketUpdater + OrderGenerator + FillUpdater,  

    Data: MarketGenerator<MarketEvent<DataKind>>,  

    Strategy: SignalGenerator,  

    Execution: ExecutionClient,  

  { /* private fields */ }
```

Expand description

Builder to construct [Trader](#) instances.

Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy,
Execution> TraderBuilder<EventTx, Statistic, Portfolio,
Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event>, Statistic:
Serialize + Send, Portfolio: MarketUpdater +
OrderGenerator + FillUpdater, Data:
MarketGenerator<MarketEvent<DataKind>> + Send,
Strategy: SignalGenerator + Send, Execution:
ExecutionClient + Send,
```

[source](#)

```
pub fn engine_id(self, value: Uuid) -> Self
```

[source](#)

```
pub fn market(self, value: Market) -> Self
```

[source](#)

```
pub fn command_rx(self, value: Receiver<Command>) -> Self
```

[source](#)

```
pub fn event_tx(self, value: EventTx) -> Self
```

[source](#)

```
pub fn portfolio(self, value: Arc<Mutex<Portfolio>>) -> Self
```

[source](#)

```
pub fn data(self, value: Data) -> Self
```

[source](#)

```
pub fn strategy(self, value: Strategy) -> Self
```

[source](#)

```
pub fn execution(self, value: Execution) -> Self
```

[source](#)

```
pub fn build( self ) -> Result<Trader<EventTx, Statistic, Portfolio, Data, Strategy, Execution>, EngineError>
```

Trait Implementations§

[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Debug for TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event> + Debug,
Statistic: Serialize + Send + Debug, Portfolio:
MarketUpdater + OrderGenerator + FillUpdater + Debug,
Data: MarketGenerator<MarketEvent<DataKind>> +
Debug, Strategy: SignalGenerator + Debug, Execution:
ExecutionClient + Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Default for TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
where EventTx: MessageTransmitter<Event> + Default,
Statistic: Serialize + Send + Default, Portfolio:
MarketUpdater + OrderGenerator + FillUpdater + Default,
Data: MarketGenerator<MarketEvent<DataKind>> +
Default, Strategy: SignalGenerator + Default, Execution:
ExecutionClient + Default,
```

[source§](#)

```
fn default( ) -> TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
```

Returns the “default value” for a type. [Read more](#)

Auto Trait Implementations§

§

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> !RefUnwindSafe for TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Send for TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>  
where Data: Send, EventTx: Send, Execution: Send,  
Portfolio: Send, Strategy: Send,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Sync for TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>  
where Data: Sync, EventTx: Sync, Execution: Sync,  
Portfolio: Send, Statistic: Sync, Strategy: Sync,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> Unpin for TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>  
where Data: Unpin, EventTx: Unpin, Execution: Unpin,  
Statistic: Unpin, Strategy: Unpin,
```

[§](#)

```
impl<EventTx, Statistic, Portfolio, Data, Strategy, Execution> !UnwindSafe for TraderBuilder<EventTx, Statistic, Portfolio, Data, Strategy, Execution>
```

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source§](#)

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source§](#)

```
impl<T, U> TryFrom<U> for T
where U: Into<T>,
```

§

```
type Error = Infallible
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<T, U> TryInto<U> for T
where U: TryFrom<T>,
```

§

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source§](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
fn with_current_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

engine - error

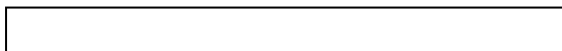


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[Module error](#)

- [Enums](#)

[In barter::engine](#)



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Module [barter::engine::error](#)

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Expand description

Barter Engine module specific errors.

Enums§

- [EngineError](#)
All errors generated in barter-engine.

[barter](#)0.8.13

[EngineError](#)

[Variants](#)

- [BuilderIncomplete](#)
- [RepositoryInteractionError](#)

[Trait Implementations](#)

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- [Error](#)
- [From<RepositoryError>](#)

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- [!RefUnwindSafe](#)
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- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)

- [Same](#)
- [ToString](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In [barter::engine::error](#)

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Enum [barter::engine::error::EngineError](#)

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[−]

```
pub enum EngineError {
    BuilderIncomplete(&'static str),
    RepositoryInteractionError(RepositoryError),
}
```

Expand description

All errors generated in barter-engine.

Variants§

§

BuilderIncomplete(&'static [str](#))

§

RepositoryInteractionError([RepositoryError](#))

Trait Implementations§

[source](#)§

impl [Debug](#) for [EngineError](#)

[source](#)§

fn [fmt](#)(&self, f: &mut [Formatter](#)<'>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl [Display](#) for [EngineError](#)

[source](#)

fn [fmt](#)(&self, __formatter: &mut [Formatter](#)<'>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl [Error](#) for [EngineError](#)

[source](#)

fn [source](#)(&self) -> [Option](#)<&(dyn [Error](#) + 'static)>

The lower-level source of this error, if any. [Read more](#)
1.0.0 · [source](#)

fn [description](#)(&self) -> &str

Deprecated since 1.42.0: use the Display impl or to_string()
[Read more](#)
1.0.0 · [source](#)

fn [cause](#)(&self) -> [Option](#)<&dyn [Error](#)>

Deprecated since 1.33.0: replaced by Error::source, which can support downcasting
[source](#)

fn [provide](#)<'a>(&'a self, request: &mut [Request](#)<'a>)

This is a nightly-only experimental API. (`error_generic_member_access`)
Provides type based access to context intended for error reports. [Read more](#)
[source](#)

impl [From](#)<[RepositoryError](#)> for [EngineError](#)

[source](#)

fn [from](#)(source: [RepositoryError](#)) -> Self

Converts to this type from the input type.

Auto Trait Implementations§

§

impl ![RefUnwindSafe](#) for [EngineError](#)

§

impl [Send](#) for [EngineError](#)

§

impl [Sync](#) for [EngineError](#)

§

impl [Unpin](#) for [EngineError](#)

§

impl ![UnwindSafe](#) for [EngineError](#)

Blanket Implementations§

[source](#)§

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)§

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)§

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)§

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToString](#) for T
where T: [Display](#) + [?Sized](#),**

[source§](#)

default fn [to_string](#)(&self) -> [String](#)

Converts the given value to a String. [Read more](#)

[source§](#)

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

```
impl<V, T> VZip<V> for T
where V: MultiLane<T>,
```

[source](#)

```
fn vzip(self) -> V
```

[source](#)

```
impl<T> WithSubscriber for T
```

[source](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

portfolio



[barter](#)0.8.13

[OrderEventBuilder](#)

[Fields](#)

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[Methods](#)

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Trait Implementations

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Blanket Implementations

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- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio

[?](#)

[Change settings](#)

Struct [barter::portfolio::OrderEventBuilder](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct OrderEventBuilder {
    pub time: Option<DateTime<Utc>>,
    pub exchange: Option<Exchange>,
```

```
pub instrument: Option<Instrument>,
pub market_meta: Option<MarketMeta>,
pub decision: Option<Decision>,
pub quantity: Option<f64>,
pub order_type: Option<OrderType>,
}
```

Expand description

Builder to construct OrderEvent instances.

Fields§

[\\$time](#): [Option](#)<[DateTime](#)<[Utc](#)>>[\\$exchange](#): [Option](#)<[Exchange](#)>[\\$instrument](#):
[Option](#)<[Instrument](#)>[\\$market_meta](#): [Option](#)<[MarketMeta](#)>[\\$decision](#):
[Option](#)<[Decision](#)>[\\$quantity](#): [Option](#)<[f64](#)>[\\$order_type](#):
[Option](#)<[OrderType](#)>

Implementations§

[source§](#)

impl [OrderEventBuilder](#)

[source](#)

pub fn [new](#)() -> Self

[source](#)

pub fn [time](#)(self, value: [DateTime](#)<[Utc](#)>) -> Self

[source](#)

pub fn [exchange](#)(self, value: [Exchange](#)) -> Self

[source](#)

pub fn [instrument](#)(self, value: [Instrument](#)) -> Self

[source](#)

pub fn [market_meta](#)(self, value: [MarketMeta](#)) -> Self

[source](#)

pub fn [decision](#)(self, value: [Decision](#)) -> Self

[source](#)

pub fn [quantity](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [order_type](#)(self, value: [OrderType](#)) -> Self

[source](#)

pub fn [build](#)(self) -> [Result](#)<[OrderEvent](#), [PortfolioError](#)>

Trait Implementations§

[source§](#)

impl [Debug](#) for [OrderEventBuilder](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [OrderEventBuilder](#)

[source§](#)

fn [default](#)() -> [OrderEventBuilder](#)

Returns the “default value” for a type. [Read more](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [OrderEventBuilder](#)

§

impl [Send](#) for [OrderEventBuilder](#)

§

impl [Sync](#) for [OrderEventBuilder](#)

§

impl [Unpin](#) for [OrderEventBuilder](#)

[§](#)

impl [UnwindSafe](#) for [OrderEventBuilder](#)

Blanket Implementations[§](#)

[source](#)[§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)[§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [From](#)<T> for T

[source](#)[§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source§](#)

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

```
fn try_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<T, U> TryInto<U> for T  
where U: TryFrom<T>,
```

[§](#)

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

```
fn with_current_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



barter0.8.13

Module portfolio

- [Modules](#)
- [Structs](#)
- [Enums](#)
- [Traits](#)
- [Type Aliases](#)

In crate barter

[?](#)

[Change settings](#)

Module [barter::portfolio](#)

Copy item path

[source](#) ·

[\[-\]](#)

Expand description

Defines useful data structures such as an `OrderEvent` and `Position`. The Portfolio must interact with `MarketEvents`, `SignalEvents`, `OrderEvents`, and `FillEvents`. The useful traits `MarketUpdater`, `OrderGenerator`, & `FillUpdater` are provided that define the interactions with these events. Contains a `MetaPortfolio` implementation that persists state in a generic `Repository`. This also contains example implementations of an `OrderAllocator` & `OrderEvaluator`, which help the Portfolio make decisions on whether to generate `OrderEvents` and of what size.

Modules§

- [allocator](#)
Logic for [OrderEvent](#) quantity allocation.
- [error](#)
Barter portfolio module specific errors.
- [portfolio](#)
Core Portfolio logic containing an implementation of [MarketUpdater](#), [OrderGenerator](#) and [FillUpdater](#). Utilises the risk and allocator logic to optimise [OrderEvent](#) generation.
- [position](#)
Data structures encapsulating the state of a trading [Position](#), as well as the logic for entering, updating and exiting them.
- [repository](#)
Repositories for persisting Portfolio state.
- [risk](#)

Logic for evaluating the risk associated with a proposed [OrderEvent](#).

Structs§

- [Balance](#)
Total and available balance at a point in time.
- [OrderEvent](#)
Orders are generated by the portfolio and details work to be done by an Execution handler to open a trade.
- [OrderEventBuilder](#)
Builder to construct OrderEvent instances.

Enums§

- [OrderType](#)
Type of order the portfolio wants the execution::handler to place.

Traits§

- [FillUpdater](#)
Updates the Portfolio from an input [FillEvent](#).
- [MarketUpdater](#)
Updates the Portfolio from an input [MarketEvent](#).
- [OrderGenerator](#)
May generate an [OrderEvent](#) from an input advisory [Signal](#).

Type Aliases§

- [BalanceId](#)
Communicates a String represents a unique identifier for an Engine's Portfolio [Balance](#).



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[FillUpdater](#)

[Required Methods](#)

- [update_from_fill](#)

[Implementors](#)

[In barter::portfolio](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::FillUpdater](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub trait FillUpdater {  
    // Required method  
    fn update\_from\_fill(  
        &mut self,  
        fill: &FillEvent  
    ) -> Result<Vec<Event>, PortfolioError>;  
}
```

Expand description

Updates the Portfolio from an input [FillEvent](#).

Required Methods§

[source](#)

```
fn update\_from\_fill( &mut self, fill: &FillEvent ) ->  
Result<Vec<Event>, PortfolioError>
```

Updates the Portfolio state using the input [FillEvent](#). The [FillEvent](#) triggers a Position entry or exit, and the Portfolio updates key fields such as `current_cash` and `current_value` accordingly.

Implementors§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
FillUpdater for MetaPortfolio<Repository, Allocator,  
RiskManager, Statistic>  
where Repository: PositionHandler + BalanceHandler +  
StatisticHandler<Statistic>, Allocator: OrderAllocator,  
RiskManager: OrderEvaluator, Statistic: Initialiser +  
PositionSummariser + Serialize,
```

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OrderGenerator

Required Methods

- [generate_exit_order](#)
- [generate_order](#)

Implementors

In barter::portfolio

[?](#)

[Change settings](#)

Trait [barter::portfolio::OrderGenerator](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub trait OrderGenerator {  
    // Required methods  
    fn generate\_order(  
        &mut self,  
        signal: &Signal  
    ) -> Result<Option<OrderEvent>, PortfolioError>;  
    fn generate\_exit\_order(  
        &mut self,  
        signal: SignalForceExit  
    ) -> Result<Option<OrderEvent>, PortfolioError>;  
}
```

Expand description

May generate an [OrderEvent](#) from an input advisory [Signal](#).

Required Methods§

[source](#)

```
fn generate\_order( &mut self, signal: &Signal ) ->  
Result<Option<OrderEvent>, PortfolioError>
```

May generate an [OrderEvent](#) after analysing an input advisory [Signal](#).

[source](#)

```
fn generate_exit_order( &mut self, signal: SignalForceExit ) ->  
Result<Option<OrderEvent>, PortfolioError>
```

Generates an exit [OrderEvent](#) if there is an open [Position](#) associated with the input [SignalForceExit](#)'s [PositionId](#).

Implementors§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
OrderGenerator for MetaPortfolio<Repository, Allocator,  
RiskManager, Statistic>  
where Repository: PositionHandler + BalanceHandler +  
StatisticHandler<Statistic>, Allocator: OrderAllocator,  
RiskManager: OrderEvaluator, Statistic: Initialiser +  
PositionSummariser,
```

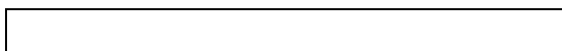


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[BalanceId](#)

[Aliased type](#)

[In barter::portfolio](#)



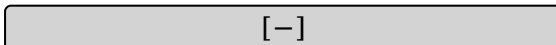
[?](#)

[Change settings](#)

Type Alias [barter::portfolio::BalanceId](#)



[source](#) ·



```
pub type BalanceId = String;
```

Expand description

Communicates a String represents a unique identifier for an Engine's Portfolio [Balance](#).

Aliased Type§

```
struct BalanceId { /* private fields */ }
```

[barter0.8.13](#)

[MarketUpdater](#)

[Required Methods](#)

- [update_from_market](#)

[Implementors](#)

[In barter::portfolio](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::MarketUpdater](#)

Copy item path

[source](#) ·

[-]

```
pub trait MarketUpdater {  
    // Required method  
    fn update\_from\_market(  
        &mut self,  
        market: &MarketEvent<DataKind>  
    ) -> Result<Option<PositionUpdate>, PortfolioError>;  
}
```

Expand description

Updates the Portfolio from an input [MarketEvent](#).

Required Methods§

[source](#)

```
fn update\_from\_market( &mut self, market:
&MarketEvent<DataKind> ) -> Result<Option<PositionUpdate>,
PortfolioError>
```

Determines if the Portfolio has an open Position relating to the input [MarketEvent](#). If so it updates it using the market data, and returns a [PositionUpdate](#) detailing the changes.

Implementors§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>
MarketUpdater for MetaPortfolio<Repository, Allocator,
RiskManager, Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic>, Allocator: OrderAllocator,
RiskManager: OrderEvaluator, Statistic: Initialiser +
PositionSummariser,
```



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[Balance](#)

[Fields](#)

- [available](#)
- [time](#)
- [total](#)

[Methods](#)

- [balance_id](#)
- [new](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize](#)<'de>
- [From](#)<[Balance](#)>
- [PartialEq](#)
- [PartialOrd](#)

- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio

[?](#)

[Change settings](#)

Struct [barter::portfolio::Balance](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct Balance {
    pub time: DateTime<Utc>,
    pub total: f64,
    pub available: f64,
}
```

Expand description

Total and available balance at a point in time.

Fields§

`$time: DateTime<Utc>$total: f64$available: f64`

Implementations§

[source§](#)

impl [Balance](#)

[source](#)

pub fn [new](#)(time: [DateTime<Utc>](#), total: [f64](#), available: [f64](#)) -> Self

Construct a new [Balance](#) using the provided total & available balance values.

[source](#)

pub fn [balance_id](#)(engine_id: [Uuid](#)) -> [BalanceId](#)

Returns the unique identifier for an Engine's [Balance](#).

Trait Implementations§

[source§](#)

impl [Clone](#) for [Balance](#)

[source§](#)

fn [clone](#)(&self) -> [Balance](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [Balance](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter<'_>](#)) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl [Default](#) for [Balance](#)

[source§](#)

fn [default](#)() -> Self

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [Balance](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [From](#)<[Balance](#)> for [EquityPoint](#)

[source§](#)

fn [from](#)(balance: [Balance](#)) -> Self

Converts to this type from the input type.
[source§](#)

impl [PartialEq](#) for [Balance](#)

[source§](#)

fn [eq](#)(&self, other: &[Balance](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [Balance](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[Balance](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source](#)

fn [ge](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source](#)

impl [Serialize](#) for [Balance](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>

where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)

[source](#)

impl [Copy](#) for [Balance](#)

[source](#)

impl [StructuralPartialEq](#) for [Balance](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Balance](#)

§

impl [Send](#) for [Balance](#)

§

impl [Sync](#) for [Balance](#)

§

impl [Unpin](#) for [Balance](#)

§

impl [UnwindSafe](#) for [Balance](#)

Blanket Implementations§

[source](#)§

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)§

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)§

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)§

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)

[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)

[source§](#)

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

**impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,**



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[OrderType](#)

[Variants](#)

- [Bracket](#)

- [Limit](#)
- [Market](#)

Trait Implementations

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize<'de>](#)
- [Eq](#)
- [Hash](#)
- [Ord](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [Comparable<K>](#)
- [DeserializeOwned](#)
- [Equivalent<K>](#)
- [Equivalent<K>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio

?

[Change settings](#)

Enum [barter::portfolio::OrderType](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub enum OrderType {  
    Market,  
    Limit,  
    Bracket,  
}
```

Expand description

Type of order the portfolio wants the `execution::handler` to place.

Variants§

§

Market

§

Limit

§

Bracket

Trait Implementations§

[source](#)§

impl [Clone](#) for [OrderType](#)

[source](#)§

fn [clone](#)(&self) -> [OrderType](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)§

impl [Debug](#) for [OrderType](#)

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'>) -> Result
```

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl [Default](#) for [OrderType](#)

[source§](#)

```
fn default() -> Self
```

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [OrderType](#)

[source§](#)

```
fn deserialize<__D>(__deserializer: __D) -> Result<Self, __D::Error>  
where __D: Deserializer<'de>,
```

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [Hash](#) for [OrderType](#)

[source§](#)

```
fn hash<__H: Hasher>(&self, state: &mut __H)
```

Feeds this value into the given [Hasher](#). [Read more](#)

1.3.0 · [source§](#)

```
fn hash\_slice<H>(data: &[Self], state: &mut H)  
where H: Hasher, Self: Sized,
```

Feeds a slice of this type into the given [Hasher](#). [Read more](#)

[source§](#)

impl [Ord](#) for [OrderType](#)

[source§](#)

fn [cmp](#)(&self, other: &[OrderType](#)) -> [Ordering](#)

This method returns an [Ordering](#) between self and other. [Read more](#)
1.21.0 · [source](#)

fn [max](#)(self, other: Self) -> Self
where Self: [Sized](#),

Compares and returns the maximum of two values. [Read more](#)
1.21.0 · [source](#)

fn [min](#)(self, other: Self) -> Self
where Self: [Sized](#),

Compares and returns the minimum of two values. [Read more](#)
1.50.0 · [source](#)

fn [clamp](#)(self, min: Self, max: Self) -> Self
where Self: [Sized](#) + [PartialOrd](#),

Restrict a value to a certain interval. [Read more](#)
[source](#)

impl [PartialEq](#) for [OrderType](#)

[source](#)

fn [eq](#)(&self, other: &[OrderType](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)

impl [PartialOrd](#) for [OrderType](#)

[source](#)

fn [partial_cmp](#)(&self, other: &[OrderType](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [OrderType](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Copy](#) for [OrderType](#)

[source§](#)

impl [Eq](#) for [OrderType](#)

[source§](#)

impl [StructuralPartialEq](#) for [OrderType](#)

Auto Trait Implementations§

[§](#)

impl [RefUnwindSafe](#) for [OrderType](#)

[§](#)

impl [Send](#) for [OrderType](#)

[§](#)

impl [Sync](#) for [OrderType](#)

[§](#)

impl [Unpin](#) for [OrderType](#)

[§](#)

impl [UnwindSafe](#) for [OrderType](#)

Blanket Implementations[§](#)

[source](#)[§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)[§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<Q, K> [Comparable](#)<K> for Q
where Q: [Ord](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [compare](#)(&self, key: [&K](#)) -> [Ordering](#)

Compare self to key and return their ordering.
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Checks if this value is equivalent to the given key. [Read more](#)
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Compare self to key and return true if they are equal.
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> **for** T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> **for** T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> **for** T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) **for** T

[source§](#)


```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
impl<T> DeserializeOwned for T  
where T: for<'de> Deserialize<'de>,
```



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[OrderEvent](#)

[Fields](#)

- [decision](#)
- [exchange](#)
- [instrument](#)
- [market_meta](#)
- [order_type](#)
- [quantity](#)
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- [Sync](#)
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[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::portfolio](#)

[?](#)

[Change settings](#)

Struct [barter::portfolio::OrderEvent](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct OrderEvent {
    pub time: DateTime<Utc>,
    pub exchange: Exchange,
    pub instrument: Instrument,
    pub market_meta: MarketMeta,
    pub decision: Decision,
    pub quantity: f64,
    pub order_type: OrderType,
}
```

Expand description

Orders are generated by the portfolio and details work to be done by an Execution handler to open a trade.

Fields§

`$time`: [DateTime<Utc>](#)`$exchange`: [Exchange](#)`$instrument`: [Instrument](#)`$market_meta`: [MarketMeta](#)

Metadata propagated from source `MarketEvent`

`$decision`: [Decision](#)

LONG, CloseLong, SHORT or CloseShort

`$quantity`: [f64](#)

+ve or -ve Quantity depending on Decision

`$order_type`: [OrderType](#)

MARKET, LIMIT etc

Implementations§

[source](#)§

impl [OrderEvent](#)

[source](#)

```
pub const ORGANIC\_ORDER: &'static str = "Order"
```

[source](#)

```
pub const FORCED\_EXIT\_ORDER: &'static str = "OrderForcedExit"
```

[source](#)

```
pub fn builder() -> OrderEventBuilder
```

Returns a `OrderEventBuilder` instance.

Trait Implementations§

[source](#)§

impl [Clone](#) for [OrderEvent](#)

[source](#)§

fn [clone](#)(&self) -> [OrderEvent](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)

impl [Debug](#) for [OrderEvent](#)

[source](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl<'de> [Deserialize](#)<'de> for [OrderEvent](#)

[source](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl [PartialEq](#) for [OrderEvent](#)

[source](#)

fn [eq](#)(&self, other: &[OrderEvent](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)

impl [PartialOrd](#) for [OrderEvent](#)

[source](#)

fn [partial_cmp](#)(&self, other: &[OrderEvent](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [OrderEvent](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [StructuralPartialEq](#) for [OrderEvent](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [OrderEvent](#)

[§](#)

impl [Send](#) for [OrderEvent](#)

[§](#)

impl [Sync](#) for [OrderEvent](#)

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impl [Unpin](#) for [OrderEvent](#)

[§](#)

impl [UnwindSafe](#) for [OrderEvent](#)

Blanket Implementations[§](#)

[source](#)[§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)[§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)


```
fn try_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

```
fn with_current_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

```
impl<T> DeserializeOwned for T  
where T: for<'de> Deserialize<'de>,
```

portfolio - risk



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[Module risk](#)

- [Structs](#)
- [Traits](#)

[In barter::portfolio](#)

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Module [barter::portfolio::risk](#)

Copy item path

[source](#) ·

[−]

Expand description

Logic for evaluating the risk associated with a proposed [OrderEvent](#).

Structs§

- [DefaultRisk](#)
Default risk manager that implements [OrderEvaluator](#).

Traits§

- [OrderEvaluator](#)
Evaluates the risk associated with an [OrderEvent](#) to determine if it should be actioned. It can also amend the order (eg/ [OrderType](#)) to better fit the risk strategy required for profitability.

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[DefaultRisk](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [Eq](#)
- [Hash](#)
- [Ord](#)
- [OrderEvaluator](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

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- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [Comparable<K>](#)
- [DeserializeOwned](#)
- [Equivalent<K>](#)
- [Equivalent<K>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio::risk

[?](#)
[Change settings](#)

Struct [barter::portfolio::risk::DefaultRisk](#)

Copy item path

[source](#) ·

[-]

```
pub struct DefaultRisk {}
```

Expand description

Default risk manager that implements [OrderEvaluator](#).

Trait Implementations§

[source§](#)

impl [Clone](#) for [DefaultRisk](#)

[source§](#)

fn [clone](#)(&self) -> [DefaultRisk](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [Self](#))

Performs copy-assignment from source. [Read more](#)

[source§](#)

impl [Debug](#) for [DefaultRisk](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [DefaultRisk](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [Hash](#) for [DefaultRisk](#)

[source§](#)

fn [hash](#)<__H: [Hasher](#)>(&self, state: &mut __H)

Feeds this value into the given [Hasher](#). [Read more](#)

1.3.0 · [source§](#)

fn [hash_slice](#)<H>(data: &[Self](#), state: &mut H)
where H: [Hasher](#), Self: [Sized](#),

Feeds a slice of this type into the given [Hasher](#). [Read more](#)

[source§](#)

impl [Ord](#) for [DefaultRisk](#)

[source§](#)

fn [cmp](#)(&self, other: &[DefaultRisk](#)) -> [Ordering](#)

This method returns an [Ordering](#) between self and other. [Read more](#)
1.21.0 · [source§](#)

fn [max](#)(self, other: Self) -> Self
where Self: [Sized](#),

Compares and returns the maximum of two values. [Read more](#)
1.21.0 · [source§](#)

fn [min](#)(self, other: Self) -> Self
where Self: [Sized](#),

Compares and returns the minimum of two values. [Read more](#)
1.50.0 · [source§](#)

fn [clamp](#)(self, min: Self, max: Self) -> Self
where Self: [Sized](#) + [PartialOrd](#),

Restrict a value to a certain interval. [Read more](#)
[source§](#)

impl [OrderEvaluator](#) for [DefaultRisk](#)

[source§](#)

const [DEFAULT_ORDER_TYPE](#): [OrderType](#) = [OrderType::Market](#)

[source§](#)

fn [evaluate_order](#)(&self, order: [OrderEvent](#)) -> [Option](#)<[OrderEvent](#)>

May return an amended [OrderEvent](#) if the associated risk is appropriate.
Returns None if the risk is too high.
[source§](#)

impl [PartialEq](#) for [DefaultRisk](#)

[source§](#)

fn [eq](#)(&self, other: &[DefaultRisk](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source](#)

impl [PartialOrd](#) for [DefaultRisk](#)

[source](#)

fn [partial_cmp](#)(&self, other: [&DefaultRisk](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source](#)

impl [Serialize](#) for [DefaultRisk](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)

[source§](#)

impl [Copy](#) for [DefaultRisk](#)

[source§](#)

impl [Eq](#) for [DefaultRisk](#)

[source§](#)

impl [StructuralPartialEq](#) for [DefaultRisk](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [DefaultRisk](#)

§

impl [Send](#) for [DefaultRisk](#)

§

impl [Sync](#) for [DefaultRisk](#)

§

impl [Unpin](#) for [DefaultRisk](#)

§

impl [UnwindSafe](#) for [DefaultRisk](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<Q, K> [Comparable](#)<K> for Q
where Q: [Ord](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [compare](#)(&self, key: [&K](#)) -> [Ordering](#)

Compare self to key and return their ordering.
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Checks if this value is equivalent to the given key. [Read more](#)
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Compare self to key and return true if they are equal.

[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



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[OrderEvaluator](#)

[Required Associated Constants](#)

- [DEFAULT_ORDER_TYPE](#)

[Required Methods](#)

- [evaluate_order](#)

Object Safety

Implementors

In `barter::portfolio::risk`

?

[Change settings](#)

Trait `barter::portfolio::risk::OrderEvaluator`

Copy item path

[source](#) ·

[−]

```
pub trait OrderEvaluator {  
    const DEFAULT\_ORDER\_TYPE: OrderType;  
  
    // Required method  
    fn evaluate\_order(&self, order: OrderEvent) -> Option<OrderEvent>;  
}
```

Expand description

Evaluates the risk associated with an [OrderEvent](#) to determine if it should be actioned. It can also amend the order (eg/ [OrderType](#)) to better fit the risk strategy required for profitability.

Required Associated Constants§

[source](#)

const [DEFAULT_ORDER_TYPE](#): [OrderType](#)

Required Methods§

[source](#)

fn [evaluate_order](#)(&self, order: [OrderEvent](#)) -> [Option](#)<[OrderEvent](#)>

May return an amended [OrderEvent](#) if the associated risk is appropriate. Returns None if the risk is too high.

Object Safety§

This trait is **not** [object safe](#).

Implementors§

[source§](#)

impl [OrderEvaluator](#) for [DefaultRisk](#)

[source§](#)

const [DEFAULT_ORDER_TYPE](#): [OrderType](#) = [OrderType::Market](#)

portfolio - allocator

[barter](#)0.8.13

[OrderAllocator](#)

[Required Methods](#)

- [allocate_order](#)

[Implementors](#)

[In barter::portfolio::allocator](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::allocator::](#)

[OrderAllocator](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub trait OrderAllocator {
    // Required method
    fn allocate\_order(
        &self,
        order: &mut OrderEvent,
        position: Option<&Position>,
        signal_strength: SignalStrength
    );
}
```

Expand description

Allocates an appropriate [OrderEvent](#) quantity.

Required Methods§

[source](#)

```
fn allocate_order( &self, order: &mut OrderEvent, position:
Option<&Position>, signal_strength: SignalStrength )
```

Returns an [OrderEvent](#) with a calculated order quantity based on the input order, [SignalStrength](#) and potential existing [Position](#).

Implementors§

[source§](#)

```
impl OrderAllocator for DefaultAllocator
```

[barter](#)**0.8.13**

[Module allocator](#)

- [Structs](#)
- [Traits](#)

[In barter::portfolio](#)

[?](#)

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Module [barter::portfolio::allocator](#)

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[-]

Expand description

Logic for [OrderEvent](#) quantity allocation.

Structs§

- [DefaultAllocator](#)
Default allocation manager that implements [OrderAllocator](#). Order size is calculated by using the default_order_value, symbol close value, and [SignalStrength](#).

Traits§

- [OrderAllocator](#)
Allocates an appropriate [OrderEvent](#) quantity.



[barter](#)0.8.13

[DefaultAllocator](#)

[Fields](#)

- [default_order_value](#)

[Trait Implementations](#)

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- [Debug](#)
- [Default](#)
- [Deserialize<'de>](#)
- [OrderAllocator](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

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- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)

- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::portfolio::allocator](#)

[?](#)

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Struct [barter::portfolio::allocator::DefaultAllocator](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct DefaultAllocator {
    pub default_order_value: f64,
}
```

Expand description

Default allocation manager that implements [OrderAllocator](#). Order size is calculated by using the default_order_value, symbol close value, and [SignalStrength](#).

Fields§

§default_order_value: [f64](#)

Trait Implementations§

[source§](#)

impl [Clone](#) for [DefaultAllocator](#)

[source§](#)

fn [clone](#)(&self) -> [DefaultAllocator](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)

impl [Debug](#) for [DefaultAllocator](#)

[source](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl [Default](#) for [DefaultAllocator](#)

[source](#)

fn [default](#)() -> [DefaultAllocator](#)

Returns the “default value” for a type. [Read more](#)
[source](#)

impl<'de> [Deserialize](#)<'de> for [DefaultAllocator](#)

[source](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl [OrderAllocator](#) for [DefaultAllocator](#)

[source](#)

fn [allocate_order](#)(&self, order: &mut [OrderEvent](#), position: [Option](#)<&[Position](#)>, signal_strength: [SignalStrength](#))

Returns an [OrderEvent](#) with a calculated order quantity based on the input order, [SignalStrength](#) and potential existing [Position](#).
[source](#)

impl [PartialEq](#) for [DefaultAllocator](#)

[source§](#)

fn [eq](#)(&self, other: &[DefaultAllocator](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [DefaultAllocator](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[DefaultAllocator](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [DefaultAllocator](#)

[source](#)§

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)§

impl [Copy](#) for [DefaultAllocator](#)

[source](#)§

impl [StructuralPartialEq](#) for [DefaultAllocator](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [DefaultAllocator](#)

§

impl [Send](#) for [DefaultAllocator](#)

§

impl [Sync](#) for [DefaultAllocator](#)

§

impl [Unpin](#) for [DefaultAllocator](#)

§

impl [UnwindSafe](#) for [DefaultAllocator](#)

Blanket Implementations§

[source](#)§

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source§](#)

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source§](#)

```
impl<T> ToOwned for T  
where T: Clone,
```

§

```
type Owned = T
```

The resulting type after obtaining ownership.

[source§](#)

```
fn to_owned(&self) -> T
```

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

```
fn clone_into(&self, target: &mut T)
```

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

```
impl<T, U> TryFrom<U> for T  
where U: Into<T>,
```

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```

portfolio - position



[barter](#)0.8.13

[PositionExiter](#)

[Required Methods](#)

- [exit](#)

[Implementors](#)

[In barter::portfolio::position](#)

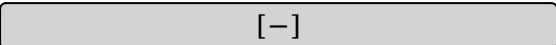


[?](#)

[Change settings](#)

Trait [barter::portfolio::position::PositionExiter](#)



[source](#) · 

```
pub trait PositionExiter {
    // Required method
    fn exit(
        &mut self,
        balance: Balance,
        fill: &FillEvent
    ) -> Result<PositionExit, PortfolioError>;
}
```

Expand description

Exits an open [Position](#).

Required Methods§

[source](#)

```
fn exit( &mut self, balance: Balance, fill: &FillEvent ) ->  
Result<PositionExit, PortfolioError>
```

Exits an open [Position](#), given the input Portfolio equity & the [FillEvent](#) returned from an Execution handler.

Implementors§

[source§](#)

impl [PositionExiter](#) for [Position](#)

[barter](#)0.8.13

[PositionUpdater](#)

[Required Methods](#)

- [update](#)

[Implementors](#)

[In barter::portfolio::position](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::position::PositionUpdater](#)

Copy item path

[source](#) ·

[−]

```
pub trait PositionUpdater {  
    // Required method  
    fn update(  
        &mut self,  
        market: &MarketEvent<DataKind>
```



```
    ) -> Option<PositionUpdate>;  
}
```

Expand description

Updates an open [Position](#).

Required Methods§

[source](#)

```
fn update(&mut self, market: &MarketEvent<DataKind>) ->  
Option<PositionUpdate>
```

Updates an open [Position](#) using the latest input [MarketEvent](#), returning a [PositionUpdate](#) that communicates the open [Position](#)'s change in state.

Implementors§

[source§](#)

```
impl PositionUpdater for Position
```



[barter](#)0.8.13

[PositionMeta](#)

[Fields](#)

- [enter_time](#)
- [exit_balance](#)
- [update_time](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize](#)<'de>
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
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- [UnwindSafe](#)

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- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::portfolio::position`

[?](#)

[Change settings](#)

Struct `barter::portfolio::position::`

`PositionMeta`

Copy item path

[source](#) ·

[-]

```
pub struct PositionMeta {
    pub enter_time: DateTime<Utc>,
    pub update_time: DateTime<Utc>,
    pub exit_balance: Option<Balance>,
}
```

Expand description

Metadata detailing the trace UUIDs & timestamps associated with entering, updating & exiting a [Position](#).

Fields§

`$enter_time: DateTime<Utc>`

[FillEvent](#) timestamp that triggered the entering of this [Position](#).

`$update_time: DateTime<Utc>`

Timestamp of the last event to trigger a [Position](#) state change (enter, update, exit).

`$exit_balance: Option<Balance>`

Portfolio [Balance](#) calculated at the point of exiting a [Position](#).

Trait Implementations§

[source§](#)

impl [Clone](#) for [PositionMeta](#)

[source§](#)

fn [clone](#)(&self) -> [PositionMeta](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)

[source§](#)

impl [Debug](#) for [PositionMeta](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter<'_,>](#)) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl [Default](#) for [PositionMeta](#)

[source§](#)

fn [default](#)() -> Self

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [PositionMeta](#)

[source§](#)

```
fn deserialize<__D>(__deserializer: __D) -> Result<Self, __D::Error>  
where __D: Deserializer<'de>,
```

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [PositionMeta](#)

[source§](#)

```
fn eq(&self, other: &PositionMeta) -> bool
```

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

```
fn ne(&self, other: &Rhs) -> bool
```

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [PositionMeta](#)

[source§](#)

```
fn partial\_cmp(&self, other: &PositionMeta) -> Option<Ordering>
```

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

```
fn lt(&self, other: &Rhs) -> bool
```

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

```
fn le(&self, other: &Rhs) -> bool
```

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Serialize](#) for [PositionMeta](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::[Ok](#), __S::[Error](#)>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

impl [Copy](#) for [PositionMeta](#)

[source](#)

impl [StructuralPartialEq](#) for [PositionMeta](#)

Auto Trait Implementations

[§](#)

impl [RefUnwindSafe](#) for [PositionMeta](#)

[§](#)

impl [Send](#) for [PositionMeta](#)

[§](#)

impl [Sync](#) for [PositionMeta](#)

[§](#)

impl [Unpin](#) for [PositionMeta](#)

[§](#)

impl [UnwindSafe](#) for [PositionMeta](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

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[In barter::portfolio::position](#)

[?](#)

[Change settings](#)

Function [barter::portfolio::position:: determine_position_id](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub fn determine_position_id(
    engine_id: Uuid,
    exchange: &Exchange,
    instrument: &Instrument
) -> PositionId
```

Expand description

Returns a unique identifier for a [Position](#) given an engine_id, [Exchange](#) & [Instrument](#).

barter0.8.13

Module position

- [Structs](#)
- [Traits](#)
- [Functions](#)
- [Type Aliases](#)

In barter::portfolio

[?](#)

[Change settings](#)

Module barter::portfolio::position

Copy item path

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[-]

Expand description

Data structures encapsulating the state of a trading [Position](#), as well as the logic for entering, updating and exiting them.

Structs§

- [Position](#)
Data encapsulating the state of an ongoing or closed [Position](#).
- [PositionBuilder](#)
Builder to construct [Position](#) instances.
- [PositionExit](#)
[Position](#) exit event. Occurs as a result of a [FillEvent](#) that exits a [Position](#).
- [PositionMeta](#)
Metadata detailing the trace UUIDs & timestamps associated with entering, updating & exiting a [Position](#).
- [PositionUpdate](#)
[Position](#) update event. Occurs as a result of receiving new [MarketEvent](#) data.

Traits§

- [PositionEnterer](#)
Enters a new [Position](#).

- [PositionExiter](#)
Exits an open [Position](#).
- [PositionUpdater](#)
Updates an open [Position](#).

Functions§

- [determine_position_id](#)
Returns a unique identifier for a [Position](#) given an engine_id, [Exchange](#) & [Instrument](#).

Type Aliases§

- [PositionId](#)
Communicates a String represents a unique [Position](#) identifier.



[barter](#)0.8.13

[Position](#)

[Fields](#)

- [current_symbol_price](#)
- [current_value_gross](#)
- [enter_avg_price_gross](#)
- [enter_fees](#)
- [enter_fees_total](#)
- [enter_value_gross](#)
- [exchange](#)
- [exit_avg_price_gross](#)
- [exit_fees](#)
- [exit_fees_total](#)
- [exit_value_gross](#)
- [instrument](#)
- [meta](#)
- [position_id](#)
- [quantity](#)
- [realised_profit_loss](#)
- [side](#)
- [unrealised_profit_loss](#)

[Methods](#)

- [builder](#)
- [calculate_avg_price_gross](#)
- [calculate_profit_loss_return](#)

- [calculate_realised_profit_loss](#)
- [calculate_unrealised_profit_loss](#)
- [determine_exit_decision](#)
- [parse_entry_side](#)

Trait Implementations

- [Clone](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [From<&mut Position>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [PositionEnterer](#)
- [PositionExiter](#)
- [PositionUpdater](#)
- [Serialize](#)
- [StructuralPartialEq](#)
- [TryFrom<&mut Position>](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio::position

[?](#)

[Change settings](#)

Struct [barter::portfolio::position::Position](#)

Copy item path

[source](#) ·

[\[-\]](#)

pub struct Position {
Show 18 fields

```
    pub position_id: PositionId,  
    pub meta: PositionMeta,  
    pub exchange: Exchange,  
    pub instrument: Instrument,  
    pub side: Side,  
    pub quantity: f64,  
    pub enter_fees: Fees,  
    pub enter_fees_total: FeeAmount,  
    pub enter_avg_price_gross: f64,  
    pub enter_value_gross: f64,  
    pub exit_fees: Fees,  
    pub exit_fees_total: FeeAmount,  
    pub exit_avg_price_gross: f64,  
    pub exit_value_gross: f64,  
    pub current_symbol_price: f64,  
    pub current_value_gross: f64,  
    pub unrealised_profit_loss: f64,  
    pub realised_profit_loss: f64,  
}
```

Expand description

Data encapsulating the state of an ongoing or closed [Position](#).

Fields

[\\$position_id](#): [PositionId](#)

Unique identifier for a [Position](#) generated from an engine_id, [Exchange](#) & [Instrument](#).

[\\$meta](#): [PositionMeta](#)

Metadata detailing trace UUIDs, timestamps & equity associated with entering, updating & exiting.

[\\$exchange](#): [Exchange](#)

[Exchange](#) associated with this [Position](#).

[\\$instrument](#): [Instrument](#)

[Instrument](#) associated with this [Position](#).

$\$$ side: [Side](#)

Buy or Sell.

Notes:

- Side::Buy considered synonymous with Long.
- Side::Sell considered synonymous with Short.

$\$$ quantity: [f64](#)

+ve or -ve quantity of symbol contracts opened.

$\$$ enter_fees: [Fees](#)

All fees types incurred from entering a [Position](#), and their associated [FeeAmount](#).

$\$$ enter_fees_total: [FeeAmount](#)

Total of enter_fees incurred. Sum of every [FeeAmount](#) in [Fees](#) when entering a [Position](#).

$\$$ enter_avg_price_gross: [f64](#)

Enter average price excluding the entry_fees_total.

$\$$ enter_value_gross: [f64](#)

$\text{abs(Quantity)} * \text{enter_avg_price_gross}$.

$\$$ exit_fees: [Fees](#)

All fees types incurred from exiting a [Position](#), and their associated [FeeAmount](#).

$\$$ exit_fees_total: [FeeAmount](#)

Total of exit_fees incurred. Sum of every [FeeAmount](#) in [Fees](#) when entering a [Position](#).

$\$$ exit_avg_price_gross: [f64](#)

Exit average price excluding the exit_fees_total.

$\$$ exit_value_gross: [f64](#)

$\text{abs(Quantity)} * \text{exit_avg_price_gross}$.

$\$$ current_symbol_price: [f64](#)

Symbol current close price.

$\$$ current_value_gross: [f64](#)

`abs(Quantity) * current_symbol_price.`

`$unrealised_profit_loss: f64`

Unrealised P&L whilst the [Position](#) is open.

`$realised_profit_loss: f64`

Realised P&L after the [Position](#) has closed.

Implementations

[source](#)

impl [Position](#)

[source](#)

pub fn [builder](#)() -> [PositionBuilder](#)

Returns a [PositionBuilder](#) instance.

[source](#)

pub fn [calculate_avg_price_gross](#)(fill: &[FillEvent](#)) -> f64

Calculates the [Position::enter_avg_price_gross](#) or [Position::exit_avg_price_gross](#) of a [FillEvent](#).

[source](#)

pub fn [parse_entry_side](#)(fill: &[FillEvent](#)) -> [Result](#)<[Side](#), [PortfolioError](#)>

Determine the [Position](#) entry [Side](#) by analysing the input [FillEvent](#).

[source](#)

pub fn [determine_exit_decision](#)(&self) -> [Decision](#)

Determines the [Decision](#) required to exit this [Side](#) (Buy or Sell) [Position](#).

[source](#)

pub fn [calculate_unrealised_profit_loss](#)(&self) -> f64

Calculate the approximate [Position::unrealised_profit_loss](#) of a [Position](#).

[source](#)

pub fn [calculate_realised_profit_loss](#)(&self) -> [f64](#)

Calculate the exact [Position::realised_profit_loss](#) of a [Position](#).

[source](#)

pub fn [calculate_profit_loss_return](#)(&self) -> [f64](#)

Calculate the PnL return of a closed [Position](#) - assumed [Position::realised_profit_loss](#) is appropriately calculated.

Trait Implementations§

[source](#)§

impl [Clone](#) for [Position](#)

[source](#)§

fn [clone](#)(&self) -> [Position](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)§

impl [Debug](#) for [Position](#)

[source](#)§

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)§

impl<'de> [Deserialize](#)<'de> for [Position](#)

[source](#)§

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)§

impl [From](#)<&mut [Position](#)> for [PositionUpdate](#)

[source§](#)

fn [from](#)(updated_position: &mut [Position](#)) -> Self

Converts to this type from the input type.

[source§](#)

impl [PartialEq](#) for [Position](#)

[source§](#)

fn [eq](#)(&self, other: &[Position](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.

1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [Position](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[Position](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [PositionEnterer](#) for [Position](#)

[source](#)

fn [enter](#)(engine_id: [Uuid](#), fill: [&FillEvent](#)) -> [Result](#)<[Position](#), [PortfolioError](#)>

Returns a new [Position](#), given an input [FillEvent](#) & an associated engine_id.
[source](#)

impl [PositionExit](#) for [Position](#)

[source](#)

fn [exit](#)(&mut self, balance: [Balance](#), fill: [&FillEvent](#)) -> [Result](#)<[PositionExit](#), [PortfolioError](#)>

Exits an open [Position](#), given the input Portfolio equity & the [FillEvent](#) returned from an Execution handler.
[source](#)

impl [PositionUpdater](#) for [Position](#)

[source](#)

fn [update](#)(&mut self, market: [&MarketEvent](#)<[DataKind](#)>) -> [Option](#)<[PositionUpdate](#)>

Updates an open [Position](#) using the latest input [MarketEvent](#), returning a [PositionUpdate](#) that communicates the open [Position](#)'s change in state.
[source](#)

impl [Serialize](#) for [Position](#)

[source](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source](#)

```
impl TryFrom<&mut Position> for PositionExit
```

[§](#)

```
type Error = PortfolioError
```

The type returned in the event of a conversion error.
[source](#)

```
fn try\_from(exited_position: &mut Position) -> Result<Self,  
Self::Error>
```

Performs the conversion.
[source](#)

```
impl StructuralPartialEq for Position
```

Auto Trait Implementations

[§](#)

```
impl RefUnwindSafe for Position
```

[§](#)

```
impl Send for Position
```

[§](#)

```
impl Sync for Position
```

[§](#)

```
impl Unpin for Position
```

[§](#)

impl [UnwindSafe](#) for [Position](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



[barter](#)0.8.13

[PositionBuilder](#)

[Fields](#)

- [current_symbol_price](#)
- [current_value_gross](#)
- [enter_avg_price_gross](#)
- [enter_fees](#)
- [enter_fees_total](#)
- [enter_value_gross](#)
- [exchange](#)
- [exit_avg_price_gross](#)
- [exit_fees](#)
- [exit_fees_total](#)
- [exit_value_gross](#)
- [instrument](#)
- [meta](#)
- [position_id](#)
- [quantity](#)
- [realised_profit_loss](#)
- [side](#)
- [unrealised_profit_loss](#)

Methods

- [build](#)
- [current_symbol_price](#)
- [current_value_gross](#)
- [enter_avg_price_gross](#)
- [enter_fees](#)
- [enter_fees_total](#)
- [enter_value_gross](#)
- [exchange](#)
- [exit_avg_price_gross](#)
- [exit_fees](#)
- [exit_fees_total](#)
- [exit_value_gross](#)
- [instrument](#)
- [meta](#)
- [new](#)
- [position_id](#)
- [quantity](#)
- [realised_profit_loss](#)
- [side](#)
- [unrealised_profit_loss](#)

Trait Implementations

- [Debug](#)
- [Default](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio::position

[?](#)

[Change settings](#)

Struct barter::portfolio::position::PositionBuilder

Copy item path

[source](#) ·

[\[-\]](#)

pub struct PositionBuilder {
Show 18 fields

```
    pub position_id: Option<PositionId>,
    pub exchange: Option<Exchange>,
    pub instrument: Option<Instrument>,
    pub meta: Option<PositionMeta>,
    pub side: Option<Side>,
    pub quantity: Option<f64>,
    pub enter_fees: Option<Fees>,
    pub enter_fees_total: Option<FeeAmount>,
    pub enter_avg_price_gross: Option<f64>,
    pub enter_value_gross: Option<f64>,
    pub exit_fees: Option<Fees>,
    pub exit_fees_total: Option<FeeAmount>,
    pub exit_avg_price_gross: Option<f64>,
    pub exit_value_gross: Option<f64>,
    pub current_symbol_price: Option<f64>,
    pub current_value_gross: Option<f64>,
    pub unrealised_profit_loss: Option<f64>,
    pub realised_profit_loss: Option<f64>,
}
```

Expand description

Builder to construct [Position](#) instances.

Fields§

[\\$position_id](#): [Option<PositionId>](#)[\\$exchange](#): [Option<Exchange>](#)[\\$instrument](#): [Option<Instrument>](#)[\\$meta](#): [Option<PositionMeta>](#)[\\$side](#): [Option<Side>](#)[\\$quantity](#): [Option<f64>](#)[\\$enter_fees](#): [Option<Fees>](#)[\\$enter_fees_total](#): [Option<FeeAmount>](#)[\\$enter_avg_price_gross](#): [Option<f64>](#)[\\$enter_value_gross](#): [Option<f64>](#)[\\$exit_fees](#): [Option<Fees>](#)[\\$exit_fees_total](#): [Option<FeeAmount>](#)[\\$exit_avg_price_gross](#): [Option<f64>](#)[\\$exit_value_gross](#): [Option<f64>](#)[\\$current_symbol_price](#): [Option<f64>](#)[\\$current_value_gross](#):

[Option<f64>](#)\$unrealised_profit_loss: [Option<f64>](#)\$realised_profit_loss:
[Option<f64>](#)

Implementations§

[source§](#)

impl [PositionBuilder](#)

[source](#)

pub fn [new](#)() -> Self

[source](#)

pub fn [position_id](#)(self, value: [PositionId](#)) -> Self

[source](#)

pub fn [exchange](#)(self, value: [Exchange](#)) -> Self

[source](#)

pub fn [instrument](#)(self, value: [Instrument](#)) -> Self

[source](#)

pub fn [meta](#)(self, value: [PositionMeta](#)) -> Self

[source](#)

pub fn [side](#)(self, value: [Side](#)) -> Self

[source](#)

pub fn [quantity](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [enter_fees](#)(self, value: [Fees](#)) -> Self

[source](#)

pub fn [enter_fees_total](#)(self, value: [FeeAmount](#)) -> Self

[source](#)

pub fn [enter_avg_price_gross](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [enter_value_gross](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [exit_fees](#)(self, value: [Fees](#)) -> Self

[source](#)

pub fn [exit_fees_total](#)(self, value: [FeeAmount](#)) -> Self

[source](#)

pub fn [exit_avg_price_gross](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [exit_value_gross](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [current_symbol_price](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [current_value_gross](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [unrealised_profit_loss](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [realised_profit_loss](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [build](#)(self) -> [Result](#)<[Position](#), [PortfolioError](#)>

Trait Implementations§

[source§](#)

impl [Debug](#) for [PositionBuilder](#)

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [PositionBuilder](#)

[source§](#)

```
fn default() -> PositionBuilder
```

Returns the “default value” for a type. [Read more](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [PositionBuilder](#)

§

impl [Send](#) for [PositionBuilder](#)

§

impl [Sync](#) for [PositionBuilder](#)

§

impl [Unpin](#) for [PositionBuilder](#)

§

impl [UnwindSafe](#) for [PositionBuilder](#)

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [From](#)<T> for T

[source](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source](#)

impl<T> [Instrument](#) for T

[source](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source§](#)

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source§](#)

```
impl<T, U> TryFrom<U> for T  
where U: Into<T>,
```

§

```
type Error = Infallible
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<T, U> TryInto<U> for T  
where U: TryFrom<T>,
```

§

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.

[source§](#)

```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source](#)

```
fn vzip(self) -> V
```

[source](#)

```
impl<T> WithSubscriber for T
```

[source](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[barter0.8.13](#)

[PositionEnterer](#)

[Required Methods](#)

- [enter](#)

[Object Safety](#)

[Implementors](#)

[In barter::portfolio::position](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::position::PositionEnterer](#)

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[–]

```
pub trait PositionEnterer {  
    // Required method  
    fn enter(  
        engine_id: Uuid,  
        fill: &FillEvent  
    ) -> Result<Position, PortfolioError>;  
}
```

Expand description

Enters a new [Position](#).

Required Methods§

[source](#)

```
fn enter(engine_id: Uuid, fill: &FillEvent) -> Result<Position,  
PortfolioError>
```

Returns a new [Position](#), given an input [FillEvent](#) & an associated engine_id.

Object Safety§

This trait is **not** [object safe](#).

Implementors§

[source§](#)

impl [PositionEnterer](#) for [Position](#)

barter0.8.13

PositionExit

Fields

- [exit_avg_price_gross](#)
- [exit_balance](#)
- [exit_fees](#)
- [exit_fees_total](#)
- [exit_time](#)
- [exit_value_gross](#)
- [position_id](#)
- [realised_profit_loss](#)

Trait Implementations

- [Clone](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)
- [TryFrom<&mut Position>](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::portfolio::position`

?

[Change settings](#)

Struct `barter::portfolio::position::PositionExit`

Copy item path

[source](#) ·

[–]

```
pub struct PositionExit {  
    pub position_id: String,  
    pub exit_time: DateTime<Utc>,  
    pub exit_balance: Balance,  
    pub exit_fees: Fees,  
    pub exit_fees_total: FeeAmount,  
    pub exit_avg_price_gross: f64,  
    pub exit_value_gross: f64,  
    pub realised_profit_loss: f64,  
}
```

Expand description

[Position](#) exit event. Occurs as a result of a [FillEvent](#) that exits a [Position](#).

Fields§

`position_id`: [String](#)

Unique identifier for a [Position](#), generated from an exchange, symbol, and `enter_time`.

`exit_time`: [DateTime<Utc>](#)

[FillEvent](#) timestamp that triggered the exiting of this [Position](#).

`exit_balance`: [Balance](#)

Portfolio [Balance](#) calculated at the point of exiting a [Position](#).

`exit_fees`: [Fees](#)

All fees types incurred from exiting a [Position](#), and their associated [FeeAmount](#).

`exit_fees_total`: [FeeAmount](#)

Total of exit_fees incurred. Sum of every [FeeAmount](#) in [Fees](#) when entering a [Position](#).

[\\$exit_avg_price_gross](#): [f64](#)

Exit average price excluding the exit_fees_total.

[\\$exit_value_gross](#): [f64](#)

abs(Quantity) * exit_avg_price_gross.

[\\$realised_profit_loss](#): [f64](#)

Realised P&L after the [Position](#) has closed.

Trait Implementations§

[source§](#)

impl [Clone](#) for [PositionExit](#)

[source§](#)

fn [clone](#)(&self) -> [PositionExit](#)

Returns a copy of the value. [Read more](#)

1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)

[source§](#)

impl [Debug](#) for [PositionExit](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl<'de> [Deserialize](#)<'de> for [PositionExit](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)

[source§](#)

impl [PartialEq](#) for [PositionExit](#)

[source§](#)

fn [eq](#)(&self, other: &[PositionExit](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [PositionExit](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[PositionExit](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source§](#)

impl [Serialize](#) for [PositionExit](#)

[source§](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [TryFrom](#)<&mut [Position](#)> for [PositionExit](#)

[§](#)

type [Error](#) = [PortfolioError](#)

The type returned in the event of a conversion error.
[source§](#)

```
fn try\_from(exited_position: &mut Position) -> Result<Self,  
Self::Error>
```

Performs the conversion.
[source§](#)

impl [StructuralPartialEq](#) for [PositionExit](#)

Auto Trait Implementations[§](#)

[§](#)

impl [RefUnwindSafe](#) for [PositionExit](#)

[§](#)

impl [Send](#) for [PositionExit](#)

[§](#)

impl [Sync](#) for [PositionExit](#)

[§](#)

impl [Unpin](#) for [PositionExit](#)

[§](#)

impl [UnwindSafe](#) for [PositionExit](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

[barter](#)0.8.13

[PositionId](#)

[Aliased type](#)

[In barter::portfolio::position](#)

[?](#)

[Change settings](#)

Type Alias [barter::portfolio::position::](#)
[PositionId](#)

Copy item path

[source](#) ·

[\[-\]](#)

pub type PositionId = [String](#);

Expand description

Communicates a String represents a unique [Position](#) identifier.

Aliased Type§

```
struct PositionId { /* private fields */ }
```



[barter](#)0.8.13

[PositionUpdate](#)

[Fields](#)

- [current_symbol_price](#)
- [current_value_gross](#)
- [position_id](#)
- [unrealised_profit_loss](#)
- [update_time](#)

[Trait Implementations](#)

- [Clone](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [From<&mut Position>](#)
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- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
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- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)

- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::portfolio::position](#)

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[Change settings](#)

Struct [barter::portfolio::position::](#)

[PositionUpdate](#)

Copy item path

[source](#) ·

[−]

```
pub struct PositionUpdate {
    pub position_id: String,
    pub update_time: DateTime<Utc>,
    pub current_symbol_price: f64,
    pub current_value_gross: f64,
    pub unrealised_profit_loss: f64,
}
```

Expand description

[Position](#) update event. Occurs as a result of receiving new [MarketEvent](#) data.

Fields§

§position_id: [String](#)

Unique identifier for a [Position](#), generated from an exchange, symbol, and enter_time.

§update_time: [DateTime<Utc>](#)

Event timestamp of the last event to trigger a [Position](#) update.

§current_symbol_price: [f64](#)

Symbol current close price.

§current_value_gross: [f64](#)

abs(Quantity) * current_symbol_price.

§unrealised_profit_loss: [f64](#)

Unrealised P&L whilst the [Position](#) is open.

Trait Implementations§

[source§](#)

impl [Clone](#) for [PositionUpdate](#)

[source§](#)

fn [clone](#)(&self) -> [PositionUpdate](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [PositionUpdate](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_,>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [PositionUpdate](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [From](#)<&mut [Position](#)> for [PositionUpdate](#)

[source§](#)

fn [from](#)(updated_position: &mut [Position](#)) -> Self

Converts to this type from the input type.
[source§](#)

impl [PartialEq](#) for [PositionUpdate](#)

[source§](#)

fn [eq](#)(&self, other: &[PositionUpdate](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)

impl [PartialOrd](#) for [PositionUpdate](#)

[source](#)

fn [partial_cmp](#)(&self, other: &[PositionUpdate](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn [le](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source](#)

fn [ge](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source](#)

impl [Serialize](#) for [PositionUpdate](#)

[source](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more source§](#)

```
impl StructuralPartialEq for PositionUpdate
```

Auto Trait Implementations§

§

```
impl RefUnwindSafe for PositionUpdate
```

§

```
impl Send for PositionUpdate
```

§

```
impl Sync for PositionUpdate
```

§

```
impl Unpin for PositionUpdate
```

§

```
impl UnwindSafe for PositionUpdate
```

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

```
fn type_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)§

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be `Self`

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

portfolio - portfolio



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[MetaPortfolio](#)

[Methods](#)

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- [Same](#)
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- [WithSubscriber](#)

[In barter::portfolio::portfolio](#)

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Struct [barter::portfolio::portfolio::](#)

[MetaPortfolio](#)

Copy item path

[source](#) ·

[−]

```
pub struct MetaPortfolio<Repository, Allocator, RiskManager, Statistic>
where
    Repository: PositionHandler + BalanceHandler + StatisticHandler<Statistic>,
    Allocator: OrderAllocator,
    RiskManager: OrderEvaluator,
    Statistic: Initialiser + PositionSummariser,
{ /* private fields */ }
```

Expand description

Portfolio with state persisted in a repository. [MarketUpdater](#), [OrderGenerator](#), [FillUpdater](#) and [PositionHandler](#).

Implementations§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>
MetaPortfolio<Repository, Allocator, RiskManager,
Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic>, Allocator: OrderAllocator,
RiskManager: OrderEvaluator, Statistic: Initialiser +
PositionSummariser,
```

[source](#)

```
pub fn init( lego: PortfolioLego<Repository, Allocator, RiskManager,
Statistic> ) -> Result<Self, PortfolioError>
```

Constructs a new [MetaPortfolio](#) using the provided [PortfolioLego](#) components, and persists the initial [MetaPortfolio](#) state in the Repository.

[source](#)

```
pub fn bootstrap\_repository<Markets, Id>( &mut self, starting_cash:
f64, markets: Markets, statistic_config: Statistic::Config ) ->
Result<(), PortfolioError>
where Markets: IntoIterator<Item = Id>, Id: Into<MarketId>,
```

Persist initial [MetaPortfolio](#) state in the repository. This includes initialised Statistics every market provided, as well as starting AvailableCash & TotalEquity.

[source](#)

```
pub fn builder( ) -> MetaPortfolioBuilder<Repository, Allocator,
RiskManager, Statistic>
```

Returns a [MetaPortfolioBuilder](#) instance.

Trait Implementations§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic> Debug
for MetaPortfolio<Repository, Allocator, RiskManager,
Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic> + Debug, Allocator:
OrderAllocator + Debug, RiskManager: OrderEvaluator +
Debug, Statistic: Initialiser + PositionSummariser +
Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>
FillUpdater for MetaPortfolio<Repository, Allocator,
RiskManager, Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic>, Allocator: OrderAllocator,
RiskManager: OrderEvaluator, Statistic: Initialiser +
PositionSummariser + Serialize,
```

[source§](#)

```
fn update\_from\_fill( &mut self, fill: &FillEvent ) ->  
Result<Vec<Event>, PortfolioError>
```

Updates the Portfolio state using the input [FillEvent](#). The [FillEvent](#) triggers a Position entry or exit, and the Portfolio updates key fields such as `current_cash` and `current_value` accordingly.

[source](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
MarketUpdater for MetaPortfolio<Repository, Allocator,  
RiskManager, Statistic>  
where Repository: PositionHandler + BalanceHandler +  
StatisticHandler<Statistic>, Allocator: OrderAllocator,  
RiskManager: OrderEvaluator, Statistic: Initialiser +  
PositionSummariser,
```

[source](#)

```
fn update\_from\_market( &mut self, market:  
&MarketEvent<DataKind> ) -> Result<Option<PositionUpdate>,  
PortfolioError>
```

Determines if the Portfolio has an open Position relating to the input [MarketEvent](#). If so it updates it using the market data, and returns a [PositionUpdate](#) detailing the changes.

[source](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
OrderGenerator for MetaPortfolio<Repository, Allocator,  
RiskManager, Statistic>  
where Repository: PositionHandler + BalanceHandler +  
StatisticHandler<Statistic>, Allocator: OrderAllocator,  
RiskManager: OrderEvaluator, Statistic: Initialiser +  
PositionSummariser,
```

[source](#)

```
fn generate\_order( &mut self, signal: &Signal ) ->  
Result<Option<OrderEvent>, PortfolioError>
```

May generate an [OrderEvent](#) after analysing an input advisory [Signal](#).

[source](#)

```
fn generate\_exit\_order( &mut self, signal: SignalForceExit ) ->  
Result<Option<OrderEvent>, PortfolioError>
```

Generates an exit [OrderEvent](#) if there is an open [Position](#) associated with the input [SignalForceExit](#)'s [PositionId](#).

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
PositionHandler for MetaPortfolio<Repository, Allocator,  
RiskManager, Statistic>  
where Repository: PositionHandler + BalanceHandler +  
StatisticHandler<Statistic>, Allocator: OrderAllocator,  
RiskManager: OrderEvaluator, Statistic: Initialiser +  
PositionSummariser,
```

[source§](#)

```
fn set_open_position( &mut self, position: Position ) -> Result<(),  
RepositoryError>
```

Upsert the open [Position](#) using it's [PositionId](#).

[source§](#)

```
fn get_open_position( &mut self, position_id: &PositionId ) ->  
Result<Option<Position>, RepositoryError>
```

Get an open [Position](#) using the [PositionId](#) provided.

[source§](#)

```
fn get_open_positions<'a, Markets: Iterator<Item = &'a  
Market>>( &mut self, _: Uuid, markets: Markets ) ->  
Result<Vec<Position>, RepositoryError>
```

Get all open [Positions](#) associated with a Portfolio.

[source§](#)

```
fn remove_position( &mut self, position_id: &PositionId ) ->  
Result<Option<Position>, RepositoryError>
```

Remove the [Position](#) at the [PositionId](#).

[source§](#)

```
fn set_exited_position( &mut self, _: Uuid, position: Position ) ->  
Result<(), RepositoryError>
```

Append an exited [Position](#) to the Portfolio's exited position list.

[source§](#)

```
fn get_exited_positions( &mut self, _: Uuid ) ->  
Result<Vec<Position>, RepositoryError>
```

Get every exited [Position](#) associated with the engine_id.

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>
StatisticHandler<Statistic> for MetaPortfolio<Repository,
Allocator, RiskManager, Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic>, Allocator: OrderAllocator,
RiskManager: OrderEvaluator, Statistic: Initialiser +
PositionSummariser,
```

[source§](#)

```
fn set_statistics( &mut self, market_id: MarketId, statistic: Statistic )
-> Result<(), RepositoryError>
```

Upsert the market statistics at the **MarketId** provided.

[source§](#)

```
fn get_statistics( &mut self, market_id: &MarketId ) ->
Result<Statistic, RepositoryError>
```

Get the market statistics using the **MarketId** provided.

Auto Trait Implementations§

§

```
impl<Repository, Allocator, RiskManager, Statistic>
RefUnwindSafe for MetaPortfolio<Repository, Allocator,
RiskManager, Statistic>
where Allocator: RefUnwindSafe, Repository:
RefUnwindSafe, RiskManager: RefUnwindSafe, Statistic:
RefUnwindSafe,
```

§

```
impl<Repository, Allocator, RiskManager, Statistic> Send
for MetaPortfolio<Repository, Allocator, RiskManager,
Statistic>
where Allocator: Send, Repository: Send, RiskManager:
Send, Statistic: Send,
```

§

impl<Repository, Allocator, RiskManager, Statistic> [Sync](#)
for [MetaPortfolio](#)<Repository, Allocator, RiskManager,
Statistic>
where Allocator: [Sync](#), Repository: [Sync](#), RiskManager:
[Sync](#), Statistic: [Sync](#),

[§](#)

impl<Repository, Allocator, RiskManager, Statistic> [Unpin](#)
for [MetaPortfolio](#)<Repository, Allocator, RiskManager,
Statistic>
where Allocator: [Unpin](#), Repository: [Unpin](#), RiskManager:
[Unpin](#), Statistic: [Unpin](#),

[§](#)

impl<Repository, Allocator, RiskManager, Statistic>
[UnwindSafe](#) for [MetaPortfolio](#)<Repository, Allocator,
RiskManager, Statistic>
where Allocator: [UnwindSafe](#), Repository: [UnwindSafe](#),
RiskManager: [UnwindSafe](#), Statistic: [UnwindSafe](#),

Blanket Implementations[§](#)

[source](#)[§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)[§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)[§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)[§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)[§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



[barter](#)0.8.13

[MetaPortfolioBuilder](#)

[Methods](#)

- [allocation_manager](#)
- [build_and_init](#)
- [engine_id](#)
- [markets](#)
- [new](#)
- [repository](#)
- [risk_manager](#)
- [starting_cash](#)
- [statistic_config](#)

[Trait Implementations](#)

- [Debug](#)
- [Default](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::portfolio::portfolio`

[?](#)

[Change settings](#)

Struct `barter::portfolio::portfolio::MetaPortfolioBuilder`

Copy item path

[source](#) ·

[−]

```
pub struct MetaPortfolioBuilder<Repository, Allocator, RiskManager, Statistic>
where
```

```
    Repository: PositionHandler + BalanceHandler + StatisticHandler<Statistic>
```

```
    Allocator: OrderAllocator,
```

```
    RiskManager: OrderEvaluator,
```

```
    Statistic: Initialiser + PositionSummariser,
```

```
{ /* private fields */ }
```

Implementations§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>
MetaPortfolioBuilder<Repository, Allocator, RiskManager,
Statistic>
```

```
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic>, Allocator: OrderAllocator,
RiskManager: OrderEvaluator, Statistic: Initialiser +
PositionSummariser,
```

[source](#)

pub fn [new](#)() -> Self

[source](#)

pub fn [engine_id](#)(self, value: [Uuid](#)) -> Self

[source](#)

pub fn [markets](#)(self, value: [Vec](#)<[Market](#)>) -> Self

[source](#)

pub fn [starting_cash](#)(self, value: [f64](#)) -> Self

[source](#)

pub fn [repository](#)(self, value: Repository) -> Self

[source](#)

pub fn [allocation_manager](#)(self, value: Allocator) -> Self

[source](#)

pub fn [risk_manager](#)(self, value: RiskManager) -> Self

[source](#)

pub fn [statistic_config](#)(self, value: Statistic::[Config](#)) -> Self

[source](#)

pub fn [build_and_init](#)(self) -> [Result](#)<[MetaPortfolio](#)<Repository, Allocator, RiskManager, Statistic>, [PortfolioError](#)>

Trait Implementations§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic> Debug
for MetaPortfolioBuilder<Repository, Allocator,
RiskManager, Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic> + Debug, Allocator:
OrderAllocator + Debug, RiskManager: OrderEvaluator +
Debug, Statistic: Initialiser + PositionSummariser +
Debug, Statistic::Config: Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>
Default for MetaPortfolioBuilder<Repository, Allocator,
RiskManager, Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic> + Default, Allocator:
OrderAllocator + Default, RiskManager: OrderEvaluator +
Default, Statistic: Initialiser + PositionSummariser +
Default, Statistic::Config: Default,
```

[source§](#)

```
fn default( ) -> MetaPortfolioBuilder<Repository, Allocator,
RiskManager, Statistic>
```

Returns the “default value” for a type. [Read more](#)

Auto Trait Implementations§

§

```
impl<Repository, Allocator, RiskManager, Statistic>
RefUnwindSafe for MetaPortfolioBuilder<Repository,
Allocator, RiskManager, Statistic>
where Allocator: RefUnwindSafe, Repository:
RefUnwindSafe, RiskManager: RefUnwindSafe, Statistic:
RefUnwindSafe, <Statistic as Initialiser>::Config:
RefUnwindSafe,
```

§

```
impl<Repository, Allocator, RiskManager, Statistic> Send  
for MetaPortfolioBuilder<Repository, Allocator,  
RiskManager, Statistic>  
where Allocator: Send, Repository: Send, RiskManager:  
Send, Statistic: Send, <Statistic as Initialiser>::Config:  
Send,
```

[§](#)

```
impl<Repository, Allocator, RiskManager, Statistic> Sync  
for MetaPortfolioBuilder<Repository, Allocator,  
RiskManager, Statistic>  
where Allocator: Sync, Repository: Sync, RiskManager:  
Sync, Statistic: Sync, <Statistic as Initialiser>::Config:  
Sync,
```

[§](#)

```
impl<Repository, Allocator, RiskManager, Statistic> Unpin  
for MetaPortfolioBuilder<Repository, Allocator,  
RiskManager, Statistic>  
where Allocator: Unpin, Repository: Unpin, RiskManager:  
Unpin, Statistic: Unpin, <Statistic as Initialiser>::Config:  
Unpin,
```

[§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
UnwindSafe for MetaPortfolioBuilder<Repository,  
Allocator, RiskManager, Statistic>  
where Allocator: UnwindSafe, Repository: UnwindSafe,  
RiskManager: UnwindSafe, Statistic: UnwindSafe,  
<Statistic as Initialiser>::Config: UnwindSafe,
```

Blanket Implementations[§](#)

[source](#)[§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source](#)[§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)

[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)§

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be `Self`

[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

§

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source](#)§

```
fn try_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.

[source§](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

```
fn with_current_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)



[barter0.8.13](#)

[PortfolioLego](#)

[Fields](#)

- [_statistic_marker](#)
- [allocator](#)
- [engine_id](#)
- [markets](#)
- [repository](#)
- [risk](#)
- [starting_cash](#)
- [statistic_config](#)

Trait Implementations

- [Debug](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio::portfolio

?

[Change settings](#)

Struct barter::portfolio::portfolio::PortfolioLego

Copy item path

[source](#) ·

[–]

```
pub struct PortfolioLego<Repository, Allocator, RiskManager, Statistic>
where
    Repository: PositionHandler + BalanceHandler + StatisticHandler<Statistic>,
    Allocator: OrderAllocator,
    RiskManager: OrderEvaluator,
    Statistic: Initialiser + PositionSummariser,
{
    pub engine_id: Uuid,
    pub markets: Vec<Market>,
    pub repository: Repository,
    pub allocator: Allocator,
    pub risk: RiskManager,
```

```
pub starting_cash: f64,  
pub statistic_config: Statistic::Config,  
pub _statistic_marker: PhantomData<Statistic>,  
}
```

Expand description

Lego components for constructing & initialising a [MetaPortfolio](#) via the `init()` constructor method.

Fields§

§engine_id: [Uuid](#)

Identifier for the [Engine](#) a [MetaPortfolio](#) is associated with (1-to-1 relationship).

§markets: [Vec<Market>](#)

[Markets](#) being tracked by a [MetaPortfolio](#).

§repository: [Repository](#)

Repository for a [MetaPortfolio](#) to persist it's state in. Implements [PositionHandler](#), [BalanceHandler](#), and [StatisticHandler](#)

§allocator: [Allocator](#)

Allocation manager implements [OrderAllocator](#).

§risk: [RiskManager](#)

Risk manager implements [OrderEvaluator](#).

§starting_cash: [f64](#)

Cash balance a [MetaPortfolio](#) starts with.

§statistic_config: [Statistic::Config](#)

Configuration used to initialise the Statistics for every Market's performance tracked by a [MetaPortfolio](#).

§_statistic_marker: [PhantomData<Statistic>](#)

Trait Implementations§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic> Debug
for PortfolioLego<Repository, Allocator, RiskManager,
Statistic>
where Repository: PositionHandler + BalanceHandler +
StatisticHandler<Statistic> + Debug, Allocator:
OrderAllocator + Debug, RiskManager: OrderEvaluator +
Debug, Statistic: Initialiser + PositionSummariser +
Debug, Statistic::Config: Debug,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

Auto Trait Implementations§

§

```
impl<Repository, Allocator, RiskManager, Statistic>
RefUnwindSafe for PortfolioLego<Repository, Allocator,
RiskManager, Statistic>
where Allocator: RefUnwindSafe, Repository:
RefUnwindSafe, RiskManager: RefUnwindSafe, Statistic:
RefUnwindSafe, <Statistic as Initialiser>::Config:
RefUnwindSafe,
```

§

```
impl<Repository, Allocator, RiskManager, Statistic> Send
for PortfolioLego<Repository, Allocator, RiskManager,
Statistic>
where Allocator: Send, Repository: Send, RiskManager:
Send, Statistic: Send, <Statistic as Initialiser>::Config:
Send,
```

§

```
impl<Repository, Allocator, RiskManager, Statistic> Sync
for PortfolioLego<Repository, Allocator, RiskManager,
Statistic>
where Allocator: Sync, Repository: Sync, RiskManager:
Sync, Statistic: Sync, <Statistic as Initialiser>::Config:
Sync,
```

§

```
impl<Repository, Allocator, RiskManager, Statistic> Unpin  
for PortfolioLego<Repository, Allocator, RiskManager,  
Statistic>  
where Allocator: Unpin, Repository: Unpin, RiskManager:  
Unpin, Statistic: Unpin, <Statistic as Initialiser>::Config:  
Unpin,
```

[§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
UnwindSafe for PortfolioLego<Repository, Allocator,  
RiskManager, Statistic>  
where Allocator: UnwindSafe, Repository: UnwindSafe,  
RiskManager: UnwindSafe, Statistic: UnwindSafe,  
<Statistic as Initialiser>::Config: UnwindSafe,
```

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source§](#)

```
impl<T> Borrow<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)


```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[barter](#)0.8.13

[Module portfolio](#)

- [Structs](#)
- [Functions](#)

[In barter::portfolio](#)

[?](#)

[Change settings](#)

Module [barter::portfolio::portfolio](#)

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Expand description

Core Portfolio logic containing an implementation of [MarketUpdater](#), [OrderGenerator](#) and [FillUpdater](#). Utilises the risk and allocator logic to optimise [OrderEvent](#) generation.

Structs

- [MetaPortfolio](#)
Portfolio with state persisted in a repository. [MarketUpdater](#), [OrderGenerator](#), [FillUpdater](#) and [PositionHandler](#).
- [MetaPortfolioBuilder](#)
- [PortfolioLego](#)
Lego components for constructing & initialising a [MetaPortfolio](#) via the `init()` constructor method.

Functions§

- [parse_signal_decisions](#)
Parses an incoming [Signal](#)'s signals map. Determines what the net signal [Decision](#) will be, and it's associated [SignalStrength](#).

[barter](#)0.8.13

[In barter::portfolio::portfolio](#)

[?](#)
[Change settings](#)

Function [barter::portfolio::portfolio::parse_signal_decisions](#)

Copy item path

[source](#) ·

[-]

```
pub fn parse_signal_decisions<'a>(
  position: &'a Option<&Position>,
  signals: &'a HashMap<Decision, SignalStrength>
) -> Option<(&'a Decision, &'a SignalStrength)>
```

Expand description

Parses an incoming [Signal](#)'s signals map. Determines what the net signal [Decision](#) will be, and it's associated [SignalStrength](#).

portfolio - repository

[barter](#)0.8.13

[Module repository](#)

- [Modules](#)
- [Traits](#)
- [Functions](#)
- [Type Aliases](#)

[In barter::portfolio](#)

[?](#)

[Change settings](#)

Module [barter::portfolio::repository](#)

Copy item path

[source](#) ·

[\[-\]](#)

Expand description

Repositories for persisting Portfolio state.

Modules§

- [error](#)
Barter repository module specific errors.
- [in_memory](#)
In-Memory repository for convenient state keeping. No fault tolerant guarantees.
- [redis](#)
Redis repository for state keeping.

Traits§

- [BalanceHandler](#)
Handles the reading & writing of a Portfolio's current balance to/from the persistence layer.
- [PositionHandler](#)
Handles the reading & writing of a [Position](#) to/from the persistence layer.
- [StatisticHandler](#)
Handles the reading & writing of a Portfolio's statistics for each of it's markets, where each market is represented by a [MarketId](#).

Functions§

- [determine_exited_positions_id](#)
Returns the unique identifier for a Portfolio's exited [Positions](#), given an engine_id.

Type Aliases§

- [ExitedPositionsId](#)

Communicates a String represents a unique identifier for all a Portfolio's exited [Positions](#). Used to append new exited [Positions](#) to the entry in the [PositionHandler](#).

[barter](#)0.8.13

[PositionHandler](#)

[Required Methods](#)

- [get_exited_positions](#)
- [get_open_position](#)
- [get_open_positions](#)
- [remove_position](#)
- [set_exited_position](#)
- [set_open_position](#)

[Object Safety](#)

[Implementors](#)

[In barter::portfolio::repository](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::repository::PositionHandler](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub trait PositionHandler {
  // Required methods
  fn set\_open\_position(
    &mut self,
    position: Position
  ) -> Result<(), RepositoryError>;
  fn get\_open\_position(
    &mut self,
    position_id: &PositionId
  ) -> Result<Option<Position>, RepositoryError>;
  fn get\_open\_positions<'a, Markets: Iterator<Item = &'a Market>>(<
```

```

        &mut self,
        engine_id: Uuid,
        markets: Markets
    ) -> Result<Vec<Position>, RepositoryError>;
    fn remove\_position(
        &mut self,
        position_id: &PositionId
    ) -> Result<Option<Position>, RepositoryError>;
    fn set\_exited\_position(
        &mut self,
        engine_id: Uuid,
        position: Position
    ) -> Result<(), RepositoryError>;
    fn get\_exited\_positions(
        &mut self,
        engine_id: Uuid
    ) -> Result<Vec<Position>, RepositoryError>;
}

```

Expand description

Handles the reading & writing of a [Position](#) to/from the persistence layer.

Required Methods§

[source](#)

```

fn set\_open\_position( &mut self, position: Position ) -> Result<(), RepositoryError>

```

Upsert the open [Position](#) using it's [PositionId](#).

[source](#)

```

fn get\_open\_position( &mut self, position_id: &PositionId ) -> Result<Option<Position>, RepositoryError>

```

Get an open [Position](#) using the [PositionId](#) provided.

[source](#)

```

fn get\_open\_positions<'a, Markets: Iterator<Item = &'a Market>>( &mut self, engine_id: Uuid, markets: Markets ) -> Result<Vec<Position>, RepositoryError>

```

Get all open [Positions](#) associated with a Portfolio.

[source](#)

```
fn remove_position( &mut self, position_id: &PositionId ) ->  
Result<Option<Position>, RepositoryError>
```

Remove the [Position](#) at the [PositionId](#).

[source](#)

```
fn set_exited_position( &mut self, engine_id: Uuid, position:  
Position ) -> Result<\(\), RepositoryError>
```

Append an exited [Position](#) to the Portfolio's exited position list.

[source](#)

```
fn get_exited_positions( &mut self, engine_id: Uuid ) ->  
Result<Vec<Position>, RepositoryError>
```

Get every exited [Position](#) associated with the engine_id.

Object Safety§

This trait is **not** [object safe](#).

Implementors§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
PositionHandler for MetaPortfolio<Repository, Allocator,  
RiskManager, Statistic>  
where Repository: PositionHandler + BalanceHandler +  
StatisticHandler<Statistic>, Allocator: OrderAllocator,  
RiskManager: OrderEvaluator, Statistic: Initialiser +  
PositionSummariser,
```

[source§](#)

```
impl<Statistic> PositionHandler for  
RedisRepository<Statistic>  
where Statistic: PositionSummariser + Serialize +  
DeserializeOwned,
```

[source§](#)

```
impl<Statistic: PositionSummariser> PositionHandler for  
InMemoryRepository<Statistic>
```

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[BalanceHandler](#)

[Required Methods](#)

- [get_balance](#)
- [set_balance](#)

[Implementors](#)

[In barter::portfolio::repository](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::repository::BalanceHandler](#)

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[-]

```
pub trait BalanceHandler {  
    // Required methods  
    fn set\_balance(  
        &mut self,  
        engine_id: Uuid,  
        balance: Balance  
    ) -> Result<(), RepositoryError>;  
    fn get\_balance(  
        &mut self,  
        engine_id: Uuid  
    ) -> Result<Balance, RepositoryError>;  
}
```

Expand description

Handles the reading & writing of a Portfolio's current balance to/from the persistence layer.

[Required Methods](#)

[source](#)

```
fn set_balance( &mut self, engine_id: Uuid, balance: Balance ) -> Result<(), RepositoryError>
```

Upsert the Portfolio [Balance](#) at the engine_id.

[source](#)

```
fn get_balance(&mut self, engine_id: Uuid) -> Result<Balance, RepositoryError>
```

Get the Portfolio [Balance](#) using the engine_id provided.

Implementors§

[source§](#)

```
impl<Statistic> BalanceHandler for RedisRepository<Statistic>
where Statistic: PositionSummariser + Serialize + DeserializeOwned,
```

[source§](#)

```
impl<Statistic: PositionSummariser> BalanceHandler for InMemoryRepository<Statistic>
```



[barter0.8.13](#)

[In barter::portfolio::repository](#)



[?](#)
[Change settings](#)

Function [barter::portfolio::repository::determine_exited_positions_id](#)



[source](#) ·

```
pub fn determine_exited_positions_id(engine_id: Uuid) -> ExitedPositionsId
```

Expand description

Returns the unique identifier for a Portfolio's exited [Positions](#), given an engine_id.

[barter](#)0.8.13

[ExitedPositionsId](#)

[Aliased type](#)

[In barter::portfolio::repository](#)

[?](#)

[Change settings](#)

Type Alias [barter::portfolio::repository::ExitedPositionsId](#)

Copy item path

[source](#) ·

[-]

```
pub type ExitedPositionsId = String;
```

Expand description

Communicates a String represents a unique identifier for all a Portfolio's exited [Positions](#). Used to append new exited [Positions](#) to the entry in the [PositionHandler](#).

Aliased Type[§](#)

```
struct ExitedPositionsId { /* private fields */ }
```

[barter](#)0.8.13

[StatisticHandler](#)

[Required Methods](#)

- [get_statistics](#)
- [set_statistics](#)

[Implementors](#)

[In barter::portfolio::repository](#)

[?](#)

[Change settings](#)

Trait [barter::portfolio::repository::StatisticHandler](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub trait StatisticHandler<Statistic> {  
    // Required methods  
    fn set\_statistics(  
        &mut self,  
        market_id: MarketId,  
        statistic: Statistic  
    ) -> Result<(), RepositoryError>;  
    fn get\_statistics(  
        &mut self,  
        market_id: &MarketId  
    ) -> Result<Statistic, RepositoryError>;  
}
```

Expand description

Handles the reading & writing of a Portfolio's statistics for each of it's markets, where each market is represented by a [MarketId](#).

Required Methods§

[source](#)

```
fn set\_statistics( &mut self, market_id: MarketId, statistic: Statistic )  
-> Result<(), RepositoryError>
```

Upsert the market statistics at the [MarketId](#) provided.

[source](#)

```
fn get\_statistics( &mut self, market_id: &MarketId ) ->  
Result<Statistic, RepositoryError>
```

Get the market statistics using the [MarketId](#) provided.

Implementors§

[source§](#)

```
impl<Repository, Allocator, RiskManager, Statistic>  
StatisticHandler<Statistic> for MetaPortfolio<Repository,  
Allocator, RiskManager, Statistic>  
where Repository: PositionHandler + BalanceHandler +  
StatisticHandler<Statistic>, Allocator: OrderAllocator,  
RiskManager: OrderEvaluator, Statistic: Initialiser +  
PositionSummariser,
```

[source§](#)

```
impl<Statistic> StatisticHandler<Statistic> for  
RedisRepository<Statistic>  
where Statistic: PositionSummariser + Serialize +  
DeserializeOwned,
```

[source§](#)

```
impl<Statistic: PositionSummariser>  
StatisticHandler<Statistic> for  
InMemoryRepository<Statistic>
```

portfolio - repository - in_memory

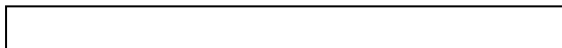


[barter0.8.13](#)

[Module in_memory](#)

- [Structs](#)

[In barter::portfolio::repository](#)



[?](#)

[Change settings](#)

Module [barter::portfolio::repository::in_memory](#)

Copy item path

[source](#) · [\[-\]](#)

Expand description

In-Memory repository for convenient state keeping. No fault tolerant guarantees.

Structs§

- [InMemoryRepository](#)

In-Memory repository for Proof Of Concepts. Implements [PositionHandler](#), [BalanceHandler](#) & [StatisticHandler](#). Used by a Proof Of Concept Portfolio implementation to save the current equity, available cash, Positions, and market pair statistics. **Careful in production - no fault tolerant guarantees!**

[barter](#)0.8.13

[InMemoryRepository](#)

[Methods](#)

- [new](#)

[Trait Implementations](#)

- [BalanceHandler](#)
- [Debug](#)
- [Default](#)
- [PositionHandler](#)
- [StatisticHandler<Statistic>](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

[Blanket Implementations](#)

- [Any](#)

- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::portfolio::repository::in_memory](#)

?

[Change settings](#)

Struct [barter::portfolio::repository::in_memory::InMemoryRepository](#)

Copy item path

[source](#) ·

[−]

```
pub struct InMemoryRepository<Statistic: PositionSummariser> { /* private
```

Expand description

In-Memory repository for Proof Of Concepts. Implements [PositionHandler](#), [BalanceHandler](#) & [StatisticHandler](#). Used by a Proof Of Concept Portfolio implementation to save the current equity, available cash, Positions, and market pair statistics. **Careful in production - no fault tolerant guarantees!**

Implementations§

[source§](#)

```
impl<Statistic: PositionSummariser>
InMemoryRepository<Statistic>
```

[source](#)

```
pub fn new() -> Self
```

Constructs a new [InMemoryRepository](#) component.

Trait Implementations§

[source§](#)

impl<Statistic: [PositionSummariser](#)> [BalanceHandler](#) for [InMemoryRepository](#)<Statistic>

[source§](#)

fn [set_balance](#)(&mut self, engine_id: [Uuid](#), balance: [Balance](#)) -> [Result](#)<[\(\)](#), [RepositoryError](#)>

Upsert the Portfolio [Balance](#) at the engine_id.

[source§](#)

fn [get_balance](#)(&mut self, engine_id: [Uuid](#)) -> [Result](#)<[Balance](#), [RepositoryError](#)>

Get the Portfolio [Balance](#) using the engine_id provided.

[source§](#)

impl<Statistic: [Debug](#) + [PositionSummariser](#)> [Debug](#) for [InMemoryRepository](#)<Statistic>

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)

[source§](#)

impl<Statistic: [Default](#) + [PositionSummariser](#)> [Default](#) for [InMemoryRepository](#)<Statistic>

[source§](#)

fn [default](#)() -> [InMemoryRepository](#)<Statistic>

Returns the “default value” for a type. [Read more](#)

[source§](#)

impl<Statistic: [PositionSummariser](#)> [PositionHandler](#) for [InMemoryRepository](#)<Statistic>

[source§](#)

```
fn set_open_position( &mut self, position: Position ) -> Result<\(\), RepositoryError>
```

Upsert the open [Position](#) using it's [PositionId](#).
[source](#)

```
fn get_open_position( &mut self, position_id: &PositionId ) ->  
Result<Option<Position>, RepositoryError>
```

Get an open [Position](#) using the [PositionId](#) provided.
[source](#)

```
fn get_open_positions<'a, Markets: Iterator<Item = &'a  
Market>>( &mut self, engine_id: Uuid, markets: Markets ) ->  
Result<Vec<Position>, RepositoryError>
```

Get all open [Positions](#) associated with a Portfolio.
[source](#)

```
fn remove_position( &mut self, position_id: &String ) ->  
Result<Option<Position>, RepositoryError>
```

Remove the [Position](#) at the [PositionId](#).
[source](#)

```
fn set_exited_position( &mut self, engine_id: Uuid, position:  
Position ) -> Result<\(\), RepositoryError>
```

Append an exited [Position](#) to the Portfolio's exited position list.
[source](#)

```
fn get_exited_positions( &mut self, engine_id: Uuid ) ->  
Result<Vec<Position>, RepositoryError>
```

Get every exited [Position](#) associated with the engine_id.
[source](#)

```
impl<Statistic: PositionSummariser>  
StatisticHandler<Statistic> for  
InMemoryRepository<Statistic>
```

[source](#)

```
fn set_statistics( &mut self, market_id: MarketId, statistic: Statistic )  
-> Result<\(\), RepositoryError>
```

Upsert the market statistics at the [MarketId](#) provided.
[source](#)

```
fn get\_statistics( &mut self, market_id: &MarketId ) ->  
Result<Statistic, RepositoryError>
```

Get the market statistics using the [MarketId](#) provided.

Auto Trait Implementations§

§

```
impl<Statistic> RefUnwindSafe for  
InMemoryRepository<Statistic>  
where Statistic: RefUnwindSafe,
```

§

```
impl<Statistic> Send for InMemoryRepository<Statistic>  
where Statistic: Send,
```

§

```
impl<Statistic> Sync for InMemoryRepository<Statistic>  
where Statistic: Sync,
```

§

```
impl<Statistic> Unpin for InMemoryRepository<Statistic>  
where Statistic: Unpin,
```

§

```
impl<Statistic> UnwindSafe for  
InMemoryRepository<Statistic>  
where Statistic: UnwindSafe,
```

Blanket Implementations§

[source](#)§

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source](#)§

```
fn type\_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)
[source](#)§

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T
where U: From<T>,
```

[source](#)§

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source](#)§

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source](#)§

```
impl<T, U> TryFrom<U> for T
where U: Into<T>,
```

§

```
type Error = Infallible
```

The type returned in the event of a conversion error.

[source](#)§

```
fn try_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source](#)§

```
impl<T, U> TryInto<U> for T
where U: TryFrom<T>,
```

§

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.

[source](#)§

```
fn try_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source§](#)

```
impl<V, T> VZip<V> for T  
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
fn with_current_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

portfolio - repository - error

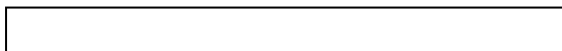


[barter0.8.13](#)

[Module error](#)

- [Enums](#)

[In barter::portfolio::repository](#)



[?](#)
[Change settings](#)

Module [barter::portfolio::repository::error](#)

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Expand description

Barter repository module specific errors.

Enums§

- [RepositoryError](#)

All errors generated in the barter::portfolio::repository module.

[barter](#)0.8.13

[RepositoryError](#)

[Variants](#)

- [DeleteError](#)
- [ExpectedDataNotPresentError](#)
- [JsonSerDeError](#)
- [ReadError](#)
- [WriteError](#)

[Trait Implementations](#)

- [Debug](#)
- [Display](#)
- [Error](#)
- [From<Error>](#)
- [From<RepositoryError>](#)
- [From<RepositoryError>](#)

[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

[Blanket Implementations](#)

- [Any](#)

- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToString](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In `barter::portfolio::repository::error`](#)

[?](#)

[Change settings](#)

Enum [barter::portfolio::repository::error::RepositoryError](#)

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[source](#) ·

[\[-\]](#)

```
pub enum RepositoryError {
    JsonSerDeError(Error),
    WriteError,
    ReadError,
    DeleteError,
    ExpectedDataNotPresentError,
}
```

Expand description

All errors generated in the `barter::portfolio::repository` module.

Variants§

§

JsonSerDeError([Error](#))

§

WriteError

§

ReadError

§

DeleteError

§

ExpectedDataNotPresentError

Trait Implementations§

[source§](#)

impl [Debug](#) for [RepositoryError](#)

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Display](#) for [RepositoryError](#)

[source§](#)

```
fn fmt(&self, __formatter: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Error](#) for [RepositoryError](#)

[source§](#)

```
fn source(&self) -> Option<&(dyn Error + 'static)>
```

The lower-level source of this error, if any. [Read more](#)
1.0.0 · [source§](#)

```
fn description(&self) -> &str
```

Deprecated since 1.42.0: use the Display impl or to_string()
[Read more](#)
1.0.0 · [source§](#)

fn [cause](#)(&self) -> [Option](#)<&dyn [Error](#)>

Deprecated since 1.33.0: replaced by `Error::source`, which can support downcasting

[source§](#)

fn [provide](#)<'a>(&'a self, request: &mut [Request](#)<'a>)

This is a nightly-only experimental API. (`error_generic_member_access`)
Provides type based access to context intended for error reports. [Read more](#)
[source§](#)

impl [From](#)<[Error](#)> for [RepositoryError](#)

[source§](#)

fn [from](#)(source: [Error](#)) -> Self

Converts to this type from the input type.

[source§](#)

impl [From](#)<[RepositoryError](#)> for [EngineError](#)

[source§](#)

fn [from](#)(source: [RepositoryError](#)) -> Self

Converts to this type from the input type.

[source§](#)

impl [From](#)<[RepositoryError](#)> for [PortfolioError](#)

[source§](#)

fn [from](#)(source: [RepositoryError](#)) -> Self

Converts to this type from the input type.

Auto Trait Implementations§

§

impl ![RefUnwindSafe](#) for [RepositoryError](#)

§

impl [Send](#) for [RepositoryError](#)

§

impl [Sync](#) for [RepositoryError](#)

[§](#)

impl [Unpin](#) for [RepositoryError](#)

[§](#)

impl [!UnwindSafe](#) for [RepositoryError](#)

Blanket Implementations

[source](#)

impl<T> [Any](#) for T
where T: 'static + [?Sized](#),

[source](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: [?Sized](#),

[source](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [BorrowMut](#)<T> for T
where T: [?Sized](#),

[source](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [From](#)<T> for T

[source](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source](#)

impl<T> [Instrument](#) for T

[source](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source](#)

**impl<T> [ToString](#) for T
where T: [Display](#) + [?Sized](#),**

[source](#)

default fn [to_string](#)(&self) -> [String](#)

Converts the given value to a String. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>  
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

portfolio - repository - redis



[barter](#)0.8.13

[RedisRepositoryBuilder](#)

[Methods](#)

- [build](#)
- [conn](#)
- [new](#)

[Trait Implementations](#)

- [Debug](#)
- [Default](#)

[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

[Blanket Implementations](#)

- [Any](#)
- [Borrow](#)<T>
- [BorrowMut](#)<T>
- [From](#)<T>
- [Instrument](#)
- [Into](#)<U>

- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::portfolio::repository::redis](#)

?

[Change settings](#)

Struct [barter::portfolio::repository::redis::RedisRepositoryBuilder](#)

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[source](#) ·

[−]

```
pub struct RedisRepositoryBuilder<Statistic>
where
    Statistic: PositionSummariser + Serialize + DeserializeOwned,
{ /* private fields */ }
```

Expand description

Builder to construct [RedisRepository](#) instances.

Implementations§

[source§](#)

```
impl<Statistic> RedisRepositoryBuilder<Statistic>
where Statistic: PositionSummariser + Serialize +
DeserializeOwned,
```

[source](#)

```
pub fn new() -> Self
```

[source](#)

```
pub fn conn(self, value: Connection) -> Self
```

[source](#)

```
pub fn build(self) -> Result<RedisRepository<Statistic>,
PortfolioError>
```

Trait Implementations§

[source§](#)

```
impl<Statistic> Debug for
RedisRepositoryBuilder<Statistic>
where Statistic: PositionSummariser + Serialize +
DeserializeOwned,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

```
impl<Statistic> Default for
RedisRepositoryBuilder<Statistic>
where Statistic: PositionSummariser + Serialize +
DeserializeOwned + Default,
```

[source§](#)

```
fn default() -> RedisRepositoryBuilder<Statistic>
```

Returns the “default value” for a type. [Read more](#)

Auto Trait Implementations§

§

```
impl<Statistic> !RefUnwindSafe for
RedisRepositoryBuilder<Statistic>
```

§

```
impl<Statistic> Send for
RedisRepositoryBuilder<Statistic>
where Statistic: Send,
```

§

impl<Statistic> [Sync](#) for
[RedisRepositoryBuilder](#)<Statistic>
where Statistic: [Sync](#),

[§](#)

impl<Statistic> [Unpin](#) for
[RedisRepositoryBuilder](#)<Statistic>
where Statistic: [Unpin](#),

[§](#)

impl<Statistic> [!UnwindSafe](#) for
[RedisRepositoryBuilder](#)<Statistic>

Blanket Implementations[§](#)

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

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[Module redis](#)

- [Structs](#)

[In barter::portfolio::repository](#)

[?](#)
[Change settings](#)

Module [barter::portfolio::repository::redis](#)

Copy item path

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Expand description

Redis repository for state keeping.

Structs[§](#)

- [Config](#)
Configuration for constructing a [RedisRepository](#) via the new() constructor method.
- [RedisRepository](#)
Redis persisted repository that implements [PositionHandler](#), [BalanceHandler](#), & [PositionSummariser](#). Used by a Portfolio implementation to persist the Portfolio state, including total equity, available cash & Positions.
- [RedisRepositoryBuilder](#)
Builder to construct [RedisRepository](#) instances.

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RedisRepository

Methods

- [builder](#)
- [new](#)
- [setup_redis_connection](#)

Trait Implementations

- [BalanceHandler](#)
- [Debug](#)
- [PositionHandler](#)
- [StatisticHandler<Statistic>](#)

Auto Trait Implementations

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::portfolio::repository::redis

[?](#)

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Struct [barter::portfolio::repository::redis::RedisRepository](#)

Copy item path

[source](#) ·

[–]

```
pub struct RedisRepository<Statistic>
where
    Statistic: PositionSummariser + Serialize + DeserializeOwned,
{ /* private fields */ }
```

Expand description

Redis persisted repository that implements [PositionHandler](#), [BalanceHandler](#), & [PositionSummariser](#). Used by a Portfolio implementation to persist the Portfolio state, including total equity, available cash & Positions.

Implementations§

[source§](#)

```
impl<Statistic> RedisRepository<Statistic>
where Statistic: PositionSummariser + Serialize +
DeserializeOwned,
```

[source](#)

```
pub fn new(connection: Connection) -> Self
```

Constructs a new [RedisRepository](#) component using the provided Redis connection struct.

[source](#)

```
pub fn builder() -> RedisRepositoryBuilder<Statistic>
```

Returns a [RedisRepositoryBuilder](#) instance.

[source](#)

```
pub fn setup\_redis\_connection(cfg: Config) -> Connection
```

Establish & return a Redis connection.

Trait Implementations§

[source§](#)

```
impl<Statistic> BalanceHandler for  
RedisRepository<Statistic>  
where Statistic: PositionSummariser + Serialize +  
DeserializeOwned,
```

[source§](#)

```
fn set\_balance( &mut self, engine_id: Uuid, balance: Balance ) ->  
Result<\(\), RepositoryError>
```

Upsert the Portfolio [Balance](#) at the engine_id.

[source§](#)

```
fn get\_balance(&mut self, engine_id: Uuid) -> Result<Balance,  
RepositoryError>
```

Get the Portfolio [Balance](#) using the engine_id provided.

[source§](#)

```
impl<Statistic> Debug for RedisRepository<Statistic>  
where Statistic: PositionSummariser + Serialize +  
DeserializeOwned,
```

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)

[source§](#)

```
impl<Statistic> PositionHandler for  
RedisRepository<Statistic>  
where Statistic: PositionSummariser + Serialize +  
DeserializeOwned,
```

[source§](#)

```
fn set\_open\_position( &mut self, position: Position ) -> Result<\(\),  
RepositoryError>
```

Upsert the open [Position](#) using it's [PositionId](#).

[source§](#)

```
fn get\_open\_position( &mut self, position_id: &PositionId ) ->  
Result<Option<Position>, RepositoryError>
```

Get an open [Position](#) using the [PositionId](#) provided.
[source](#)

```
fn get\_open\_positions<'a, Markets: Iterator<Item = &'a  
Market>>( &mut self, engine_id: Uuid, markets: Markets ) ->  
Result<Vec<Position>, RepositoryError>
```

Get all open [Positions](#) associated with a Portfolio.
[source](#)

```
fn remove\_position( &mut self, position_id: &String ) ->  
Result<Option<Position>, RepositoryError>
```

Remove the [Position](#) at the [PositionId](#).
[source](#)

```
fn set\_exited\_position( &mut self, engine_id: Uuid, position:  
Position ) -> Result<(), RepositoryError>
```

Append an exited [Position](#) to the Portfolio's exited position list.
[source](#)

```
fn get\_exited\_positions( &mut self, engine_id: Uuid ) ->  
Result<Vec<Position>, RepositoryError>
```

Get every exited [Position](#) associated with the engine_id.
[source](#)

```
impl<Statistic> StatisticHandler<Statistic> for  
RedisRepository<Statistic>  
where Statistic: PositionSummariser + Serialize +  
DeserializeOwned,
```

[source](#)

```
fn set\_statistics( &mut self, market_id: MarketId, statistic: Statistic )  
-> Result<(), RepositoryError>
```

Upsert the market statistics at the [MarketId](#) provided.
[source](#)

```
fn get\_statistics( &mut self, market_id: &MarketId ) ->  
Result<Statistic, RepositoryError>
```

Get the market statistics using the [MarketId](#) provided.

Auto Trait Implementations§

§

impl<Statistic> ![RefUnwindSafe](#) for [RedisRepository](#)<Statistic>

§

**impl<Statistic> [Send](#) for [RedisRepository](#)<Statistic>
where Statistic: [Send](#),**

§

**impl<Statistic> [Sync](#) for [RedisRepository](#)<Statistic>
where Statistic: [Sync](#),**

§

**impl<Statistic> [Unpin](#) for [RedisRepository](#)<Statistic>
where Statistic: [Unpin](#),**

§

impl<Statistic> ![UnwindSafe](#) for [RedisRepository](#)<Statistic>

Blanket Implementations§

[source](#)§

**impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),**

[source](#)§

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)§

**impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),**

[source](#)§

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From<T>](#) for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be Self

[source](#)§

**impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,**

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

§

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.

[source](#)§

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)§

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

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[Config](#)

[Fields](#)

- [uri](#)

[Trait Implementations](#)

- [Clone](#)
- [Debug](#)
- [Default](#)
- [Deserialize](#)<'de>
- [Eq](#)
- [Hash](#)
- [Ord](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

[Auto Trait Implementations](#)

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)

- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [Comparable<K>](#)
- [DeserializeOwned](#)
- [Equivalent<K>](#)
- [Equivalent<K>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In [barter::portfolio::repository::redis](#)

[?](#)

[Change settings](#)

Struct [barter::portfolio::repository::redis::Config](#)

Copy item path

[source](#) ·

```
pub struct Config {
    pub uri: String,
}
```

Expand description

Configuration for constructing a [RedisRepository](#) via the new() constructor method.

Fields§

[§uri](#): [String](#)

Trait Implementations§

[source§](#)

impl [Clone](#) for [Config](#)

[source§](#)

fn [clone](#)(&self) -> [Config](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [Config](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [Config](#)

[source§](#)

fn [default](#)() -> [Config](#)

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [Config](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [Hash](#) for [Config](#)

[source§](#)

fn [hash](#)<_H: [Hasher](#)>(&self, state: [&mut __H](#))

Feeds this value into the given [Hasher](#). [Read more](#)
1.3.0 · [source](#)

fn [hash_slice](#)<H>(data: [&\[Self\]](#), state: [&mut H](#))
where H: [Hasher](#), Self: [Sized](#),

Feeds a slice of this type into the given [Hasher](#). [Read more](#)
[source](#)

impl [Ord](#) for [Config](#)

[source](#)

fn [cmp](#)(&self, other: [&Config](#)) -> [Ordering](#)

This method returns an [Ordering](#) between self and other. [Read more](#)
1.21.0 · [source](#)

fn [max](#)(self, other: Self) -> Self
where Self: [Sized](#),

Compares and returns the maximum of two values. [Read more](#)
1.21.0 · [source](#)

fn [min](#)(self, other: Self) -> Self
where Self: [Sized](#),

Compares and returns the minimum of two values. [Read more](#)
1.50.0 · [source](#)

fn [clamp](#)(self, min: Self, max: Self) -> Self
where Self: [Sized](#) + [PartialOrd](#),

Restrict a value to a certain interval. [Read more](#)
[source](#)

impl [PartialEq](#) for [Config](#)

[source](#)

fn [eq](#)(&self, other: [&Config](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source](#)

impl [PartialOrd](#) for [Config](#)

[source](#)

fn [partial_cmp](#)(&self, other: [&Config](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source](#)

impl [Serialize](#) for [Config](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)

[source](#)

impl [Eq](#) for [Config](#)

[source](#)

impl [StructuralPartialEq](#) for [Config](#)

Auto Trait Implementations

[§](#)

impl [RefUnwindSafe](#) for [Config](#)

[§](#)

impl [Send](#) for [Config](#)

[§](#)

impl [Sync](#) for [Config](#)

[§](#)

impl [Unpin](#) for [Config](#)

[§](#)

impl [UnwindSafe](#) for [Config](#)

Blanket Implementations

[source](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more source§](#)

```
impl<Q, K> Comparable<K> for Q  
where Q: Ord + ?Sized, K: Borrow<Q> + ?Sized,
```

[source§](#)

```
fn compare(&self, key: &K) -> Ordering
```

Compare self to key and return their ordering. [source§](#)

```
impl<Q, K> Equivalent<K> for Q  
where Q: Eq + ?Sized, K: Borrow<Q> + ?Sized,
```

[source§](#)

```
fn equivalent(&self, key: &K) -> bool
```

Checks if this value is equivalent to the given key. [Read more source§](#)

```
impl<Q, K> Equivalent<K> for Q  
where Q: Eq + ?Sized, K: Borrow<Q> + ?Sized,
```

[source§](#)

```
fn equivalent(&self, key: &K) -> bool
```

Compare self to key and return true if they are equal. [source§](#)

```
impl<T> From<T> for T
```

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

portfolio - error

[barter](#)0.8.13

[Module error](#)

- [Enums](#)

[In barter::portfolio](#)

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Module [barter::portfolio::error](#)

Copy item path

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Expand description

Barter portfolio module specific errors.

Enums§

- [PortfolioError](#)
All errors generated in the barter::portfolio module.



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[PortfolioError](#)

[Variants](#)

- [BuilderIncomplete](#)
- [CannotEnterPositionWithExitFill](#)
- [CannotExitPositionWithEntryFill](#)
- [ParseEntrySide](#)
- [PositionExit](#)
- [RepositoryInteraction](#)

[Trait Implementations](#)

- [Debug](#)
- [Display](#)
- [Error](#)
- [From<RepositoryError>](#)

[Auto Trait Implementations](#)

- [!RefUnwindSafe](#)
- [!UnwindSafe](#)
- [Send](#)
- [Sync](#)
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[Blanket Implementations](#)

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToString](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)

- [VZip<V>](#)
- [WithSubscriber](#)

[In barter::portfolio::error](#)

[?](#)

[Change settings](#)

Enum [barter::portfolio::error::PortfolioError](#)

Copy item path

[source](#) ·

[−]

```
pub enum PortfolioError {  
    BuilderIncomplete(&'static str),  
    ParseEntrySide,  
    CannotEnterPositionWithExitFill,  
    CannotExitPositionWithEntryFill,  
    PositionExit,  
    RepositoryInteraction(RepositoryError),  
}
```

Expand description

All errors generated in the barter::portfolio module.

Variants§

§

BuilderIncomplete(&'static [str](#))

§

ParseEntrySide

§

CannotEnterPositionWithExitFill

§

CannotExitPositionWithEntryFill

§

PositionExit

§

RepositoryInteraction([RepositoryError](#))

Trait Implementations§

[source§](#)

impl [Debug](#) for [PortfolioError](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Display](#) for [PortfolioError](#)

[source§](#)

fn [fmt](#)(&self, __formatter: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Error](#) for [PortfolioError](#)

[source§](#)

fn [source](#)(&self) -> [Option](#)<&(dyn [Error](#) + 'static)>

The lower-level source of this error, if any. [Read more](#)
1.0.0 · [source§](#)

fn [description](#)(&self) -> &str

Deprecated since 1.42.0: use the Display impl or to_string()
[Read more](#)
1.0.0 · [source§](#)

fn [cause](#)(&self) -> [Option](#)<&dyn [Error](#)>

Deprecated since 1.33.0: replaced by Error::source, which can support downcasting
[source§](#)

```
fn provide<'a>(&'a self, request: &mut Request<'a>)
```

This is a nightly-only experimental API. (error_generic_member_access)
Provides type based access to context intended for error reports. [Read more source§](#)

```
impl From<RepositoryError> for PortfolioError
```

[source§](#)

```
fn from(source: RepositoryError) -> Self
```

Converts to this type from the input type.

Auto Trait Implementations§

§

```
impl !RefUnwindSafe for PortfolioError
```

§

```
impl Send for PortfolioError
```

§

```
impl Sync for PortfolioError
```

§

```
impl Unpin for PortfolioError
```

§

```
impl !UnwindSafe for PortfolioError
```

Blanket Implementations§

[source§](#)

```
impl<T> Any for T  
where T: 'static + ?Sized,
```

[source§](#)

```
fn type_id(&self) -> TypeId
```

Gets the TypeId of self. [Read more](#)

[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T
where U: From<T>,
```

[source](#)§

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source](#)§

```
impl<T> Same for T
```

§

```
type Output = T
```

Should always be `Self`

[source](#)§

```
impl<T> ToString for T
where T: Display + ?Sized,
```

[source](#)§

```
default fn to\_string(&self) -> String
```

Converts the given value to a `String`. [Read more](#)
[source](#)§

```
impl<T, U> TryFrom<U> for T
where U: Into<T>,
```

§

```
type Error = Infallible
```

The type returned in the event of a conversion error.

[source](#)§

```
fn try\_from(value: U) -> Result<T, <T as TryFrom<U>>::Error>
```

Performs the conversion.

[source](#)§

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

strategy



[barter](#)0.8.13

[SignalGenerator](#)

[Required Methods](#)

- [generate_signal](#)

[Implementors](#)

[In barter::strategy](#)

[?](#)

[Change settings](#)

Trait [barter::strategy::SignalGenerator](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub trait SignalGenerator {  
    // Required method  
    fn generate\_signal(  
        &mut self,  
        market: &MarketEvent<DataKind>  
    ) -> Option<Signal>;  
}
```

Expand description

May generate an advisory [Signal](#) as a result of analysing an input [MarketEvent](#).

Required Methods§

[source](#)

```
fn generate\_signal(&mut self, market: &MarketEvent<DataKind>) ->  
Option<Signal>
```

Optionally return a [Signal](#) given input [MarketEvent](#).

Implementors§

[source§](#)

impl [SignalGenerator](#) for [RSIStrategy](#)

[barter](#)0.8.13

[Module strategy](#)

- [Modules](#)
- [Structs](#)
- [Enums](#)
- [Traits](#)

[In crate barter](#)

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Module [barter::strategy](#)

Copy item path

[source](#) ·

[-]

Expand description

Defines a [SignalEvent](#) and [SignalForceExit](#), as well as the [SignalGenerator](#) trait for handling the generation of them. Contains an example [RSIStrategy](#) implementation that analyses a [MarketEvent](#) and may generate a new advisory [SignalEvent](#) to be analysed by the [Portfolio OrderGenerator](#).

Modules§

- [example](#)
Barter example RSI strategy [SignalGenerator](#) implementation.

Structs§

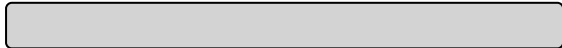
- [Signal](#)
Advisory [Signal](#) for a [Market](#) detailing the [SignalStrength](#) associated with each possible [Decision](#). Interpreted by an [OrderGenerator](#).
- [SignalForceExit](#)
Force exit [Signal](#) produced after an [Engine](#) receives a [Command::ExitPosition](#) from an external source.
- [SignalStrength](#)
Strength of an advisory [Signal](#) decision produced by [SignalGenerator](#) strategy.

Enums§

- [Decision](#)
Describes the type of advisory signal the strategy is endorsing.

Traits§

- [SignalGenerator](#)
May generate an advisory [Signal](#) as a result of analysing an input [MarketEvent](#).



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[Decision](#)

[Variants](#)

- [CloseLong](#)
- [CloseShort](#)
- [Long](#)
- [Short](#)

[Methods](#)

- [is_entry](#)
- [is_exit](#)
- [is_long](#)
- [is_short](#)

[Trait Implementations](#)

- [Clone](#)
- [Copy](#)
- [Debug](#)
- [Default](#)
- [Deserialize<'de>](#)
- [Eq](#)
- [Hash](#)
- [Ord](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [Comparable<K>](#)
- [DeserializeOwned](#)
- [Equivalent<K>](#)
- [Equivalent<K>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::strategy`

[?](#)

[Change settings](#)

Enum [`barter::strategy::Decision`](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub enum Decision {  
    Long,  
    CloseLong,  
    Short,  
    CloseShort,  
}
```

Expand description

Describes the type of advisory signal the strategy is endorsing.

Variants§

§

Long

§

CloseLong

§

Short

§

CloseShort

Implementations§

[source§](#)

impl [Decision](#)

[source](#)

pub fn [is_long](#)(&self) -> [bool](#)

Determines if a [Decision](#) is Long.

[source](#)

pub fn [is_short](#)(&self) -> [bool](#)

Determines if a [Decision](#) is Short.

[source](#)

pub fn [is_entry](#)(&self) -> [bool](#)

Determines if a [Decision](#) is an entry (long or short).

[source](#)

pub fn [is_exit](#)(&self) -> [bool](#)

Determines if a [Decision](#) is an exit (close_long or close_short).

Trait Implementations§

[source§](#)

impl [Clone](#) for [Decision](#)

[source§](#)

fn [clone](#)(&self) -> [Decision](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [Decision](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [Default](#) for [Decision](#)

[source§](#)

fn [default](#)() -> Self

Returns the “default value” for a type. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [Decision](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::[Error](#)>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [Hash](#) for [Decision](#)

[source§](#)

fn [hash](#)<__H: [Hasher](#)>(&self, state: [&mut __H](#))

Feeds this value into the given [Hasher](#). [Read more](#)
1.3.0 · [source](#)

**fn [hash_slice](#)<H>(data: [&\[Self\]](#), state: [&mut H](#))
where H: [Hasher](#), Self: [Sized](#),**

Feeds a slice of this type into the given [Hasher](#). [Read more](#)
[source](#)

impl [Ord](#) for [Decision](#)

[source](#)

fn [cmp](#)(&self, other: [&Decision](#)) -> [Ordering](#)

This method returns an [Ordering](#) between self and other. [Read more](#)
1.21.0 · [source](#)

**fn [max](#)(self, other: Self) -> Self
where Self: [Sized](#),**

Compares and returns the maximum of two values. [Read more](#)
1.21.0 · [source](#)

**fn [min](#)(self, other: Self) -> Self
where Self: [Sized](#),**

Compares and returns the minimum of two values. [Read more](#)
1.50.0 · [source](#)

**fn [clamp](#)(self, min: Self, max: Self) -> Self
where Self: [Sized](#) + [PartialOrd](#),**

Restrict a value to a certain interval. [Read more](#)
[source](#)

impl [PartialEq](#) for [Decision](#)

[source](#)

fn [eq](#)(&self, other: [&Decision](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source](#)

impl [PartialOrd](#) for [Decision](#)

[source](#)

fn [partial_cmp](#)(&self, other: [&Decision](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source](#)

impl [Serialize](#) for [Decision](#)

[source](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)

[source§](#)

impl [Copy](#) for [Decision](#)

[source§](#)

impl [Eq](#) for [Decision](#)

[source§](#)

impl [StructuralPartialEq](#) for [Decision](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Decision](#)

§

impl [Send](#) for [Decision](#)

§

impl [Sync](#) for [Decision](#)

§

impl [Unpin](#) for [Decision](#)

§

impl [UnwindSafe](#) for [Decision](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<Q, K> [Comparable](#)<K> for Q
where Q: [Ord](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [compare](#)(&self, key: [&K](#)) -> [Ordering](#)

Compare self to key and return their ordering.
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Checks if this value is equivalent to the given key. [Read more](#)
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Compare self to key and return true if they are equal.

[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



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[SignalStrength](#)

[Tuple Fields](#)

- [0](#)

[Trait Implementations](#)

- [Clone](#)
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- [Debug](#)
- [Deserialize](#)<'de>
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)

- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
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- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In barter::strategy

[?](#)

[Change settings](#)

Struct [barter::strategy::SignalStrength](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct SignalStrength(pub f64);
```

Expand description

Strength of an advisory [Signal](#) decision produced by [SignalGenerator](#) strategy.

Tuple Fields§

§0: [f64](#)

Trait Implementations§

[source§](#)

impl [Clone](#) for [SignalStrength](#)

[source§](#)

fn [clone](#)(&self) -> [SignalStrength](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [SignalStrength](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> [Deserialize](#)<'de> for [SignalStrength](#)

[source§](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

impl [PartialEq](#) for [SignalStrength](#)

[source§](#)

fn [eq](#)(&self, other: &[SignalStrength](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.

[source§](#)

impl [PartialOrd](#) for [SignalStrength](#)

[source§](#)

fn [partial_cmp](#)(&self, other: [&SignalStrength](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)

1.0.0 · [source§](#)

fn [lt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)

1.0.0 · [source§](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)

1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)

1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)

[source§](#)

impl [Serialize](#) for [SignalStrength](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)

[source§](#)

impl [Copy](#) for [SignalStrength](#)

[source§](#)

impl [StructuralPartialEq](#) for [SignalStrength](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [SignalStrength](#)

§

impl [Send](#) for [SignalStrength](#)

§

impl [Sync](#) for [SignalStrength](#)

§

impl [Unpin](#) for [SignalStrength](#)

§

impl [UnwindSafe](#) for [SignalStrength](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

```
fn borrow(&self) -> &T
```

Immutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> BorrowMut<T> for T  
where T: ?Sized,
```

[source§](#)

```
fn borrow_mut(&mut self) -> &mut T
```

Mutably borrows from an owned value. [Read more](#)
[source§](#)

```
impl<T> From<T> for T
```

[source§](#)

```
fn from(t: T) -> T
```

Returns the argument unchanged.

[source§](#)

```
impl<T> Instrument for T
```

[source§](#)

```
fn instrument(self, span: Span) -> Instrumented<Self>
```

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
fn in_current_span(self) -> Instrumented<Self>
```

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

```
impl<T, U> Into<U> for T  
where U: From<T>,
```

[source§](#)

```
fn into(self) -> U
```

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From<T>](#) for U chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be Self

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)

[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)

[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.

[source](#)§

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.

[source](#)§

```
impl<T, U> TryInto<U> for T
where U: TryFrom<T>,
```

[§](#)

```
type Error = <U as TryFrom<T>>::Error
```

The type returned in the event of a conversion error.
[source§](#)

```
fn try\_into(self) -> Result<U, <U as TryFrom<T>>::Error>
```

Performs the conversion.
[source§](#)

```
impl<V, T> VZip<V> for T
where V: MultiLane<T>,
```

[source§](#)

```
fn vzip(self) -> V
```

[source§](#)

```
impl<T> WithSubscriber for T
```

[source§](#)

```
fn with\_subscriber<S>(self, subscriber: S) -> WithDispatch<Self>
where S: Into<Dispatch>,
```

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
fn with\_current\_subscriber(self) -> WithDispatch<Self>
```

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source§](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```



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SignalForceExit

Fields

- [exchange](#)
- [instrument](#)
- [time](#)

Associated Constants

- [FORCED_EXIT_SIGNAL](#)

Methods

- [new](#)

Trait Implementations

- [Clone](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [Eq](#)
- [From<M>](#)
- [PartialEq](#)
- [PartialOrd](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [Equivalent<K>](#)
- [Equivalent<K>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)

- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In [barter::strategy](#)

[?](#)

[Change settings](#)

Struct [barter::strategy::SignalForceExit](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct SignalForceExit {
    pub time: DateTime<Utc>,
    pub exchange: Exchange,
    pub instrument: Instrument,
}
```

Expand description

Force exit Signal produced after an [Engine](#) receives a [Command::ExitPosition](#) from an external source.

Fields§

§time: [DateTime<Utc>](#)§exchange: [Exchange](#)§instrument: [Instrument](#)

Implementations§

[source](#)§

impl [SignalForceExit](#)

[source](#)

pub const [FORCED_EXIT_SIGNAL](#): &'static **str** = "SignalForcedExit"

[source](#)

```
pub fn new<E, I>(exchange: E, instrument: I) -> Self  
where E: Into<Exchange>, I: Into<Instrument>,
```

Constructs a new [Self](#) using the configuration provided.

Trait Implementations§

[source§](#)

impl **Clone** for **SignalForceExit**

[source§](#)

```
fn clone(&self) -> SignalForceExit
```

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

```
fn clone_from(&mut self, source: &Self)
```

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl **Debug** for **SignalForceExit**

[source§](#)

```
fn fmt(&self, f: &mut Formatter<'_>) -> Result
```

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl<'de> **Deserialize**<'de> for **SignalForceExit**

[source§](#)

```
fn deserialize<__D>(__deserializer: __D) -> Result<Self, __D::Error>  
where __D: Deserializer<'de>,
```

Deserialize this value from the given Serde deserializer. [Read more](#)
[source§](#)

```
impl<M> From<M> for SignalForceExit  
where M: Into<Market>,
```

[source§](#)

fn [from](#)(market: M) -> Self

Converts to this type from the input type.
[source§](#)

impl [PartialEq](#) for [SignalForceExit](#)

[source§](#)

fn [eq](#)(&self, other: &[SignalForceExit](#)) -> bool

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> bool

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [PartialOrd](#) for [SignalForceExit](#)

[source§](#)

fn [partial_cmp](#)(&self, other: &[SignalForceExit](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source§](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> bool

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source§](#)

fn [le](#)(&self, other: &[Rhs](#)) -> bool

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: &[Rhs](#)) -> bool

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [SignalForceExit](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Eq](#) for [SignalForceExit](#)

[source§](#)

impl [StructuralPartialEq](#) for [SignalForceExit](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [SignalForceExit](#)

§

impl [Send](#) for [SignalForceExit](#)

§

impl [Sync](#) for [SignalForceExit](#)

§

impl [Unpin](#) for [SignalForceExit](#)

§

impl [UnwindSafe](#) for [SignalForceExit](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Checks if this value is equivalent to the given key. [Read more](#)
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Compare self to key and return true if they are equal.

[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

impl<T> [ToOwned](#) for T
where T: [Clone](#),

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,



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[Signal](#)

[Fields](#)

- [exchange](#)
- [instrument](#)
- [market_meta](#)
- [signals](#)
- [time](#)

[Trait Implementations](#)

- [Clone](#)
- [Debug](#)
- [Deserialize](#)<'de>

- [PartialEq](#)
- [Serialize](#)
- [StructuralPartialEq](#)

Auto Trait Implementations

- [RefUnwindSafe](#)
- [Send](#)
- [Sync](#)
- [Unpin](#)
- [UnwindSafe](#)

Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [DeserializeOwned](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::strategy`

?

[Change settings](#)

Struct [barter::strategy::Signal](#)

Copy item path

[source](#) ·

[\[-\]](#)

```
pub struct Signal {
    pub time: DateTime<Utc>,
    pub exchange: Exchange,
    pub instrument: Instrument,
    pub signals: HashMap<Decision, SignalStrength>,
    pub market_meta: MarketMeta,
}
```

Expand description

Advisory [Signal](#) for a [Market](#) detailing the [SignalStrength](#) associated with each possible [Decision](#). Interpreted by an [OrderGenerator](#).

Fields

`$time: DateTime<Utc>$exchange: Exchange$instrument: Instrument$signals: HashMap<Decision, SignalStrength>$market_meta: MarketMeta`

Metadata propagated from the [MarketEvent](#) that yielded this [Signal](#).

Trait Implementations

[source](#)

impl [Clone](#) for [Signal](#)

[source](#)

fn [clone](#)(&self) -> [Signal](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)

impl [Debug](#) for [Signal](#)

[source](#)

fn [fmt](#)(&self, f: &mut [Formatter<'_,>](#)) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl<'de> [Deserialize<'de>](#) for [Signal](#)

[source](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer<'de>](#),

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl [PartialEq](#) for [Signal](#)

[source§](#)

```
fn eq(&self, other: &Signal) -> bool
```

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source§](#)

```
fn ne(&self, other: &Rhs) -> bool
```

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source§](#)

impl [Serialize](#) for [Signal](#)

[source§](#)

```
fn serialize<__S>(&self, __serializer: __S) -> Result<__S::Ok,  
__S::Error>  
where __S: Serializer,
```

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [StructuralPartialEq](#) for [Signal](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Signal](#)

§

impl [Send](#) for [Signal](#)

§

impl [Sync](#) for [Signal](#)

§

impl [Unpin](#) for [Signal](#)

§

impl [UnwindSafe](#) for [Signal](#)

Blanket Implementations§

[source§](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source§](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source§](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source§](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.

[source§](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more source§](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more source§](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source§](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source§](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source§](#)

fn [vzip](#)(self) -> V

[source§](#)

impl<T> [WithSubscriber](#) for T

[source§](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source§](#)

impl<T> [DeserializeOwned](#) for T
where T: for<'de> [Deserialize](#)<'de>,

strategy - example



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[RSIStrategy](#)

[Methods](#)

- [new](#)

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- [Debug](#)
- [SignalGenerator](#)

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Blanket Implementations

- [Any](#)
- [Borrow<T>](#)
- [BorrowMut<T>](#)
- [From<T>](#)
- [Instrument](#)
- [Into<U>](#)
- [Same](#)
- [ToOwned](#)
- [TryFrom<U>](#)
- [TryInto<U>](#)
- [VZip<V>](#)
- [WithSubscriber](#)

In `barter::strategy::example`

[?](#)

[Change settings](#)

Struct [barter::strategy::example::RSIStrategy](#)

Copy item path

[source](#) ·

[−]

```
pub struct RSIStrategy { /* private fields */ }
```

Expand description

Example RSI based strategy that implements [SignalGenerator](#).

Implementations§

[source§](#)

impl [RSIStrategy](#)

[source](#)

```
pub fn new(config: Config) -> Self
```

Constructs a new [RSIStrategy](#) component using the provided configuration struct.

Trait Implementations§

[source§](#)

impl [Clone](#) for [RSIStrategy](#)

[source§](#)

fn [clone](#)(&self) -> [RSIStrategy](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source§](#)

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source§](#)

impl [Debug](#) for [RSIStrategy](#)

[source§](#)

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source§](#)

impl [SignalGenerator](#) for [RSIStrategy](#)

[source§](#)

fn [generate_signal](#)(&mut self, market: &[MarketEvent](#)<[DataKind](#)>) -> [Option](#)<[Signal](#)>

Optionally return a [Signal](#) given input [MarketEvent](#).

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [RSIStrategy](#)

§

impl [Send](#) for [RSIStrategy](#)

§

impl [Sync](#) for [RSIStrategy](#)

[§](#)

impl [Unpin](#) for [RSIStrategy](#)

[§](#)

impl [UnwindSafe](#) for [RSIStrategy](#)

Blanket Implementations

[source](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [From](#)<T> for T

[source](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)

[source§](#)

**impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,**

[source§](#)

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of [From](#)<T> for U chooses to do.

[source§](#)

impl<T> [Same](#) for T

[§](#)

type [Output](#) = T

Should always be Self

[source§](#)

**impl<T> [ToOwned](#) for T
where T: [Clone](#),**

[§](#)

type [Owned](#) = T

The resulting type after obtaining ownership.
[source](#)

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

[§](#)

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

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[Module example](#)

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Expand description

Barter example RSI strategy [SignalGenerator](#) implementation.

Structs

- [Config](#)

Configuration for constructing a [RSIStrategy](#) via the new() constructor method.

- [RSIStrategy](#)
Example RSI based strategy that implements [SignalGenerator](#).



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[Config](#)

[Fields](#)

- [rsi_period](#)

[Trait Implementations](#)

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- [Copy](#)
- [Debug](#)
- [Deserialize<'de>](#)
- [Eq](#)
- [PartialEq](#)
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- [VZip<V>](#)

- [WithSubscriber](#)

[In barter::strategy::example](#)

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Struct [barter::strategy::example::Config](#)

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```
pub struct Config {  
    pub rsi_period: usize,  
}
```

Expand description

Configuration for constructing a [RSIStrategy](#) via the new() constructor method.

Fields§

§rsi_period: [usize](#)

Trait Implementations§

[source](#)§

impl [Clone](#) for [Config](#)

[source](#)§

fn [clone](#)(&self) -> [Config](#)

Returns a copy of the value. [Read more](#)
1.0.0 · [source](#)§

fn [clone_from](#)(&mut self, source: [&Self](#))

Performs copy-assignment from source. [Read more](#)
[source](#)§

impl [Debug](#) for [Config](#)

[source](#)§

fn [fmt](#)(&self, f: &mut [Formatter](#)<'_>) -> [Result](#)

Formats the value using the given formatter. [Read more](#)
[source](#)

impl<'de> [Deserialize](#)<'de> for [Config](#)

[source](#)

fn [deserialize](#)<__D>(__deserializer: __D) -> [Result](#)<Self, __D::Error>
where __D: [Deserializer](#)<'de>,

Deserialize this value from the given Serde deserializer. [Read more](#)
[source](#)

impl [PartialEq](#) for [Config](#)

[source](#)

fn [eq](#)(&self, other: &[Config](#)) -> [bool](#)

This method tests for self and other values to be equal, and is used by ==.
1.0.0 · [source](#)

fn [ne](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests for !=. The default implementation is almost always sufficient, and should not be overridden without very good reason.
[source](#)

impl [PartialOrd](#) for [Config](#)

[source](#)

fn [partial_cmp](#)(&self, other: &[Config](#)) -> [Option](#)<[Ordering](#)>

This method returns an ordering between self and other values if one exists. [Read more](#)
1.0.0 · [source](#)

fn [lt](#)(&self, other: &[Rhs](#)) -> [bool](#)

This method tests less than (for self and other) and is used by the < operator. [Read more](#)
1.0.0 · [source](#)

fn [le](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests less than or equal to (for self and other) and is used by the <= operator. [Read more](#)
1.0.0 · [source§](#)

fn [gt](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than (for self and other) and is used by the > operator. [Read more](#)
1.0.0 · [source§](#)

fn [ge](#)(&self, other: [&Rhs](#)) -> [bool](#)

This method tests greater than or equal to (for self and other) and is used by the >= operator. [Read more](#)
[source§](#)

impl [Serialize](#) for [Config](#)

[source§](#)

fn [serialize](#)<__S>(&self, __serializer: __S) -> [Result](#)<__S::Ok, __S::Error>
where __S: [Serializer](#),

Serialize this value into the given Serde serializer. [Read more](#)
[source§](#)

impl [Copy](#) for [Config](#)

[source§](#)

impl [Eq](#) for [Config](#)

[source§](#)

impl [StructuralPartialEq](#) for [Config](#)

Auto Trait Implementations§

§

impl [RefUnwindSafe](#) for [Config](#)

§

impl [Send](#) for [Config](#)

[§](#)

impl [Sync](#) for [Config](#)

[§](#)

impl [Unpin](#) for [Config](#)

[§](#)

impl [UnwindSafe](#) for [Config](#)

Blanket Implementations

[source](#)

impl<T> [Any](#) for T
where T: 'static + ?[Sized](#),

[source](#)

fn [type_id](#)(&self) -> [TypeId](#)

Gets the TypeId of self. [Read more](#)
[source](#)

impl<T> [Borrow](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow](#)(&self) -> [&T](#)

Immutably borrows from an owned value. [Read more](#)
[source](#)

impl<T> [BorrowMut](#)<T> for T
where T: ?[Sized](#),

[source](#)

fn [borrow_mut](#)(&mut self) -> [&mut T](#)

Mutably borrows from an owned value. [Read more](#)
[source](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Checks if this value is equivalent to the given key. [Read more](#)
[source§](#)

impl<Q, K> [Equivalent](#)<K> for Q
where Q: [Eq](#) + ?[Sized](#), K: [Borrow](#)<Q> + ?[Sized](#),

[source§](#)

fn [equivalent](#)(&self, key: [&K](#)) -> [bool](#)

Compare self to key and return true if they are equal.
[source§](#)

impl<T> [From](#)<T> for T

[source§](#)

fn [from](#)(t: T) -> T

Returns the argument unchanged.

[source§](#)

impl<T> [Instrument](#) for T

[source§](#)

fn [instrument](#)(self, span: [Span](#)) -> [Instrumented](#)<Self>

Instruments this type with the provided [Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

fn [in_current_span](#)(self) -> [Instrumented](#)<Self>

Instruments this type with the [current Span](#), returning an Instrumented wrapper. [Read more](#)
[source§](#)

impl<T, U> [Into](#)<U> for T
where U: [From](#)<T>,

[source](#)§

fn [into](#)(self) -> U

Calls `U::from(self)`.

That is, this conversion is whatever the implementation of `From<T>` for `U` chooses to do.

[source](#)§

impl<T> [Same](#) for T

§

type [Output](#) = T

Should always be `Self`

[source](#)§

impl<T> [ToOwned](#) for T
where T: [Clone](#),

§

type [Owned](#) = T

The resulting type after obtaining ownership.

[source](#)§

fn [to_owned](#)(&self) -> T

Creates owned data from borrowed data, usually by cloning. [Read more](#)
[source](#)§

fn [clone_into](#)(&self, target: [&mut T](#))

Uses borrowed data to replace owned data, usually by cloning. [Read more](#)
[source](#)§

impl<T, U> [TryFrom](#)<U> for T
where U: [Into](#)<T>,

§

type [Error](#) = [Infallible](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_from](#)(value: U) -> [Result](#)<T, <T as [TryFrom](#)<U>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<T, U> [TryInto](#)<U> for T
where U: [TryFrom](#)<T>,**

[§](#)

type [Error](#) = <U as [TryFrom](#)<T>>::[Error](#)

The type returned in the event of a conversion error.
[source](#)

fn [try_into](#)(self) -> [Result](#)<U, <U as [TryFrom](#)<T>>::[Error](#)>

Performs the conversion.
[source](#)

**impl<V, T> [VZip](#)<V> for T
where V: [MultiLane](#)<T>,**

[source](#)

fn [vzip](#)(self) -> V

[source](#)

impl<T> [WithSubscriber](#) for T

[source](#)

**fn [with_subscriber](#)<S>(self, subscriber: S) -> [WithDispatch](#)<Self>
where S: [Into](#)<[Dispatch](#)>,**

Attaches the provided [Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)
[source](#)

fn [with_current_subscriber](#)(self) -> [WithDispatch](#)<Self>

Attaches the current [default Subscriber](#) to this type, returning a [WithDispatch](#) wrapper. [Read more](#)

[source](#)

```
impl<T> DeserializeOwned for T
where T: for<'de> Deserialize<'de>,
```

test_util

[barter](#)0.8.13

[In barter::test_util](#)

[?](#)
[Change settings](#)

Function [barter::test_util::market_event_candle](#)

Copy item path

[source](#) · [\[-\]](#)

```
pub fn market_event_candle() -> MarketEvent<DataKind>
```

Expand description

Build a [MarketEvent](#) of [DataKind::Candle](#).

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[Module test_util](#)

- [Functions](#)

[In crate barter](#)

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Module [barter::test_util](#)

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Functions§

- [fill_event](#)
Build a [FillEvent](#) for a single bought contract.
- [market_event_candle](#)
Build a [MarketEvent](#) of [DataKind::Candle](#).
- [market_event_trade](#)
Build a [MarketEvent](#) of [DataKind::PublicTrade](#) with the provided [Side](#).
- [order_event](#)
Build an [OrderEvent](#) to buy 1.0 contract.
- [position](#)
Build a [Position](#).
- [signal](#)
Build a [Signal](#).

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[In barter::test_util](#)

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[Change settings](#)

Function [barter::test_util::market_event_trade](#)

Copy item path

[source](#) ·

[−]

```
pub fn market_event_trade(side: Side) -> MarketEvent<DataKind>
```

Expand description

Build a [MarketEvent](#) of [DataKind::PublicTrade](#) with the provided [Side](#).

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[In barter::test_util](#)

[?](#)
[Change settings](#)

Function [barter::test_util::position](#)

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pub fn position() -> [Position](#)

Expand description

Build a [Position](#).

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[In barter::test_util](#)

[?](#)
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Function [barter::test_util::fill_event](#)

Copy item path

[source](#) ·

[-]

pub fn fill_event() -> [FillEvent](#)

Expand description

Build a [FillEvent](#) for a single bought contract.

[barter](#)0.8.13

[In barter::test_util](#)

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Function [barter::test_util::order_event](#)

Copy item path

[source](#) ·

pub fn order_event() -> [OrderEvent](#)

Expand description

Build an [OrderEvent](#) to buy 1.0 contract.

[barter](#)0.8.13

[In barter::test_util](#)

?

[Change settings](#)

Function [barter::test_util::signal](#)

Copy item path

[source](#) ·

pub fn signal() -> [Signal](#)

Expand description

Build a [Signal](#).